TLx49012

User manual

Scope and purpose

This is the user manual of the XENSIV™ TLx49012 magnetic angle sensor family. It contains detailed instructions on how to configure the device and how to use it in an application. The document also offers reference measurements and implementation guidance. The system integrator must ensure that the sensor configuration is fully evaluated and verified for the intended environmental conditions.

For the device's specification, please refer to the device’s data sheet.

Intended audience

Hardware and software integrators using one of the related device variants of the digital angle sensor TLx49012.

Related devices

The table below lists the devices covered by this document. Interfaces highlighted in bold define the pre-configured interface of the corresponding sensor product.

Name	Description and features	Package	Marking
TLE49012-S0001	Interface: **SPI**, ABZ, UVW, PWM Resolution: 16-bit Temperature: -40°C to 150°C ISO 26262-compliant AEC-Q100 qualified	PG-VSON-8	E12S1
TLE49012-S0002	Interface: **ABZ**, PWM Resolution: 14-bit Temperature: -40°C to 150°C ISO 26262-compliant AEC-Q100 qualified	PG-VSON-8	E12S2
TLE49012-E0001	Interface: **UVW**, PWM Temperature: -40°C to 125°C AEC-Q100 qualified	PG-VSON-8	E12E1
TLI49012-E0001	Interface: **SPI**, ABZ, UVW, PWM Resolution: 16-bit Temperature: -40°C to 125°C	PG-VSON-8	I12E1

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1 Magnet sensor system

The TLx49012 sensor family supports flexible mechanical integration while keeping calibration effort to a minimum. This chapter outlines magnet-to-sensor alignments enabled by the family. The widely used end-of-shaft setup offers the most robust position for magnetic sensing. In some applications, however, the end of the motor shaft must stay accessible, and a sensor at the shaft end would interfere. In these cases, an out-of-shaft (off-axis) placement is advantageous. This non-ideal position is subject to intrinsic measurement errors, making error compensation mandatory. The TLx49012 family integrates an on-device system self-calibration feature (see Chapter 3.3) that automatically compensates mechanical alignment errors. This flexibility also eases integration of dual-sensor setups used to increase safety and availability of motor systems.

The sensor delivers high-precision angle sensing, supporting magnetic field ranges from 20 mT to 120 mT. The product offers a small magnetic field range (up to 70mT) and a large magnetic field range (up to 120mT) to ensure application-optimized angle sensing. The magnetic sensing range can be configured by setting "adc_max_rng_en" in <usr_config_1_reg>. To ensure measurement accuracy, it's essential to apply the magnetic field within this specified/configured range, as outlined in the product datasheet.

Typically, diametrically magnetized dipole disc magnets are used for angle sensing. Depending on the system context other magnet types might be in use. For instance, diametrically magnetized ring magnets are used to access the shaft, while multi-pole magnets, paired with the UVW interface, enable trapezoidal motor commutation. Table 1 provides an overview of recommended magnet dimensions to ensure accurate and stable angle sensing. For individual engineered system designs, Infineon provides an online simulation tool to evaluate the magnetic field based on magnet-sensor-system. Table 2 provides an example of the magnetic field strength seen by the sensor based on the magnet-sensor alignment specified in Table 1. The radial field component (X) is denoted by Br, while the tangential field component (Y) is denoted by Bt. Depending on the application requirements different types of magnet materials are available. Table 3 gives a basic overview of magnet materials and their properties.

Table 1 Setup example: magnet dimensions and air gap

Magnet type	typ. diameter	typ. height	Air gap
Disc magnet	8 mm	2.5 mm	3 mm
Ring magnet	15 mm outer 8 mm inner	5 mm	

Table 2 Magnetic field at sensor over temperature based on setup examples (NdFeB)

Configuration	Field component	-40°C	25°C	150°C
Disc magnet - EoS	B_r and B_t	67 mT	62 mT	53 mT
Ring magnet - OoS	B_r	107 mT	99 mT	84 mT
	B_t	45 mT	41 mT	35 mT

Table 3 General characteristics of magnet types

Property	Hard Ferrite	Plastic bonded NdFeB	Neodym (NdFeB)	Samarium Cobalt (SmCo)	Aluminum Nickel Cobalt (AlNiCo)
Temperature coefficient	high typ. -0.2%/K	medium typ. -0.18%/K	medium typ. -0.12%/K	low typ. -0.035%/K	low typ. -0.02%/K
Remanence	345 - 550 mT	250 - 690 mT	1050 - 1400 mT	880 - 1130 mT	600 - 1080 mT
Costs	low	medium/low	medium/high	high	high
Notes	lowest costs	variety of magnetic shapes	good magnetic characteristic	low thermal drift	low thermal drift

A typical target use case of angle sensors is the accurate and reliable sensing of a motor rotor shaft to improve efficiency and robustness of the drive system. The angle sensor thereby senses the rotating magnetic field of a magnet mounted on the motor's shaft. A distinction is made between two basic magnet-sensor-alignments.

- End of Shaft (EoS): the sensor is located on the rotation-axis of the magnet.
- Out of Shaft (OoS): the sensor is located on the horizontal symmetry-plane of the magnet (beside the magnet).

The EoS position represents the optimum sensor placement for measuring accurate B-field components and therefore an accurate angle. In OoS the measured field amplitudes mismatch by a factor of 2 to 3 which introduces an intrinsic angle error due to the magnet-sensor alignment.

In both EoS and OoS configurations, the sensor can be positioned off-axis to the major axis (EoS: axis of rotation, OoS: axial symmetry plane). This off-axis misalignment introduces an additional system-related imperfection (e.g. amplitude mismatch) in the measured B-field.

In OoS the direction of rotation is inverse to EoS due to the magnetic field characteristic. This means if the application rotates clock-wise (CW), the sensor measures a counter clock-wise (CCW) rotation in OoS alignment. The direction rotation change can be corrected by setting the "angle_dir" bit in the register <usr_config_7_reg>. The default rotation direction is counter clock-wise.

1.1 End of Shaft (EoS)

Using a disc magnet with end of shaft sensor alignment is the most prominent use case in rotor position sensing applications. The setup is the most robust magnet-sensor alignment and is clearly recommended for highly accurate angle measurement.

Infineon supports module integration of End of Shaft alignment with ring magnet, known as "integrated end-of-shaft". The sensor is placed within the ring magnet benefiting from homogeneous magnetic field characteristic within the ring magnet and therefore providing highly accurate and robust measurement results.

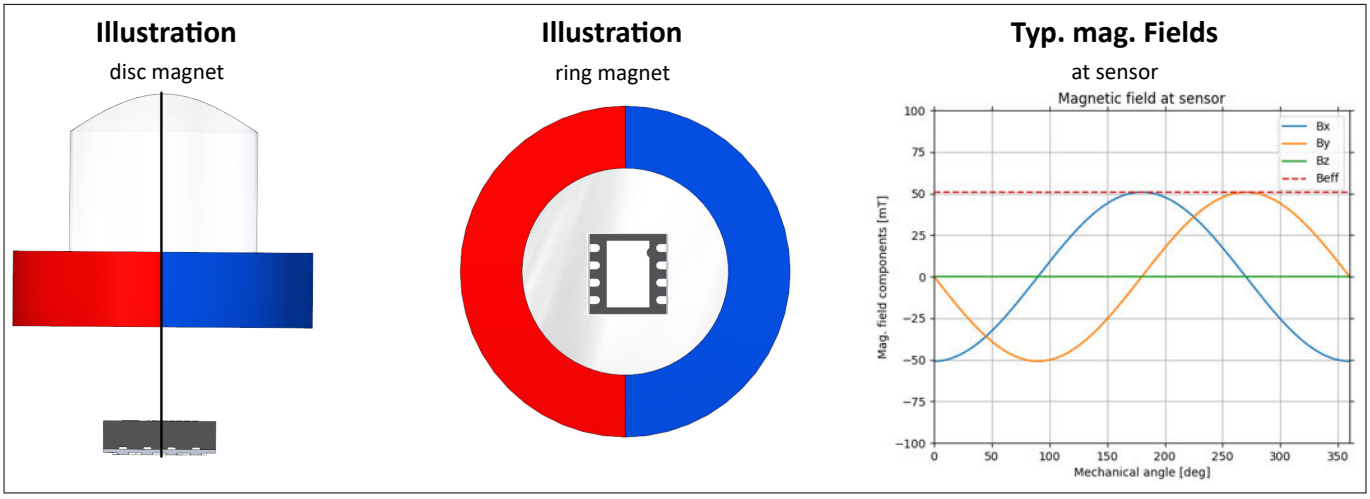


Figure 1 End of Shaft Magnet-Sensor-Alignment and typical magnetic field shape at the sensor

A misalignment in end of shaft outside the rotation axis (off axis) introduces an amplitude mismatch of the measured magnetic field which in turn causes an angle error (~2° to 5°). The introduced amplitude mismatch requires an active compensation to get an accurate angle measurement.

EoS - off axis alignment is typically used if there is very little design space e.g. due to a bearing ring or access to end of shaft is required.

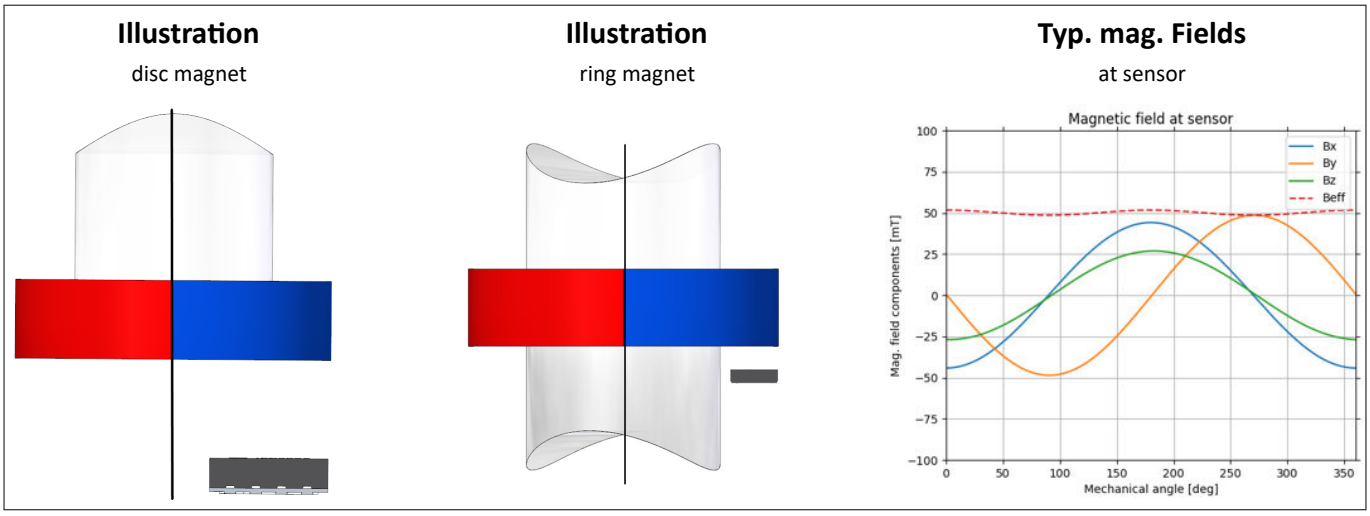


Figure 2 End of Shaft off axis Magnet-Sensor-Alignment and typical magnetic field shape at the sensor

1.2 Out of Shaft (OoS)

The out-of-shaft misalignment introduces a high amplitude mismatch of the measured magnetic field. The introduced amplitude mismatch requires an active compensation to get an accurate angle measurement.

In this setup, the planar field components (X/Y) are sensed. This setup is a typical OoS rotor position measurement e.g. as also used in trapezoidal commutated drives with Hall latches.

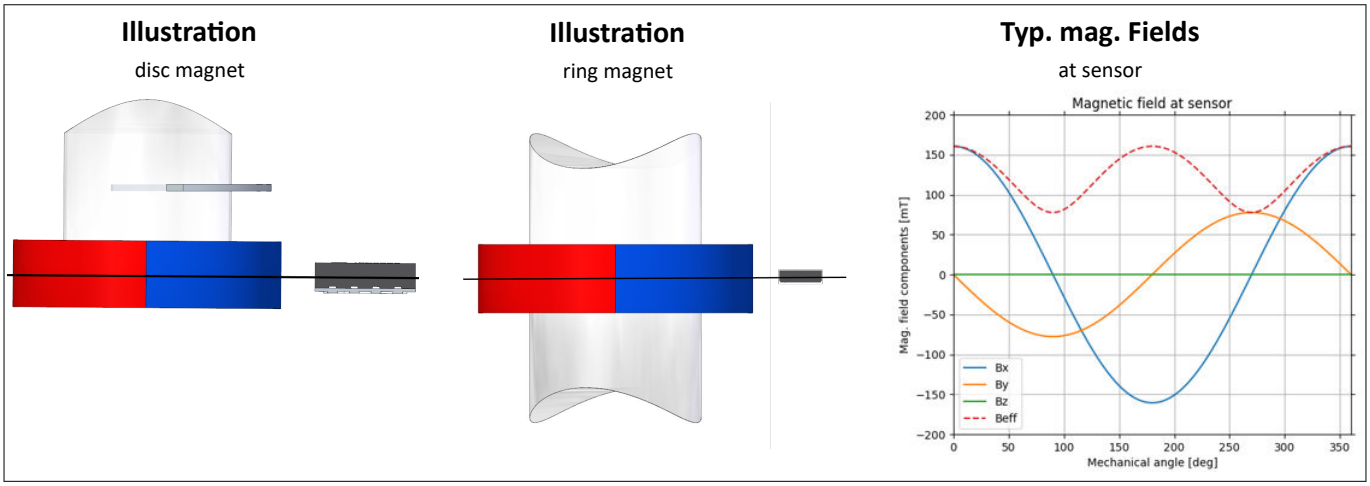


Figure 3 Out of Shaft Magnet-Sensor-Alignment and typical magnetic field shape at the sensor

An out-of-shaft off-axis position of the sensor can help to reduce measurement error by reducing the amplitude mismatch. In out of shaft off axis the sensor is displaced outside of the magnet mid-plane. Depending on the displacement range the setup can be prone to system tolerances due to the steep magnetic field gradient.

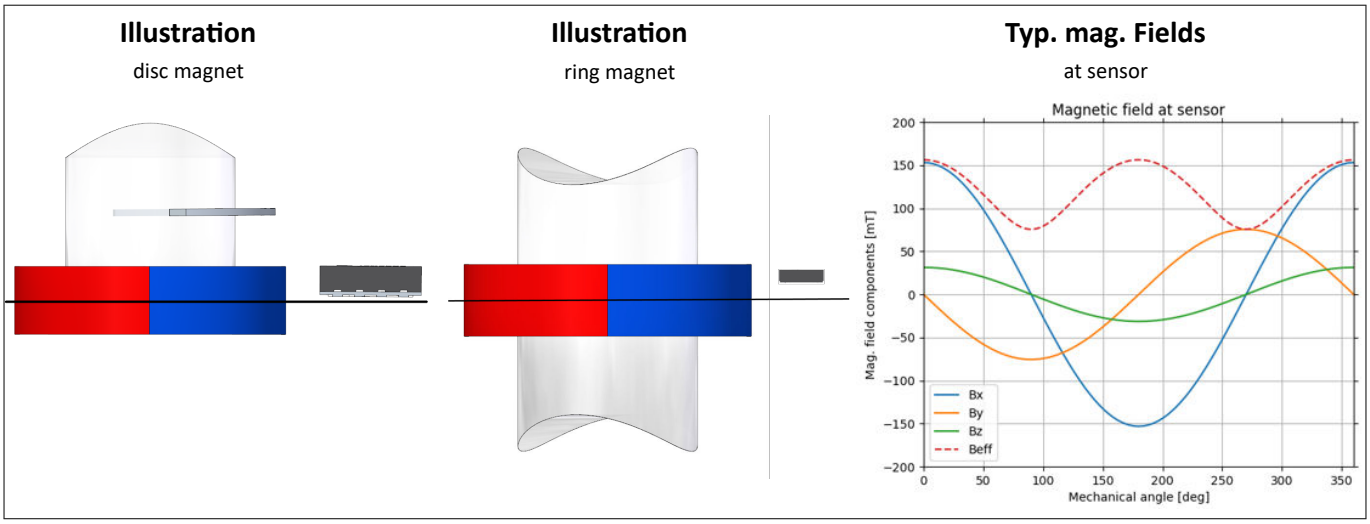


Figure 4 **Out of Shaft off axis Magnet-Sensor-Alignment and typical magnetic field shape at the sensor**

An out-of-shaft setup with vertical sensor orientation, requires sensing of the in-plane and out-of-plane field components (X/Z) to measure an angle. However, TLx49012 is optimized to sense in-plane field components (X/Y). The sensed X component of the magnetic field is constantly zero by design, thus no angle calculation is possible.

Use of Infineon's 3D magnetic sensor portfolio or using TLx49012 in "Out-of-Shaft off-axis (vertical)" alignment are alternative design options.

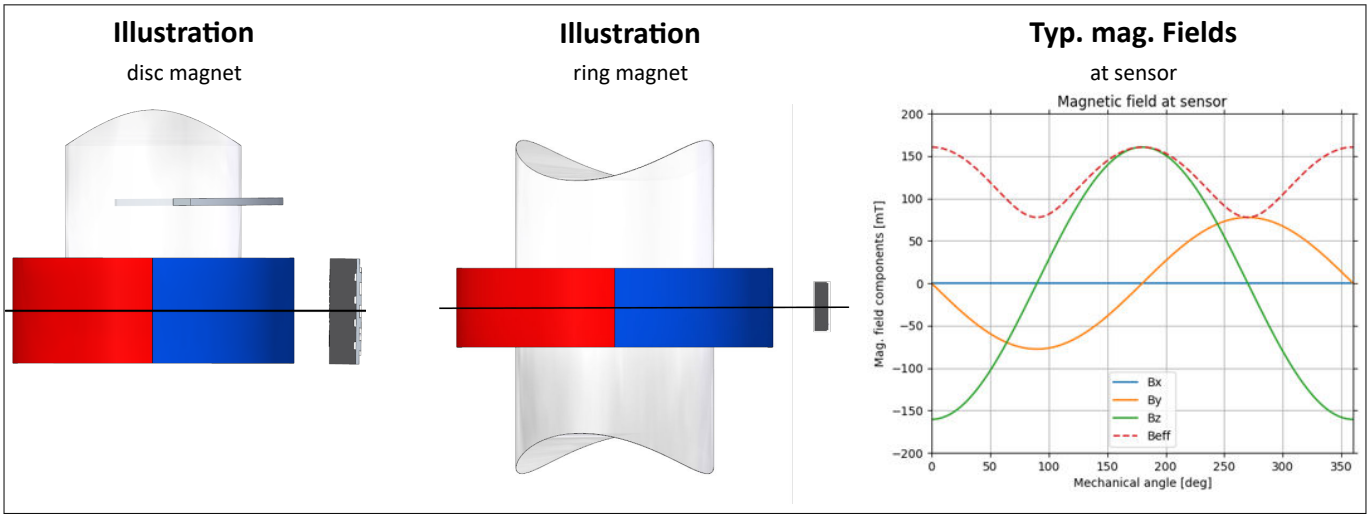


Figure 5 **Out of Shaft (vertical) Magnet-Sensor-Alignment and typical magnetic field shape at the sensor**

The out of shaft off-axis alignment enables a measurement of in-plane field components (X/Y). An angle calculation is possible with a typical angle sensor measuring the X-Y field components. This setup is required to measure the angle in out-of-shaft with vertical orientation of TLx49012.

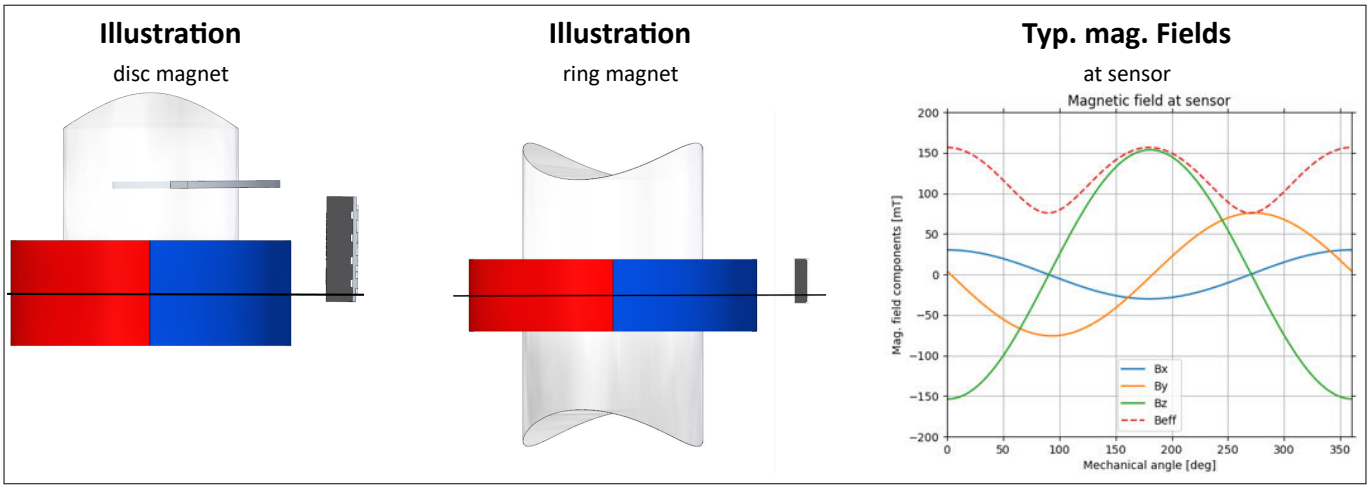


Figure 6 **Out of Shaft off axis (vertical) Magnet-Sensor-Alignment and typical magnetic field shape at the sensor**

1.3 Dual Sensor Setup

The flexibility to use the TLx49012 sensor family in both end-of-shaft (EoS) and out-of-shaft (OoS) configurations simplifies the integration of dual-sensor systems. The design flexibility allows easy integration of dual-sensor setups to improve functional safety or measurement performance using sensor fusion. Figure 7 illustrates options for magnet-sensor-alignments with two sensors. Figure 8 illustrates an example in which one sensor measures the rotor position for efficient motor commutation, while a second sensor measures the application-side position after a transmission. Such dual-sensor configurations are common in wiper and robotic systems and enable additional post-processing features, including torque estimation. Figure 9 and Figure 10 provide guidance on electrical connections for different interface types when using two sensors in parallel.

***Note**: For interface-specific application diagrams with the correct pin connections, refer to the product datasheet.*

When the synchronization feature is used, both sensors deliver time-aligned position measurements. As a result, complex timing compensation is not required before performing plausibility checks on the two independently measured positions.

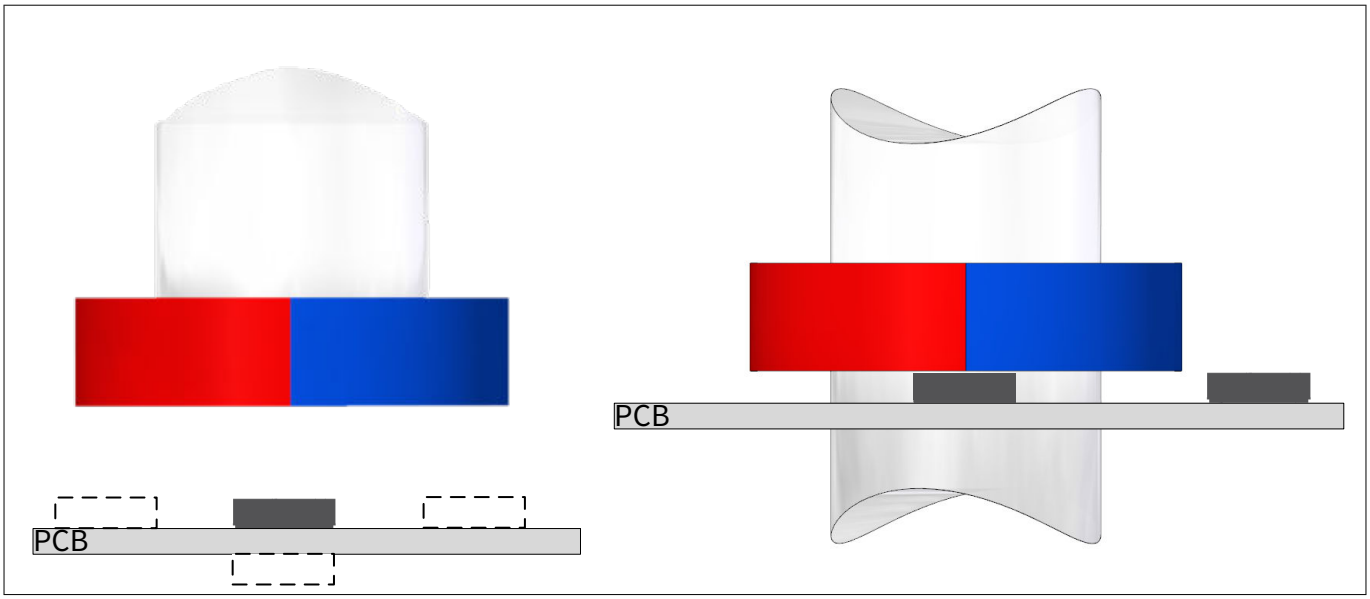


Figure 7 **Examples of magnet-to-sensor alignments using two TLx49012 sensors to increase system safety through independent redundant measurement paths**

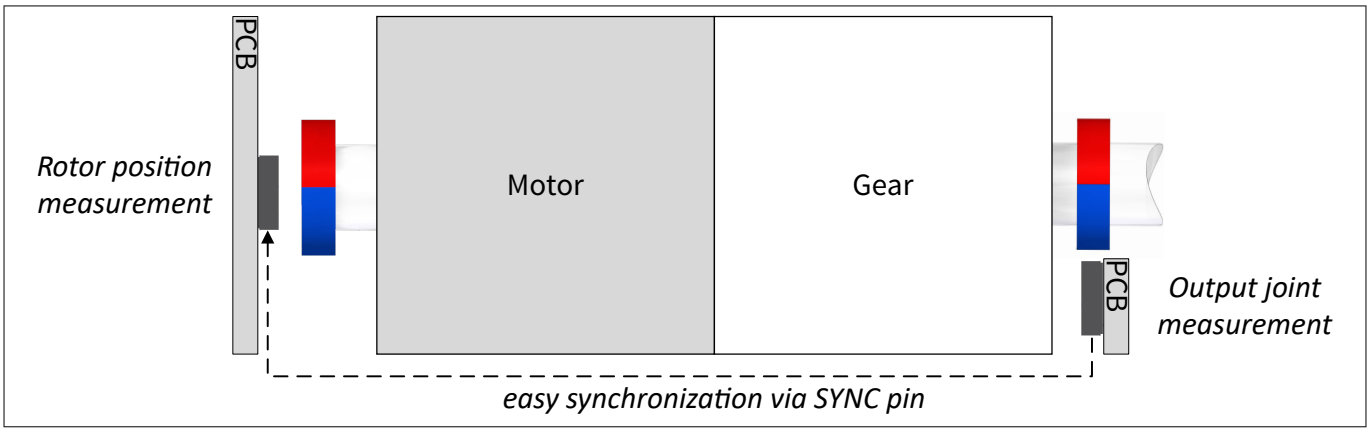


Figure 8 **Application example: Synchronization simplifies the integration of rotor-position and output-joint-position measurements for advanced sensor fusion**

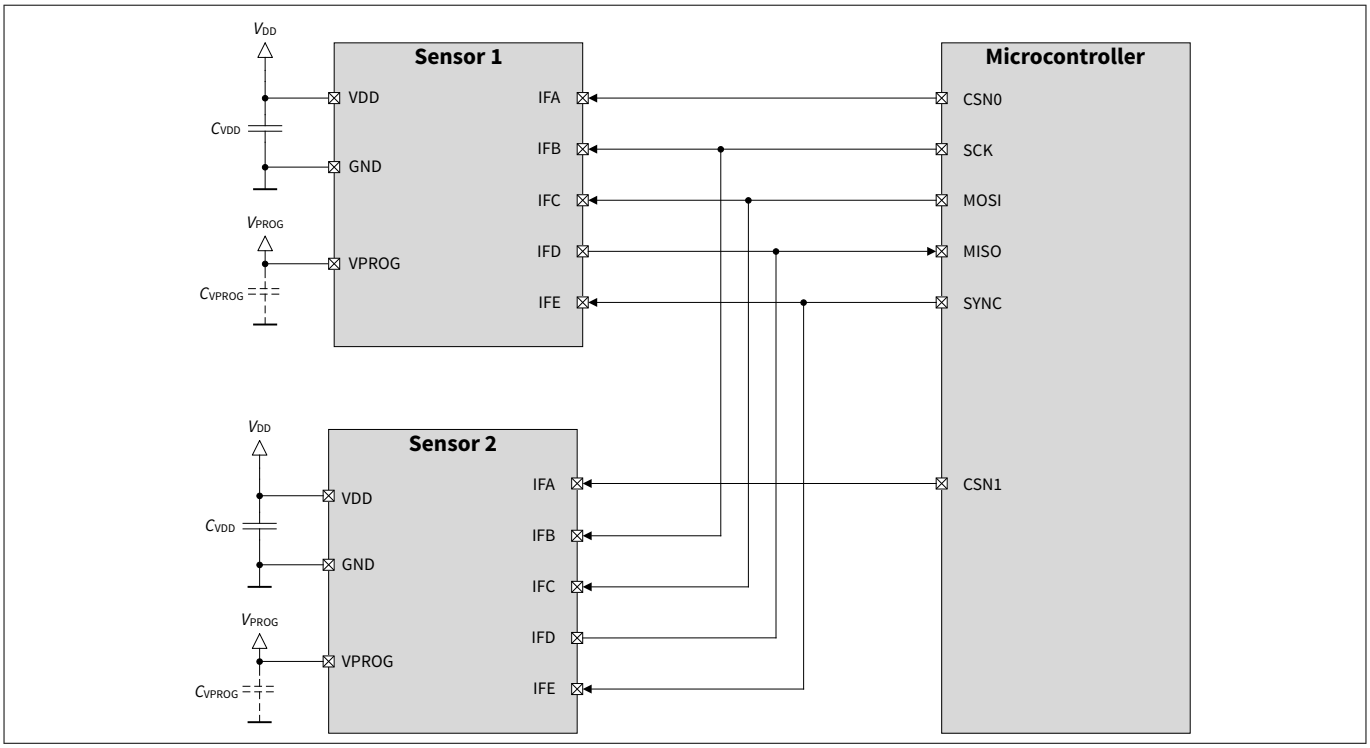


Figure 9 Dual-sensor application diagram with SPI

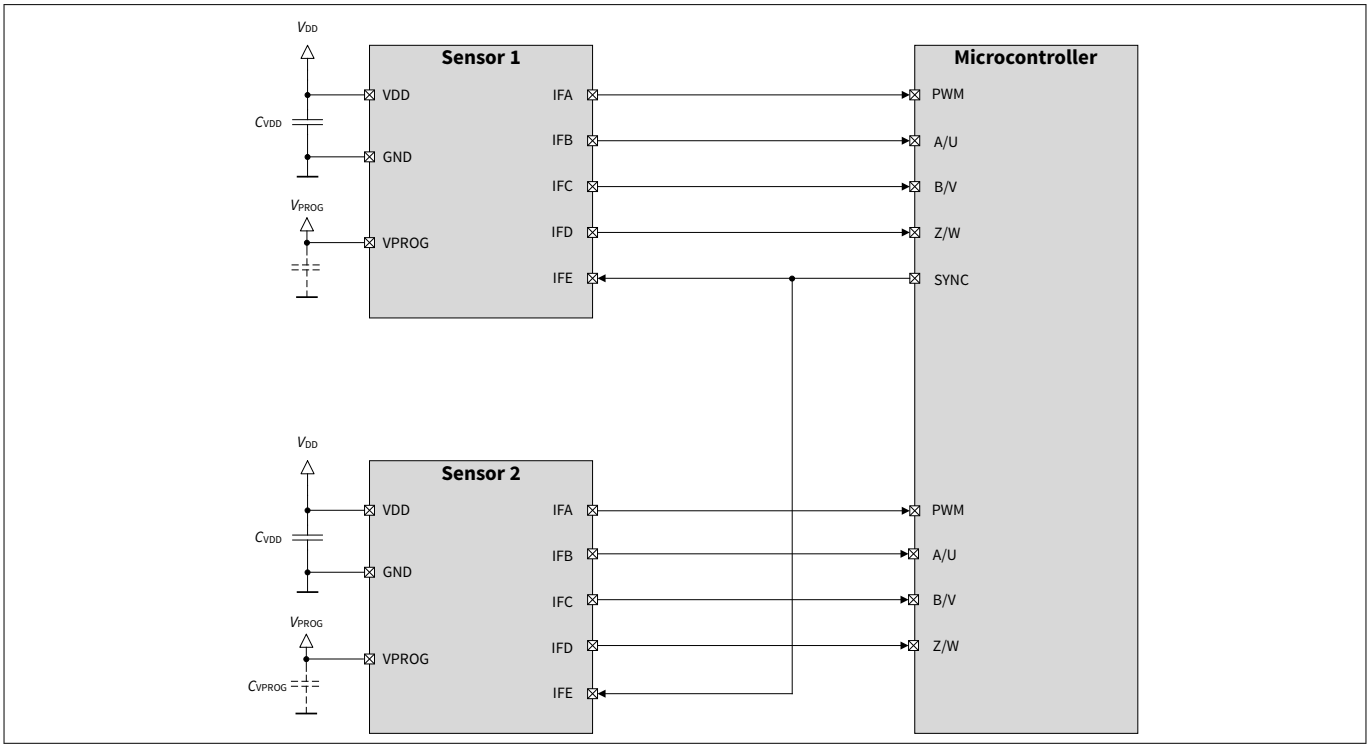
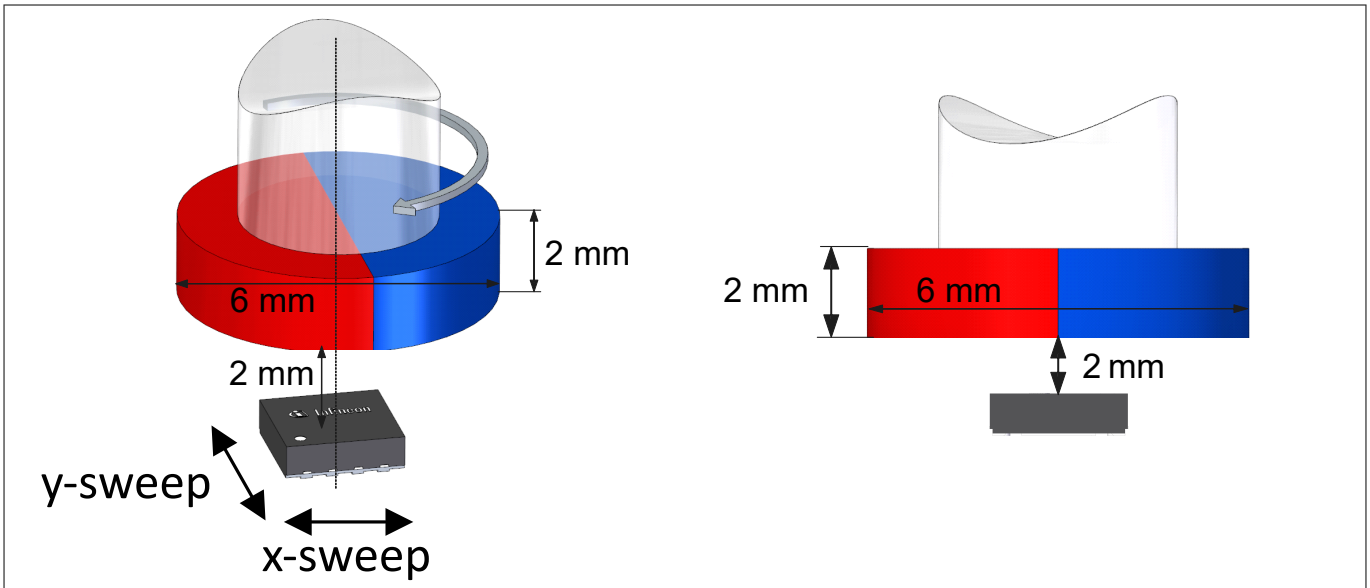


Figure 10 Dual Sensor setup with ABZ/UVW (optional PWM) interface

2 Angle accuracy performance

The XENSIV™ TLx49012 magnetic angle sensor family integrates a so-called system self-calibration feature that is capable of compensating for angle errors caused by system-related imperfections. Chapter 1 provides a basic overview of those typical system misalignments. The system self-calibration minimizes the measure angle error continuously and therefore also compensates for aging-related system drifts.

Figure 11 illustrates the measurement accuracy and robustness against mechanical displacements. The resulting measurement matrix reveals the angle accuracy achieved over mechanical displacements within the EoS plane. The integrated System self-calibration (SysCal) fully automatically provides highly accurate and robust angle measurements.



AG: 2.00mm		SysCal compensated angle error [°]								
		X Displacement [µm]								
		-200	-150	-100	-50	0	50	100	150	200
Z Displacement [µm]	-200	0.160	0.147	0.140	0.137	0.110	0.101	0.114	0.112	0.118
	-150	0.157	0.138	0.129	0.122	0.108	0.120	0.090	0.092	0.100
	-100	0.152	0.134	0.120	0.110	0.099	0.089	0.104	0.075	0.076
	-50	0.149	0.130	0.111	0.104	0.090	0.075	0.079	0.063	0.065
	0	0.130	0.122	0.109	0.097	**0.083**	0.069	0.067	0.050	0.045
	50	0.131	0.121	0.115	0.090	0.078	0.068	0.060	0.048	0.035
	100	0.132	0.127	0.112	0.089	0.083	0.069	0.056	0.054	0.031
	150	0.130	0.122	0.122	0.087	0.079	0.070	0.064	0.070	0.054
	200	0.152	0.131	0.116	0.096	0.093	0.092	0.084	0.074	0.083

Figure 11 System measurement results of TLE49012 with ±200 µm displacement range. The system self-calibration (SysCal) provides highly robust measurements against mechanical tolerances with up to 0.08° angle accuracy

3 Static error compensation

The measurement performance of the angle sensor is strongly dependent on the system setup such as mechanical tolerances, magnetic field distortions, and sensor-to-magnet alignment. One common source of error arises from out-of-shaft positioning, where the sensor is not perfectly aligned with the center of the rotating magnet. This misalignment can introduce non-linearities and offset errors in the measured angle.

The sensor supports two powerful calibration methods to compensate the system-induced error sources:

- **Look-Up Table (LUT) - one time calibration**: This method allows users to correct systematic errors by mapping raw angle readings to corrected values. It is particularly useful in systems where the error profile is consistent and can be characterized during production or setup.
- **System Self-Calibration - continuous calibration**: This feature enables the sensor to continuously self-calibrate by automatically calculating compensation values. It is ideal for dynamic environments or systems where manual calibration is impractical.

By applying one or both calibration techniques, users can significantly improve the accuracy and reliability of angle measurements, even in mechanically imperfect setups

For sensor calibration besides use of automatically running SysCal, following sequence is recommended:

- Program the angle direction conventions (CW/CCW) by setting "angle_dir" in <usr_config_7_reg>
- Evaluate and program the angle base/zero angle to "angle_base" in <usr_config_7_reg>
- Perform the angle calibration, e.g. LUT

3.1 Angle base calibration

The sensor can be configured with a so-called angle base to account for mechanical angle between 0° of the application and the 0° of the magnetic measurement. The angle base can be programmed in register <usr_config_7_reg> in bitfield "angle_base". The angular error caused by system inaccuracies (e..g OoS) is usually not constant over a full rotation. Therefore it should be avoided to use only a single measurement for the angle base calibration. Evaluating an average angle base over a full rotation minimizes the potential error due to angle base setting. When using the SysCal the angle base can be automatically corrected over a full rotation. Following procedures to evaluate and program the angle base are recommended.

3.1.1 Angle base without SysCal

Angle base evaluation recommended procedure - without SysCal:

- Simple procedure (only recommended for EoS):
  1. move the mechanical assembly to desired 0°-position
  2. read the angle from register <angle_pred_reg> - ideally average [1] over several consecutive reads to reduce noise
  3. shift "angle_pred" one bit right and write value to <usr_config_7_reg>."angle_base"
- Mean error procedure (mandatory for OoS):
  1. store the sensor angles and the position setpoint over a full rotation turn, e.g. with a step-size of 1°
  2. calculate the average angle error [1] over a full rotation by subtracting the targeted position from the measured angles over a full rotation
  3. shift average angle error one bit right and write value to <usr_config_7_reg>."angle_base"

[1] Ideally calculate as circular mean angle error to account for periodicity of angle values. Also refer to Python SciPy (circmean) documentation for more details.

3.1.2 Angle base with SysCal

Angle base evaluation recommended procedure - with SysCal:

1. move the mechanical assembly to desired 0°-position
2. read-back the angle3 [2], x3 and y3 values and write them
   - for volatile configuration to
     - <usr_config_5_reg>."x3_base" = <x3_reg>."x3_spi"
     - <usr_config_6_reg>."y3_base" = <y3_reg>."y3_spi"
     - <usr_config_7_reg>."angle_base" = <angle3_reg>."angle3" >> 1
   - or for non-volatile configuration to:
     - <cmtp_word_37_reg>."cmtp_x3_base" = <x3_reg>."x3_spi"
     - <cmtp_word_38_reg>."cmtp_y3_base" = <y3_reg>."y3_spi"
     - <cmtp_word_39_reg>."angle_base" = <angle3_reg>."angle3" >> 1

[2] optionally, use mean angle error procedure as described in [1]

3.2 Look-up table (LUT) - one time calibration

The sensor provides an on-device look-up table (LUT) to linearize for any system-induced measurement inaccuracies such as magnet-sensor-misalignments. The 32-point LUT supports two modes that can be selected in register <usr_config_10_reg> in bitfield "lut_mode"

- equidistant mode: 32 correction values can be programmed whereas each is located at a fixed location with a fixed distance 360°/32 = 11.25° to each other
- non-equidistant mode: 16 correction values can be programmed whereas each can be located at individual programmed location

Before programming the LUT, the desired LUT mode must be selected. The LUT correction values to configure the LUT transfer characteristics are located in the registers <cmtp_word_0_reg> through <cmtp_word_31_reg>. Figure 12 illustrates the two LUT modes. The LUT is disabled by default. The default transfer characteristic is linear such that the measurement results are not modified in non-equidistant mode.

***Note**: Please note that the LUT values are CRC protected to ensure data integrity. For more details refer to Chapter 7.1.2.*

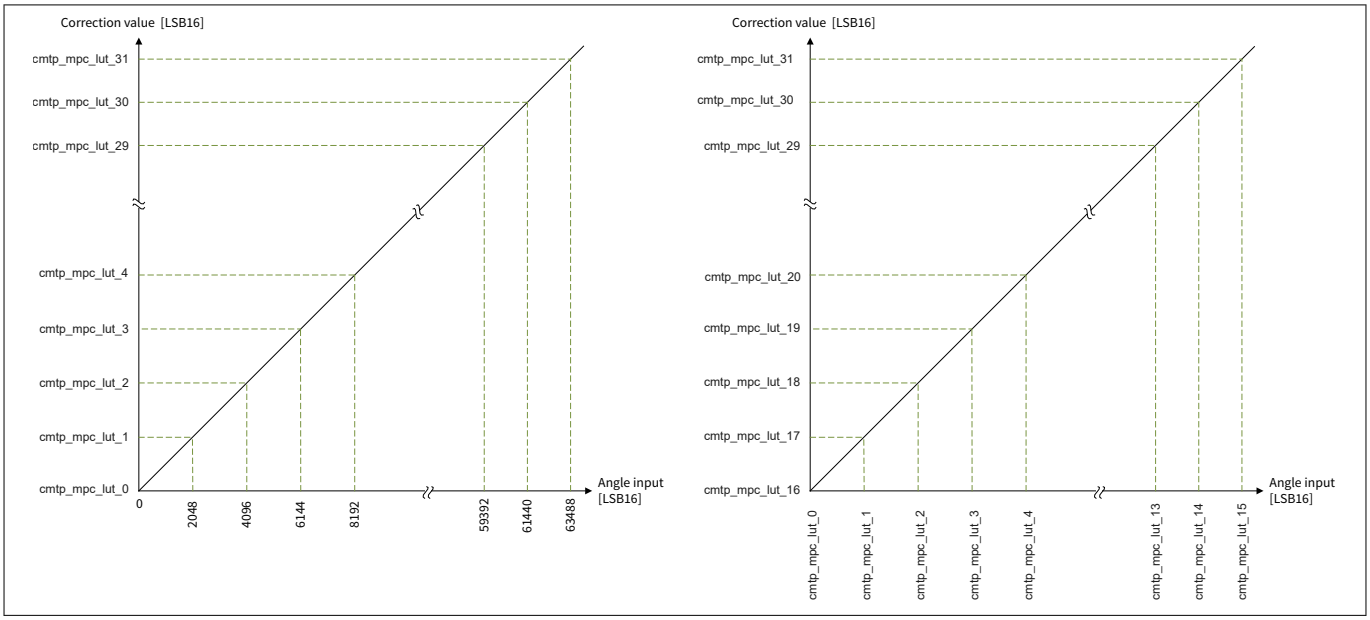


Figure 12 LUT transfer characteristic in equidistant mode (left) and non-equidistant mode (right)

***Note**: In particular in OoS setups, the angle error can be large and highly nonlinear, limiting the effectiveness of LUT compensation alone. Best accuracy is achieved by using SysCal and LUT together: SysCal pre-compensates the dominant angle error contributors (low-order harmonics), while the LUT corrects the remaining higher-order harmonics. For SysCal initialization, refer to Chapter 3.3.*

Following points must be considered when programming a LUT correction curve:

- The equidistant compensation values are distributed over a full 360° rotation. After compensation point 31 (equidistant mode) or point 15 (non-equidistant mode) compensation point 0 is used again.
- For non-equidistant LUT, the supporting points shall fulfill following constraints:
  - cmtp_mpc_lut_0 ≥ 0
  - cmtp_mpc_lut_15 < 65535_{LSB16}
  - cmtp_mpc_lut_[n+1] > cmtp_mpc_lut_[n] (for 0 ≤ n < 15)
- The slope between two subsequent correction points is limited to 2. In OoS it is therefore recommended to use the LUT in con-junction with SysCal if such steep slopes might be present. In the following formula, the slope of the LUT is denoted by g, intervals of the angle input by Δx, intervals of the correction value by Δy.

$$\left|g\right| = \left|\frac{\Delta y}{\Delta x}\right| = \left|\frac{\text{cmtp_mpc_lut_[i+1] - cmtp_mpc_lut_[i]}}{\text{ref_angle[i+1] - ref_angle[i]}}\right| < 2$$

for non-equidistant LUT:

$$16 \leq \text{i} < 31, \quad \textit{and} \quad \text{ref_angle[i]} \quad = \quad \text{cmtp_mpc_lut[i-16]} \tag{1}$$

for equidistant LUT:

$$0 \leq \text{i} < 31, \quad \textit{and} \quad \text{ref_angle[i]} \quad = \text{i} \cdot \frac{360°}{32}, \quad \textit{thus}$$

$$\Delta x = \frac{360°}{32} \quad \Rightarrow \Delta y < 2 \cdot \frac{360°}{32}$$

LUT programming

There are two options available to program the LUT:

- *volatile*: LUT values can be newly configured after each sensor reset, e.g. in evaluation phase to determine the best fitting LUT values
- *non-volatile*: final evaluated LUT values are permanently programmed to the sensor, e.g. in production phase

The LUT correction values can be directly programmed to the NVM register ranging from "cmtp_word_0_reg" to "cmtp_word_31_reg" for non-volatile programming of the LUT.

For volatile LUT configuration an "indexing flow" must be followed. Via the register <mpc_lut_idx_reg> the position within the LUT where the correction values should be written to can be indexed. The correction values can be provided in a set of four correction values placed in the registers "mpc_lut_data_0" to "mpc_lut_data_3". Figure 13 illustrates the indexing mechanism to set the desired transfer characteristic of the LUT.

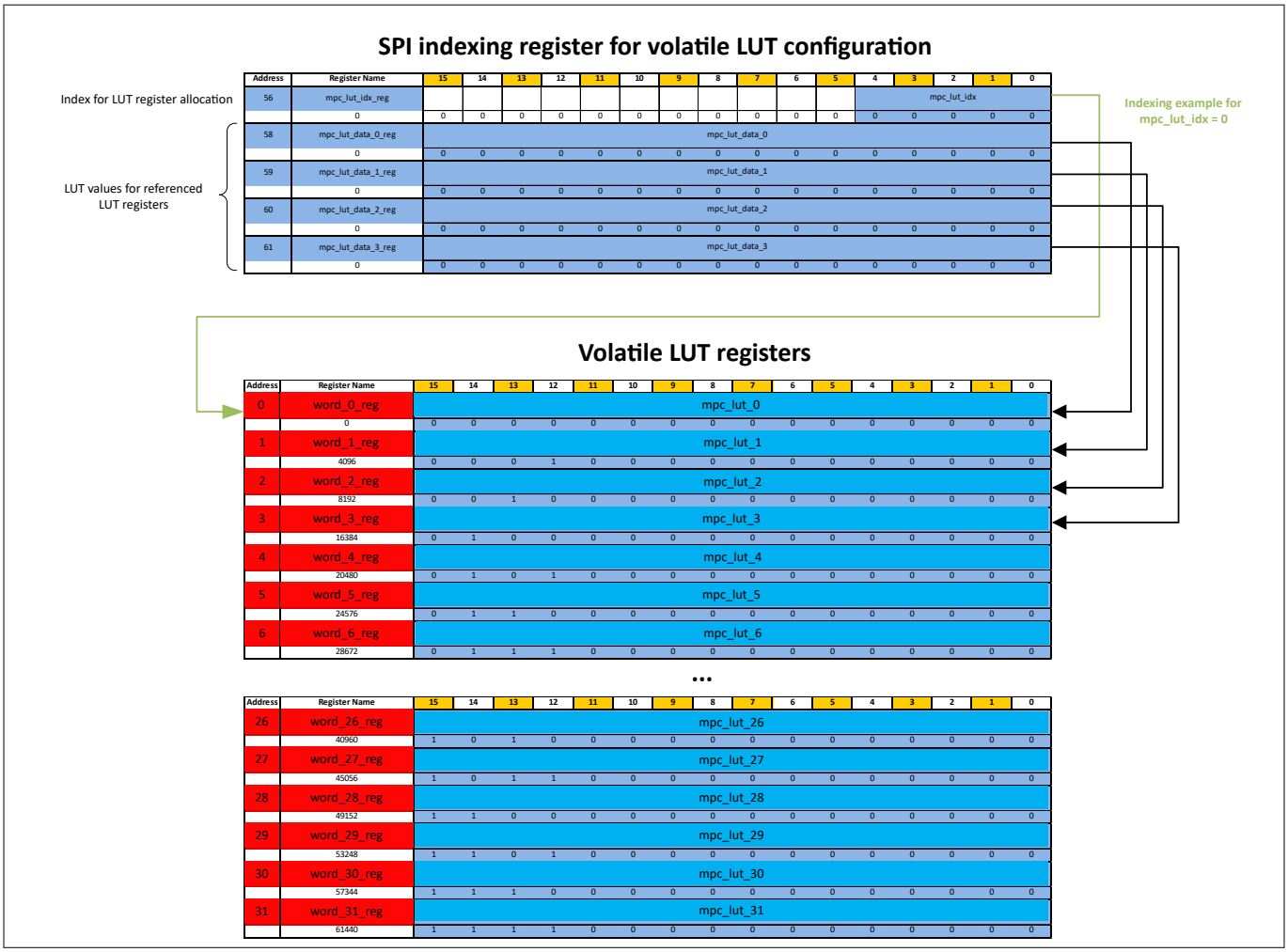


Figure 13 Illustration of LUT register indexing required for volatile LUT configuration (soft configuration)

3.3 System self-calibration (SysCal) - continuous calibration

The system self-calibration (SysCal) is an on-device compensation feature that automatically evaluates system-induced measurement errors (e.g. magnet-sensor-misalignments), calculates compensation parameters and corrects the measurement error. The compensation parameters are used to compensate measured magnetic field components X and Y which in turn are used to calculate the angle. The compensation values calculated by SysCal are available for read-out in following registers:

- Offset O_x, O_y : <syscal_offset_x_reg>, <syscal_offset_y_reg>
- Ellipticity *E:* <syscal_ellipticity_reg>
- Rotation *R:* <syscal_oos_rotation_reg>

The basic SysCal signal flow denoting the signal and register names for reading them back via SPI is shown in Figure 26. Access the compensated angle in <angle_pred_reg>, the compensated angular velocity in <angle_vel_reg>, and compensated magnetic field strength measurement in <vec_len_reg>. All other intermediate signals are available as optional add-ons for specific use cases when needed.

The SysCal is activated by default and does not require any further user input. If required, the SysCal can be deactivated or its compensation algorithm can be changed by programming <syscal_mode> in register <usr_config_10_reg>. There are two main types of modes with additional sub-options available. The quadrature modes (default) are most robust whereas the vector length modes are faster. Table 4 gives an overview of the SysCal modes.

Table 4 SysCal Modes

Bitfield "syscal_mode"	Name	Update	Comments
0	disabled	-	SysCal feature deactivated
2	Quadrature	after 3 turns	very accurate and robust - **default**
4	Quadrature with stray-field	after 3 turns	additionally to mode 2 any offsets caused by DC stray-fields are compensated
5	Vector length	after 1 turn	faster, but less accurate than quadrature modes
1	Vector length without filter	after 1 turn	similar to mode 5, but without filtering of compensation parameters - not recommended in operational use

***Note**: The chosen SysCal mode shall always be verified within the system context to avoid unintended sensor characteristics.*

Initialization

The time after sensor-startup and fully active SysCal is called initialization phase. The initial compensation values are available after the update rate which is depending on the used SysCal mode (refer to Table 4).

Please note that correct SysCal initialization **does not** impose dynamic limitations on the application - for example, a constant-speed condition during initialization turns **is not** required. Once the initialization phase is finished the sensor indicates this in the device status bit "CAL" via SPI.

During initialization phase the angle error is not compensated which lead specifically in out-of-shaft setups to large angle errors. There are following options to account for the SysCal initialization phase:

- discard measurement results until SysCal is active, if it is possible in the application
- initialize the SysCal with calibration values based on the system inaccuracies

There are three ways to determine the compensation parameters to initialize SysCal for accurate angle sensing. This is in particular useful for out-of-shaft applications or systems where no full turning is possible.

- **Simulation**: Use Infineon´s online simulation tool to simulate the resulting angle error based on typical setup condition (e.g. OoS-misalignment) and calculate a LUT.

- **Reference measurement:** With a reference setup, measure the magnetic field and angle characteristics over a full rotation and calculate the compensation parameters.
- **Sensor read-back:** Rotate the application until the SysCal is initialized (recommended: at least 10 turns) and read back the compensation parameters automatically calculated by SysCal. Use these values to program the sensor.

SysCal compensation parameters are stored in two register sets. One set contains the compensation values that SysCal calculates, continuously updates, and makes available for readout (Syscal readout values). The other set is used to write the initial values that SysCal uses for effective compensation during initialization phase (SysCal init. values).

Table 5 Compensation parameter register assignment

	SysCal readout values		SysCal init. values	
Parameter	**Register**	**Bitfield**	**Register**	**Bitfield**
Offset X	<syscal_offset_x_reg>	"syscal_offset_x"	<usr_config_8_reg>	"offset_app_x"
Offset Y	<syscal_offset_y_reg>	"syscal_offset_y"	<usr_config_9_reg>	"offset_app_y"
Ellipticity	<syscal_ellipticity_reg>	"syscal_ellipticity"	<usr_config_12_reg>	"ellipticity"
Rotation	<syscal_oos_rotation_reg>	"syscal_oos_rotation"	<usr_config_13_reg>	"oos_rotation"

The initialization values for SysCal can be either derived from the sensor by reading back the SysCal calculated compensation parameters or derived from physical units that can be evaluated with system simulations or measurements. Table 6 provides the correlation between the SysCal initializations values and the compensation parameters calculated by SysCal. Table 7 shows how the initialization values can be calculated based on physical/simulated units.

Table 6 SysCal compensation parameter transfer formulas SysCal calculated vs. readout value

Compensation parameter	init. compensation value ⇔ readout value	Comment
Offset X	"offset_app_x" = "syscal_offset_x"	
Offset Y	"offset_app_y" = "syscal_offset_y"	
Ellipticity	"ellipticity" = (E - 1) * 100 E = "syscal_ellipticity" / $2^{(13 - \text{"syscal_ellipticity_scale"})}$	E: ellipticity, ratio between minor and major axis
Rotation	"oos_rotation" = "syscal_oos_rotation" / 2^5	

Table 7 SysCal compensation parameter transfer formulas SysCal init value vs. physical value

Compensation parameter	init. compensation value ⇔ readout value	Comment
Offset X	"offset_app_x" = B_{radial}/S_{XY}	$1/S_{XY,typ}$ = 6.4 µT/LSB refer to datasheet
Offset Y	"offset_app_y" = $B_{tangential}/S_{XY}$	
Ellipticity	$E = B_{radial}/B_{tangential}$ "syscal_ellipticity" = $E * 2^{(13 - \text{"syscal_ellipticity_scale"})}$	E: ellipticity, ratio between minor and major axis
Rotation	"oos_rotation" = ρ / 360° * 2^{10}	ρ: angle between ellipse major axis and zero angle

The SysCal is deactivated as long as the magnetic field is out of range to avoid wrong compensation. Furthermore high temperature changes can cause faulty compensations. Therefore the SysCal is automatically

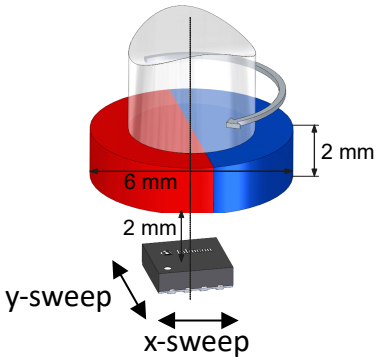
restarted if the temperature limit configured in "syscal_t_lin_lim" in the register <usr_config_14_reg> is exceeded. The temperature change limit is defined over 1 revolution for SysCal mode 5 and 1 and over 3 revolutions for SysCal mode 2 and 4.

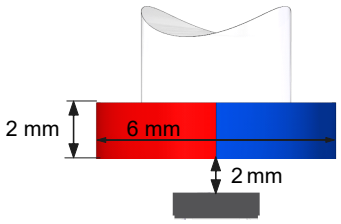
Some applications just have a limited range of rotation below 360°. The SysCal cannot calculate any compensation values without a full rotation. Depending on the rotation range available in the assembly line following options are possible to keep the effort as low as possible using SysCal:

- full turn capability in assembly line:
  - turn the application several times, e.g. 10 turns
  - read-back the compensation parameters calculated by the SysCal
- no 360° capability at all:
  - use a reference setup during design phase to evaluate compensation parameters via SysCal
  - use Infineon's online simulation tool to obtain the compensation parameters for typ. misalignments

Figure 14 shows angle accuracy measurements over sensor displaced positions in EoS. The upper measurement matrix are LUT-compensated values and the lower matrix provides the angle accuracy with SysCal applied. The greenish area indicates that the SysCal is much more robust against any mechanical displacements due to its ongoing evaluation of system-related inaccuracies and update of compensation parameters. The LUT in turn requires external reference encoders and is only working well for specific positions visible in the high accuracy degradation outside the optimal position. The SysCal is also much more robust against any other drift effects impacting the angle accuracy such as temperature or aging effects.

3 Static error compensation





AG: 2.00mm	LUT compensated angle error [°]																				
	X Displacement [mm]																				
Y Displacement [mm]	2.50	2.25	2.00	1.75	1.50	1.25	1.00	0.75	0.50	0.25	0.00	-0.25	-0.50	-0.75	-1.00	-1.25	-1.50	-1.75	-2.00	-2.25	-2.50
2.50	55.8	46.9	39.9	34.3	30.0	26.3	23.5	21.3	20.1	19.3	18.8	19.0	19.7	21.0	22.8	25.5	28.8	33.1	38.2	44.7	52.6
2.25	47.4	39.6	33.2	28.2	24.2	21.0	18.5	16.6	15.4	14.6	14.3	14.5	15.2	16.3	18.0	20.3	23.2	26.9	31.7	37.5	44.7
2.00	40.7	33.6	27.9	23.3	19.7	16.8	14.5	12.8	11.7	11.0	10.8	10.9	11.5	12.5	14.0	16.1	18.7	22.1	26.4	31.7	38.2
1.75	35.2	28.7	23.4	19.2	15.9	13.3	11.2	9.7	8.7	8.0	7.8	7.9	8.5	9.4	10.8	12.6	15.1	18.2	22.1	26.9	32.8
1.50	30.8	24.8	19.9	16.0	12.97	10.56	8.63	7.20	6.25	5.68	5.48	5.61	6.10	6.92	8.18	9.89	12.2	15.0	18.6	23.1	28.6
1.25	27.3	21.7	17.1	13.5	10.66	8.40	6.60	5.26	4.39	3.84	3.64	3.79	4.24	5.02	6.19	7.83	9.9	12.5	15.9	20.1	25.2
1.00	24.8	19.4	15.1	11.7	8.97	6.83	5.13	3.85	3.02	2.51	2.33	2.45	2.88	3.61	4.71	6.20	8.2	10.7	13.9	17.9	22.7
0.75	22.8	17.7	13.5	10.2	7.66	5.61	3.98	2.75	1.96	1.46	1.28	1.42	1.80	2.50	3.55	4.97	6.8	9.2	12.3	16.1	20.8
0.50	21.5	16.5	12.5	9.3	6.79	4.81	3.23	2.04	1.26	0.77	0.58	0.71	1.09	1.76	2.78	4.16	6.0	8.3	11.3	15.0	19.5
0.25	20.8	15.8	11.9	8.8	6.30	4.35	2.80	1.63	0.86	0.35	0.16	0.28	0.65	1.32	2.31	3.67	5.5	7.8	10.7	14.3	18.8
0.00	20.6	15.7	11.7	8.6	6.15	4.21	2.68	1.51	0.74	0.24	0.03	0.11	0.49	1.17	2.16	3.51	5.3	7.6	10.5	14.1	18.6
-0.25	20.9	15.9	11.9	8.8	6.35	4.40	2.85	1.68	0.91	0.41	0.22	0.30	0.66	1.34	2.33	3.69	5.5	7.8	10.7	14.4	18.9
-0.50	21.7	16.7	12.6	9.5	6.91	4.92	3.34	2.14	1.36	0.86	0.68	0.77	1.16	1.84	2.85	4.21	6.1	8.4	11.4	15.2	19.7
-0.75	23.2	17.9	13.8	10.5	7.84	5.81	4.18	2.95	2.15	1.66	1.47	1.57	1.93	2.67	3.71	5.13	7.0	9.5	12.6	16.5	21.2
-1.00	25.2	19.9	15.4	12.0	9.22	7.06	5.34	4.07	3.24	2.72	2.57	2.66	3.11	3.80	4.89	6.43	8.4	10.9	14.2	18.2	23.2
-1.25	28.1	22.3	17.7	14.1	11.14	8.86	7.04	5.68	4.80	4.26	4.07	4.19	4.58	5.42	6.58	8.19	10.3	13.0	16.5	20.8	26.0
-1.50	31.6	25.5	20.5	16.6	13.52	11.08	9.12	7.67	6.75	6.16	5.96	6.08	6.54	7.42	8.68	10.42	12.7	15.6	19.3	23.9	29.5
-1.75	36.2	29.6	24.2	20.0	16.65	13.99	11.87	10.29	9.28	8.64	8.43	8.56	9.10	10.03	11.43	13.30	15.8	18.9	22.9	27.9	33.9
-2.00	41.9	34.7	28.9	24.3	20.55	17.64	15.33	13.58	12.48	11.78	11.53	11.72	12.26	13.31	14.82	16.88	19.6	23.1	27.5	32.9	39.5
-2.25	49.0	41.0	34.6	29.5	25.45	22.22	19.66	17.69	16.45	15.65	15.38	15.64	16.23	17.35	19.03	21.37	24.4	28.2	33.1	39.0	46.2
-2.50	57.9	49.0	41.8	36.2	31.39	28.01	25.06	22.99	21.54	20.66	20.32	20.59	21.26	22.58	24.46	27.05	30.5	34.8	40.2	46.8	54.8

AG: 2.00mm	SysCal compensated angle eror [°]																				
	X Displacement [mm]																				
Y Displacement [mm]	2.50	2.25	2.00	1.75	1.50	1.25	1.00	0.75	0.50	0.25	0.00	-0.25	-0.50	-0.75	-1.00	-1.25	-1.50	-1.75	-2.00	-2.25	-2.50
2.50	55.7	46.9	39.9	25.44	23.47	20.92	17.62	13.79	9.79	5.85	2.59	1.56	4.48	7.49	10.29	12.95	15.20	17.24	38.1	44.6	52.7
2.25	47.4	39.6	15.8	3.52	2.73	2.32	1.88	1.58	1.51	1.28	1.26	1.20	1.22	1.29	1.43	1.72	2.28	2.89	16.4	37.5	44.6
2.00	40.7	15.7	3.3	2.63	2.10	1.75	1.46	1.24	1.12	1.02	0.94	0.97	0.94	1.00	1.07	1.34	1.64	2.06	1.7	12.3	38.1
1.75	35.2	3.6	2.7	2.11	1.78	1.46	1.24	1.03	0.89	0.83	0.73	0.77	0.71	0.78	0.91	1.10	1.32	1.72	2.3	2.9	12.8
1.50	11.3	2.9	2.3	1.86	1.49	1.25	1.06	0.85	0.71	0.66	0.60	0.56	0.58	0.64	0.79	0.97	1.14	1.46	2.1	2.3	3.2
1.25	3.2	2.5	2.0	1.68	1.35	1.11	0.91	0.74	0.63	0.56	0.57	0.52	0.49	0.54	0.72	0.90	1.09	1.34	1.3	2.1	2.7
1.00	2.8	2.2	1.9	1.51	1.22	1.01	0.78	0.64	0.54	0.48	0.41	0.41	0.45	0.52	0.63	0.83	1.03	1.25	1.4	1.9	2.4
0.75	2.7	2.2	1.8	1.50	1.24	1.02	0.79	0.59	0.48	0.36	0.27	0.27	0.36	0.49	0.62	0.81	1.06	1.27	1.7	1.9	2.4
0.50	2.6	2.2	1.8	1.48	1.23	0.99	0.74	0.53	0.43	0.25	0.20	0.22	0.28	0.44	0.58	0.76	1.02	1.25	1.4	1.9	2.3
0.25	2.7	2.2	1.8	1.56	1.18	0.99	0.72	0.53	0.41	0.23	0.14	0.17	0.26	0.45	0.60	0.79	1.01	1.32	1.6	1.9	2.3
0.00	2.7	2.3	1.9	1.62	1.28	1.01	0.76	0.59	0.38	0.23	0.08	0.13	0.28	0.50	0.64	0.85	1.05	1.34	1.7	2.0	2.4
-0.25	2.8	2.5	2.1	1.73	1.39	1.17	0.88	0.69	0.49	0.30	0.18	0.21	0.39	0.56	0.75	0.95	1.18	1.47	2.0	2.2	2.6
-0.50	3.1	2.7	2.3	1.84	1.61	1.33	1.03	0.80	0.63	0.47	0.39	0.42	0.52	0.68	0.87	1.08	1.33	1.58	2.1	2.3	2.8
-0.75	3.2	2.9	2.4	2.11	1.73	1.48	1.29	1.02	0.83	0.71	0.66	0.67	0.73	0.90	1.07	1.28	1.48	1.74	2.5	2.6	3.1
-1.00	3.8	3.3	2.9	2.32	1.98	1.79	1.51	1.26	1.07	0.94	0.85	0.89	0.97	1.09	1.27	1.50	1.74	1.99	2.8	2.8	3.4
-1.25	4.4	3.9	3.2	2.68	2.33	2.02	1.81	1.56	1.43	1.27	1.20	1.23	1.29	1.42	1.60	1.83	2.03	2.35	3.3	3.3	3.9
-1.50	5.2	4.6	3.8	3.05	2.65	2.35	2.08	1.92	1.74	1.56	1.50	1.57	1.62	1.77	1.90	2.11	2.32	2.65	3.7	3.7	4.5
-1.75	6.4	5.7	4.6	3.65	3.15	2.80	2.51	2.30	2.17	2.03	1.96	2.00	2.05	2.18	2.28	2.52	2.81	3.13	4.3	4.4	5.5
-2.00	8.9	7.5	5.9	4.59	3.68	3.37	3.07	2.82	2.68	2.54	2.44	2.47	2.53	2.62	2.81	3.01	3.30	3.70	5.3	5.4	7.5
-2.25	20.8	10.3	7.6	5.94	4.70	3.98	3.65	3.41	3.24	3.11	2.99	3.05	3.06	3.20	3.40	3.67	4.06	4.61	6.4	7.4	11.5
-2.50	62.3	20.2	11.1	8.29	6.21	5.23	4.60	4.21	4.03	3.88	3.70	3.78	3.78	4.00	4.18	4.55	5.06	6.05	8.8	11.8	54.6

Figure 14 **SysCal angle accuracy achieves similar performance as a LUT while being much more robust against mechanical tolerances and requiring no external compensation effort**

4 Dynamic error compensation

To improve the dynamic response of angle and speed measurements, the sensor provides two configurable filter parameters: "dyn_comp_k1" and "dyn_comp_k2" located in register <usr_config_11_reg>. These control the responsiveness of the internal dynamic angle error compensation:

- K1 affects the angle estimator.
- K2 affects the speed estimator.

Higher values result in faster response reducing dynamic angle error but increase noise. Lower values reduce noise and improve stability but slow down the measurement response. Therefore, selecting optimal filter settings requires a trade-off between dynamic accuracy and noise performance.

Additionally, the dynamic compensation mode can be switched between continuous (default) and discontinuous via "dyn_comp_mode" in register <usr_config_13_reg>. The continuous mode is mandatory for the ABZ and UVW output interfaces. However, the discontinuous mode is strongly recommended for the SPI interface because it significantly reduces the velocity noise.

In most applications, the default filter configuration is fully adequate to handle the system dynamics. If additional optimization of measurement noise or dynamic response is required, Table 8 provides a comprehensive overview of available filter settings aligned with specific application requirements. The following procedures can be used to determine appropriate filter gains:

1. **Max. acceleration known**
   - **a.** Determine the filter bandwidth based on the maximum acceptable dynamic angle error (see Figure 16)
   - **b.** Select the appropriate filter setting corresponding to the chosen bandwidth (see Figure 15)
2. **Max. desired noise / resolution known**
   - **a.** Determine the filter bandwidth based on the maximum acceptable noise or resolution (see Figure 17)
   - **b.** Select the appropriate filter setting corresponding to the chosen bandwidth (see Figure 15)
3. **Max. resolved signal frequency known**
   - **a.** Determine the filter bandwidth based on the highest signal frequency that must be resolved (e.g., torque ripple).
   - **b.** Select the appropriate filter setting corresponding to the chosen bandwidth (see Figure 15)

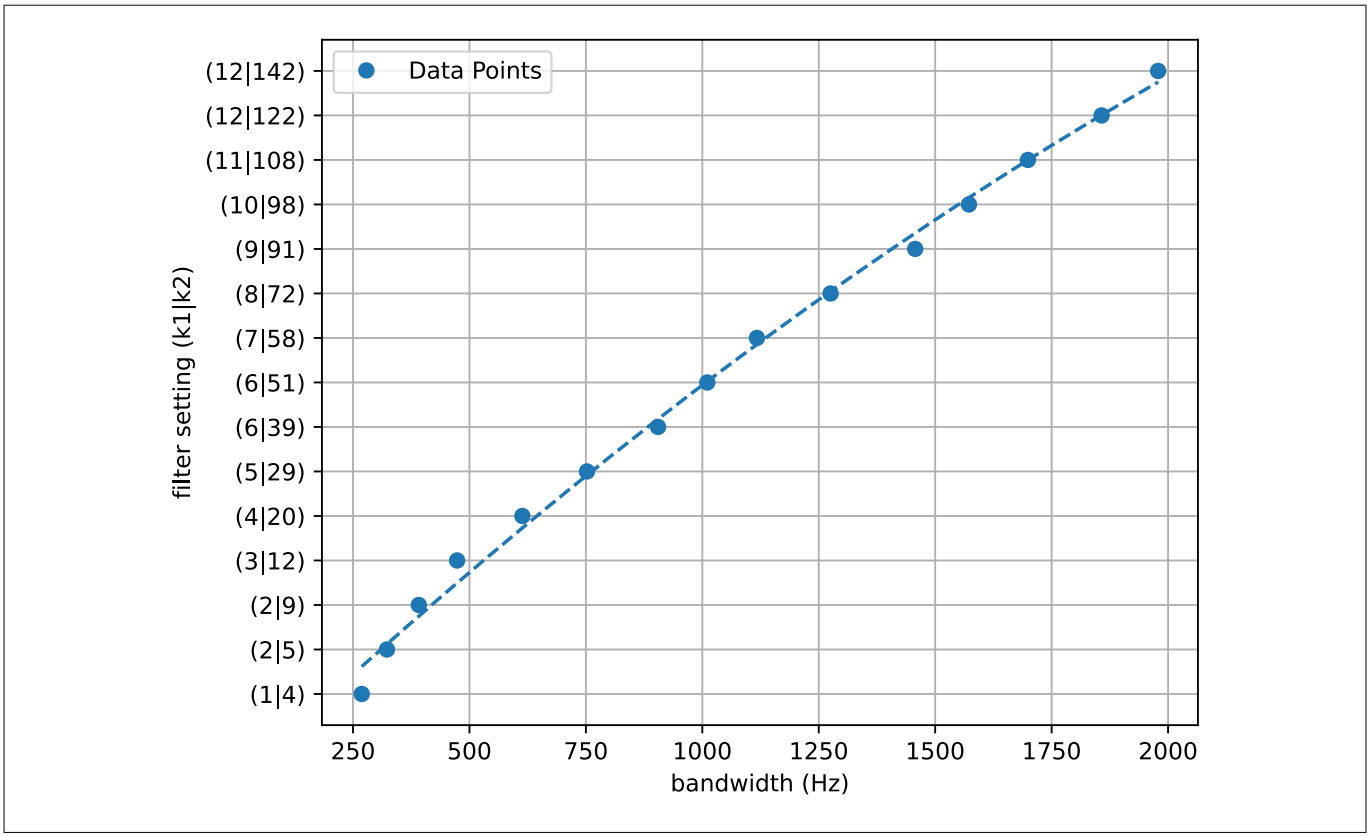


Figure 15 **Filter gain settings k1 and k2 vs. dynamic filter bandwidth**

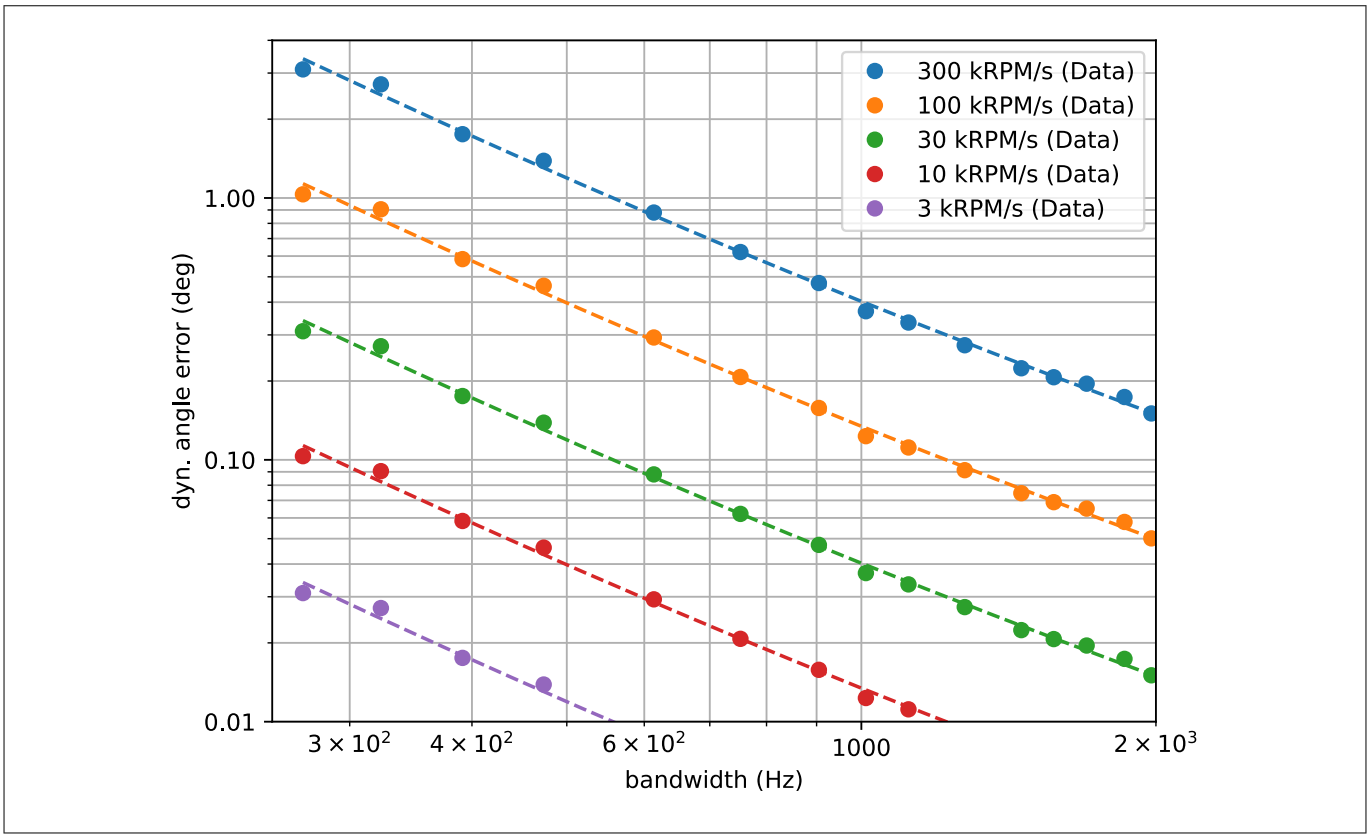


Figure 16 **Dynamic angle error vs. filter bandwidth for different application acceleration rates**

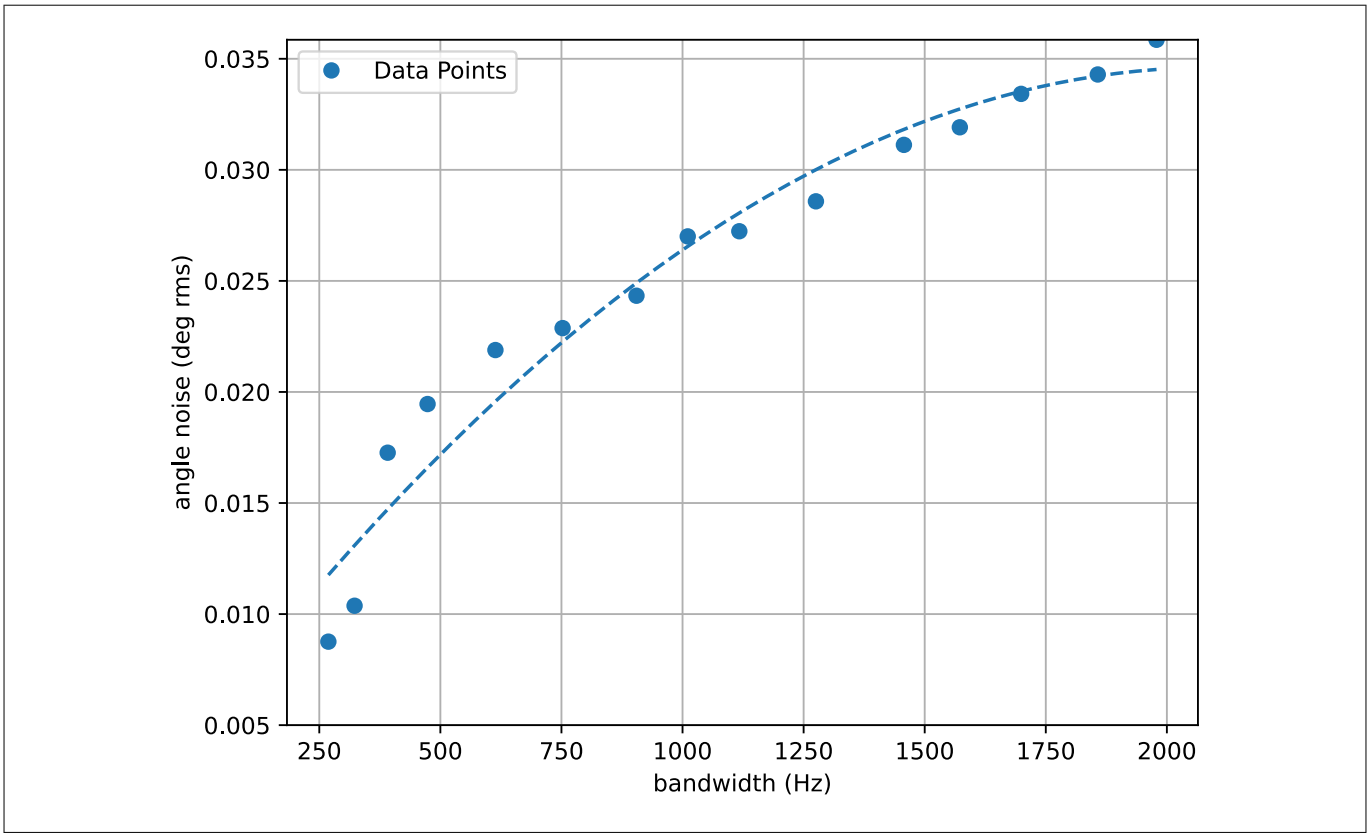


Figure 17 Angle noise over filter bandwidth

4.1 Dynamic Filter Options Table

Table 8 Filter options with according K1 & K2 gain values

ID	Filter bandwidth	K1 "dyn_comp_k1" [LSB]	K2 "dyn_comp_k2" [LSB]	max. acceleration for <0.1° error	Typ. noise
0	268 Hz	1	4	<10krpm/s	0.01° - 0.02°
1	322 Hz	2	5		
2	391 Hz	2	9		
3	473 Hz	3	12		
4	613 Hz	4	20	<30krpm/s	0.02° - 0.03°
5	752 Hz	5	29		
6	904 Hz	6	39		
7	**1010 Hz**	**6**	**51**		
8	1117 Hz	7	58		
9	1275 Hz	8	72	100krpm/s	
10	1457 Hz	9	91		
11	1572 Hz	10	98		0.03 - 0.04°
12	1699 Hz	11	108		
13	1857 Hz	12	122		
14	1978 Hz	12	142		

5 Interface

The XENSIV™ TLx49012 magnetic angle sensor family integrates four different digital interfaces: SPI, ABZ, UVW, PWM. Please refer to the latest product datasheet for specification details of each interface. Depending on the TLx49012 family product type a main interface is pre-configured and a limited amount of interfaces is available. Please refer to cover page for the TLx49012 product overview.

- SPI: main interface for customer-specific configuration setting and CRC protected angle read
- ABZ: emulates incremental quadrature signals for easy replacement of costly and bulky optical encoder systems
- UVW: generates motor-pole-pair dependent signals for direct use in trapezoidal commutation systems
- PWM: one-wire interface for global connection and/or parallel use with ABZ/UVW for redundant data transmission

***Note**: For interface-specific application diagrams with the correct pin connections, refer to the product datasheet.*

SPI is the main interface for sensor configuration such as LUT programming, interface settings or measurement settings. Depending on the sensor variant, different types of interfaces are available and specific interfaces are set by default. SPI is for every variant active right at start-up. If the sensor requires a non-default configuration setting, the SPI interface must be kept active by sending a SPI register unlock command after t_{SPI_active} and before $t_{start\text{-}up}$. Once $t_{start\text{-}up}$ has elapsed the pre-configured interface is active.

Depending on the sensor variant, different interface types are available. The interface type can be selected by programming "if_mode" in the <usr_conifg_1_reg> register.

***Note**: Please follow the sensor configuration flow as depicted in Figure 22*

5.1 SPI

The 32-bit SPI interface features two basic frame schemes:

- In-Frame: the requested data is provided in the same communication frame
- Next-Frame: the requested data is provided in the subsequent communication frame

The In-Frame architecture is designed to minimize latency in communication, ensuring that critical data is accessed quickly and efficiently. By requiring only a single communication frame to read information from the sensor, the In-Frame architecture reduces communication overhead by half compared to a standard SPI next-frame scheme. This is particularly important for time-sensitive applications, such as retrieving angle measurements without delay. The In-Frame scheme provides direct access to all timing-critical registers, enabling fast and reliable data exchange. For more information, please refer to the individual register descriptions.

The Next-Frame architecture is the frame scheme used in typical SPI protocols. In particular, non-timing critical registers with configuration settings are accessible via Next-Frame. The use of next-frame is recommended for configuration and required for programming the device. The Next-Frame communication speed can be improved by doing burst read/writes or interleaving command and read frames, i.e. sending a new command while receiving the data from the previous read command.

Chapter 5.1.1 and Chapter 5.1.2 provide examples for reading the sensed angle via SPI next-frame and SPI in-frame scheme, respectively. The examples show that the in-frame scheme can provide double as much register accesses than next-frame scheme within the same amount of communication frames.

Please refer to interface chapter in the TLx49012 datasheet for further information on the SPI frame scheme and protocol definition.

***Note**: Please note that each SPI frame is CRC protected to ensure data integrity. For more details refer to Chapter 7.1.1.*

5.1.1 SPI read next-frame - Example

Figure 18 provides an example for reading the measured angle and angular velocity without clearing the device status bit using the *SPI next-frame read command scheme*. Following registers are read:

- read angle value: <angle_pred> (address: 0x0C)
- read angular velocity: <angle_vel_reg> (address: 0x0E)

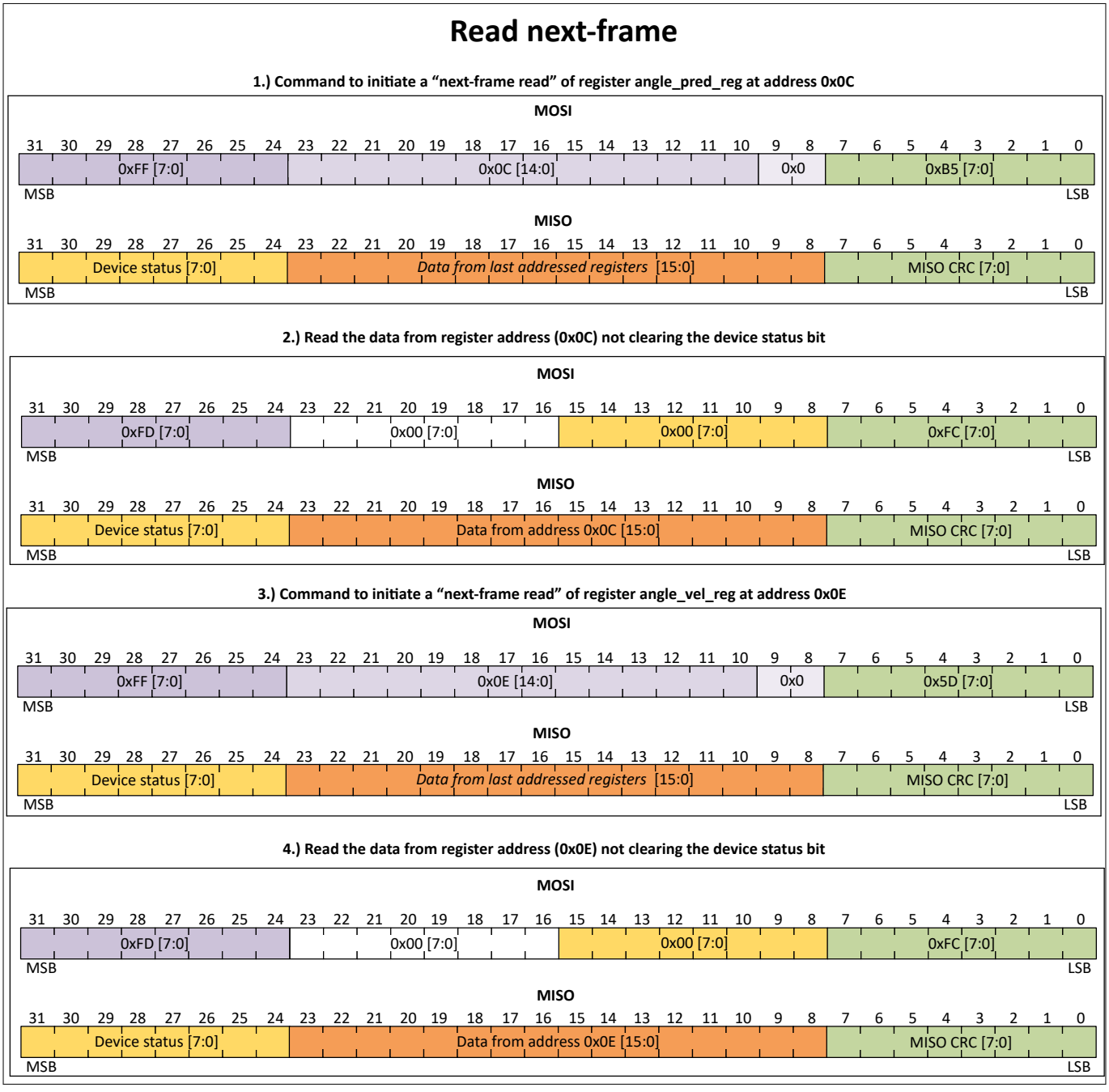


Figure 18 Read next-frame example

5.1.2 SPI read in-frame - Example

Figure 19 provides an example for reading the measured angle, angular velocity and magnetic field strength (field vector) without clearing the device status bit using the *SPI in-frame read scheme*. Following registers are read:

- read angle value: <angle_pred> (address: 0x0C)
- read angular velocity: <angle_vel_reg> (address: 0x0E)
- read actual magnetic field strength (non-compensated vector length): <vec_len3_reg> (address: 0x15)

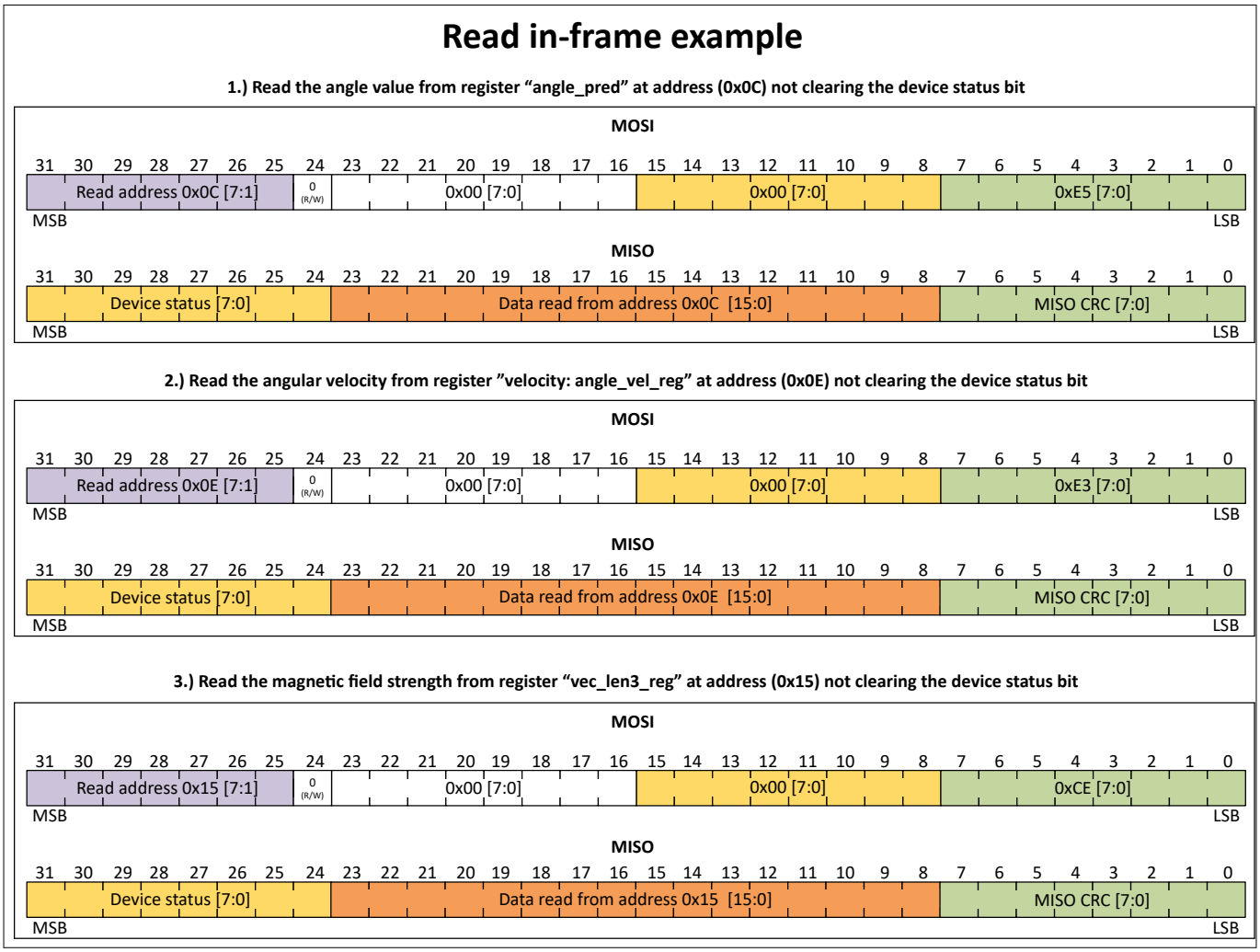


Figure 19 Read in-frame example

5.2 ABZ

After start-up the ABZ interface outputs initial pulses to provide absolute angular position information. The initial pulses can be deactivated by setting "abz_init_pos_mode" in register <usr_config_1_reg> to 1_B. Depending on the configured interface resolution and angular position of the application the amount and therefore the duration of initial pulses vary. Figure 20 illustrates two examples of initial pulse scenarios. If the application starts rotating before the initial angular position of the application has been reached, the interface immediately tries to reach the new angular position.

A hysteresis on the angular output can be configured to avoid pulsing of unintended steps due to vibrations of the mechanical system. The hysteresis value can be selected by "dyn_comp_hyst" in <usr_config_14_reg>. The integrator must consider that a hysteresis masks small angular movements by suppressing the sensor output until the configured hysteresis threshold has been exceeded.

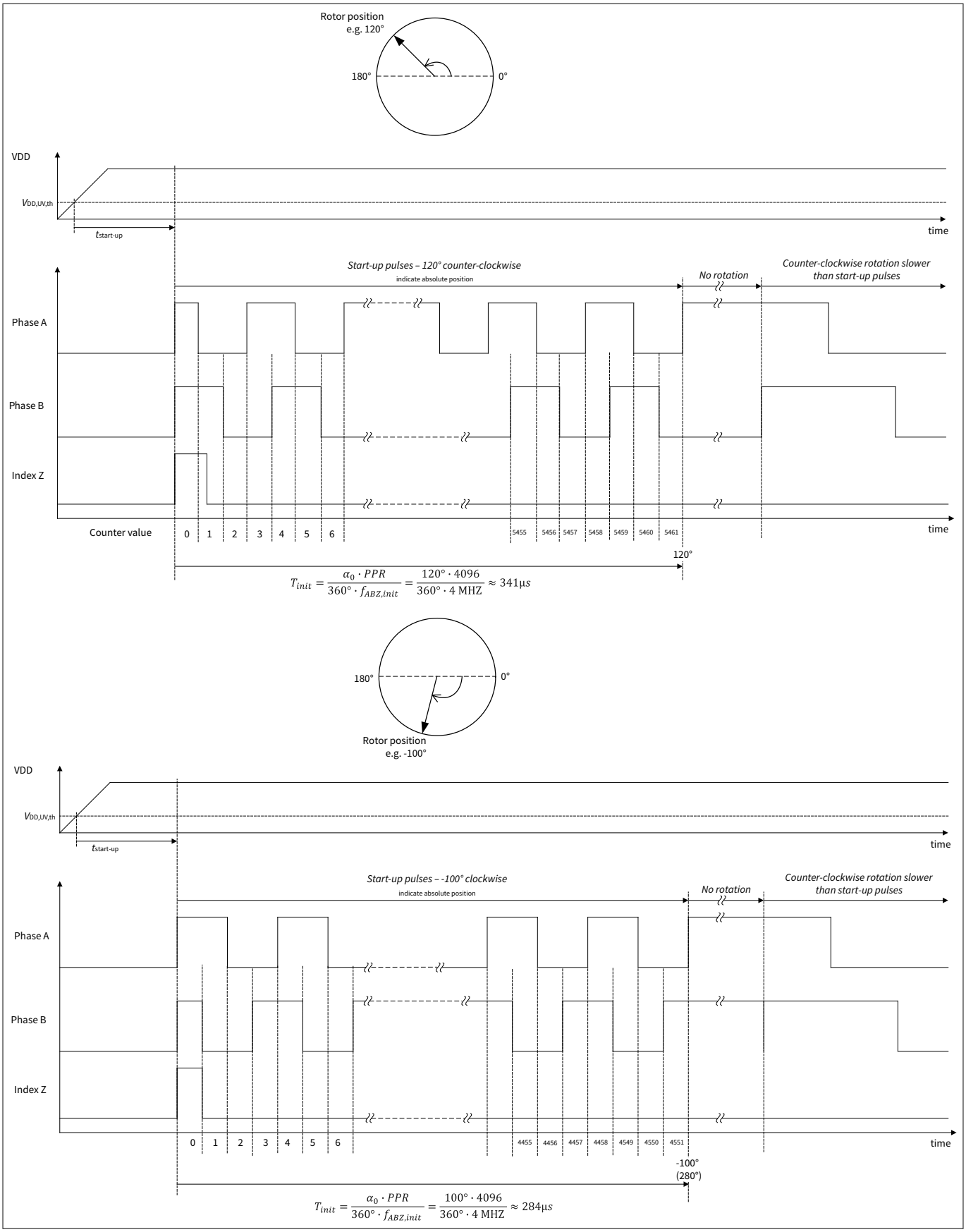


Figure 20 **After start-up initial pulses are provided to determine the absolute position. The time period of initial pulses T_{init} depends on the position α_0 and configured resolution *PPR*.**

5.3 UVW

The UVW interface emulates the use of a Hall switch based position feedback system. Therefore, this interface is in particular intended for trapezoidal commutated motor applications. The used motor type can be set by configuring the according amount of pole pairs via "abz_uvw_pulse_pp" in <usr_config_2_reg> register. The hysteresis applied on the switching points can be configured via "dyn_comp_hyst" in <usr_config_14_reg> register.

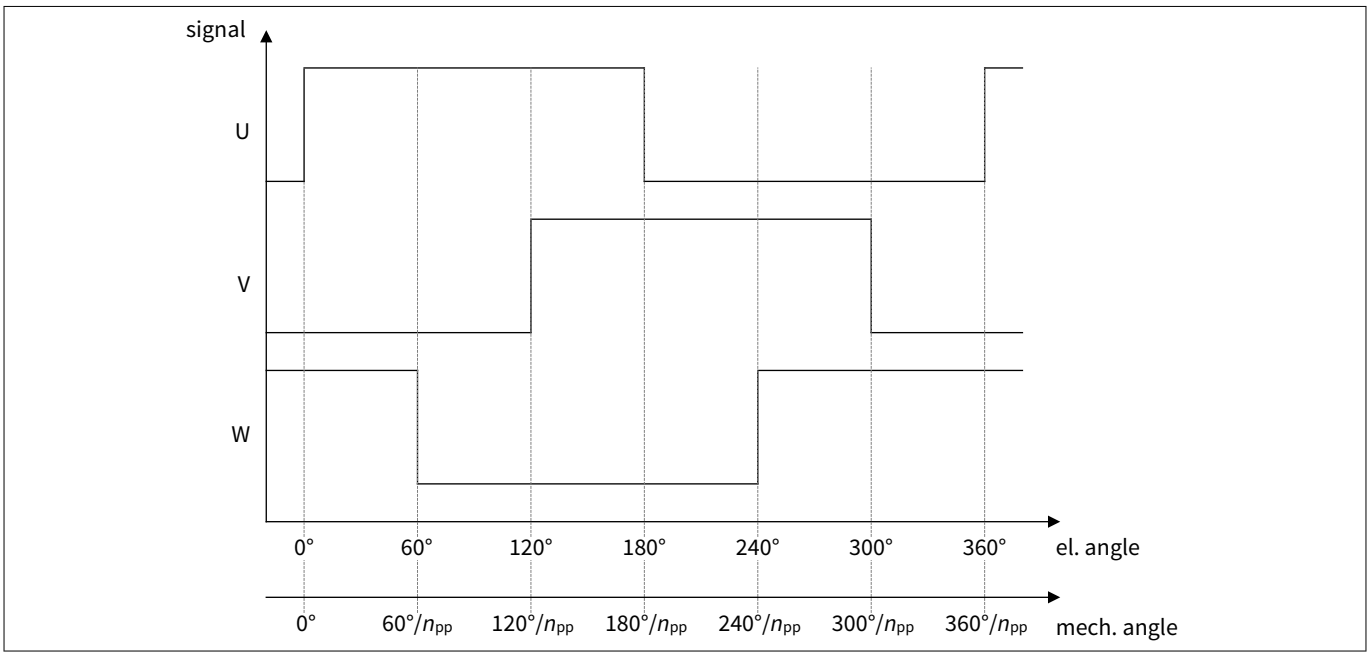


Figure 21 UVW interface switching pattern

5.4 PWM

The pulse-width modulated (PWM) output is a single-wire interface targeting typically two main use cases:

- safety: use as redundant and absolute angle measurement in parallel to ABZ/UVW interface for a safe end-to-end communication by performing angle plausibility checks on module level
- longer connection distances: use of a single wire/pin to connect the sensor module with the controller board, e.g. for HMI applications like a smart knob

The start edge direction and therefore the duty cycle polarity can be set by "pwm_start_edge" in the <usr_config_1_reg>. Additionally, the PWM frequency can be configured by "pwm_freq" in register <usr_config_1_reg>. A higher PWM frequency reduces the output resolution of the interface but increases the refresh rate of new measurement results. The refresh-rate of the outputted PWM angle value can also be controlled via the SYNC pin. This is in particular helpful if the synchronized angle with ABZ/UVW interface in safety applications is required. Please refer to Chapter 5.5 for details about synchronization and to the datasheet for specified interface characteristics.

The duty cycle of the PWM interface represents the angle measurement and can be decoded with following formula. Using the selected PWM-frequency f_{pwm} in units of Hz and the measured on-time t_{on} in units of seconds yields the angle result α in units of degree.

$$\alpha = \left(t_{on} \quad - \frac{0.05}{f_{pwm}} \right) \cdot \frac{360° \cdot f_{pwm}}{0.9} \tag{2}$$

Outside of the duty cycle range encoding the angle, additional diagnostic information is provided. A sensor reset is indicated with a high duty cycle once after a reset has occurred. A low duty cycle indicates any internal sensor failure and all external failures as represented in the SPI device status (see datasheet).

5.5 IFE - Multi-purpose input pin

Beyond the digital data interfaces, the sensor supports specific control features on the IFE input to ease sensor control across many applications. The IFE functionality can be programmed in register <usr_config_1_reg> by setting the bitfield "sync_mode". The edge sensitivity and functionality to trigger via IFE input pin is selectable via bitfield "sync_edge".

- Measurement synchronization: Measurement results are stored at the moment a trigger event is detected. This supports motor control loops or the use of setups using two sensors (redundancy).
  - SPI variant: a trigger edge stores the current set of measurement samples in according registers (<angle_sync_reg>, <angle_vel_sync_reg>). This supports field-oriented control (FOC) loops by providing angle and velocity information synchronously with current measurements, enabling a more accurate coordinate transformation and therefore higher motor efficiency.
  - PWM variant: The measurement output is updated only when a trigger edge at the IFE pin is detected. This eases timing constraints for synchronous angle readback across the ABZ/UVW and PWM interfaces.
- Initial pulse trigger: When using the ABZ interface, the initial pulses after start-up that determine the initial absolute position can be re-triggered via IFE. This ensures that the absolute position reading is independent of the microcontroller’s start-up timing. It also enables recovery of the absolute position after an error indication on the ABZ/UVW interface.
- Reset trigger: With ABZ, UVW, and PWM interfaces that do not support bidirectional communication, the IFE input provides a fast and simple way to trigger a sensor reset.

6 Package information

Package outline drawing, footprint, marking and pin-out information are given in the product datasheet. Additional information for download is available on the Infineon website.

The lead-less package supports lead tip inspection to ease monitoring of soldering quality in production. Further information about handling of leadless packages such as PG-VSON-8-2 can be found in the Infineon Board Assembly Recommendation AppNote.

The location of the sensitive Hall elements within the package is specified in the product datasheet.

Please contact your Infineon representative if further information for package related topics are required.

7 Programming and configuration

The sensor features a non-volatile memory (NVM), also referred to as customer multiple-time programmable (CMTP), which stores application-specific configuration settings and look-up table correction values. This memory allows users to customize the sensor's behavior to suit their specific needs. The NVM can be programmed with user-specific data once per line. Before programming the sensor with non-volatile settings, it is recommended to identify the sensor's target setting using a volatile configuration after each sensor reset. Once the target configuration is identified, the sensor can be programmed with the corresponding bitmap values.

SPI must be used for configuring and programing the sensor. For non-SPI variants, a command to keep SPI active right after power-on must be send to the sensor (refer to Figure 22). It is required to use SPI next-frame architecture for programming the device. A non-volatile programming of the sensor requires a voltage at VPROG.

***Note**: Infineon offers example codes as reference for fast prototyping*

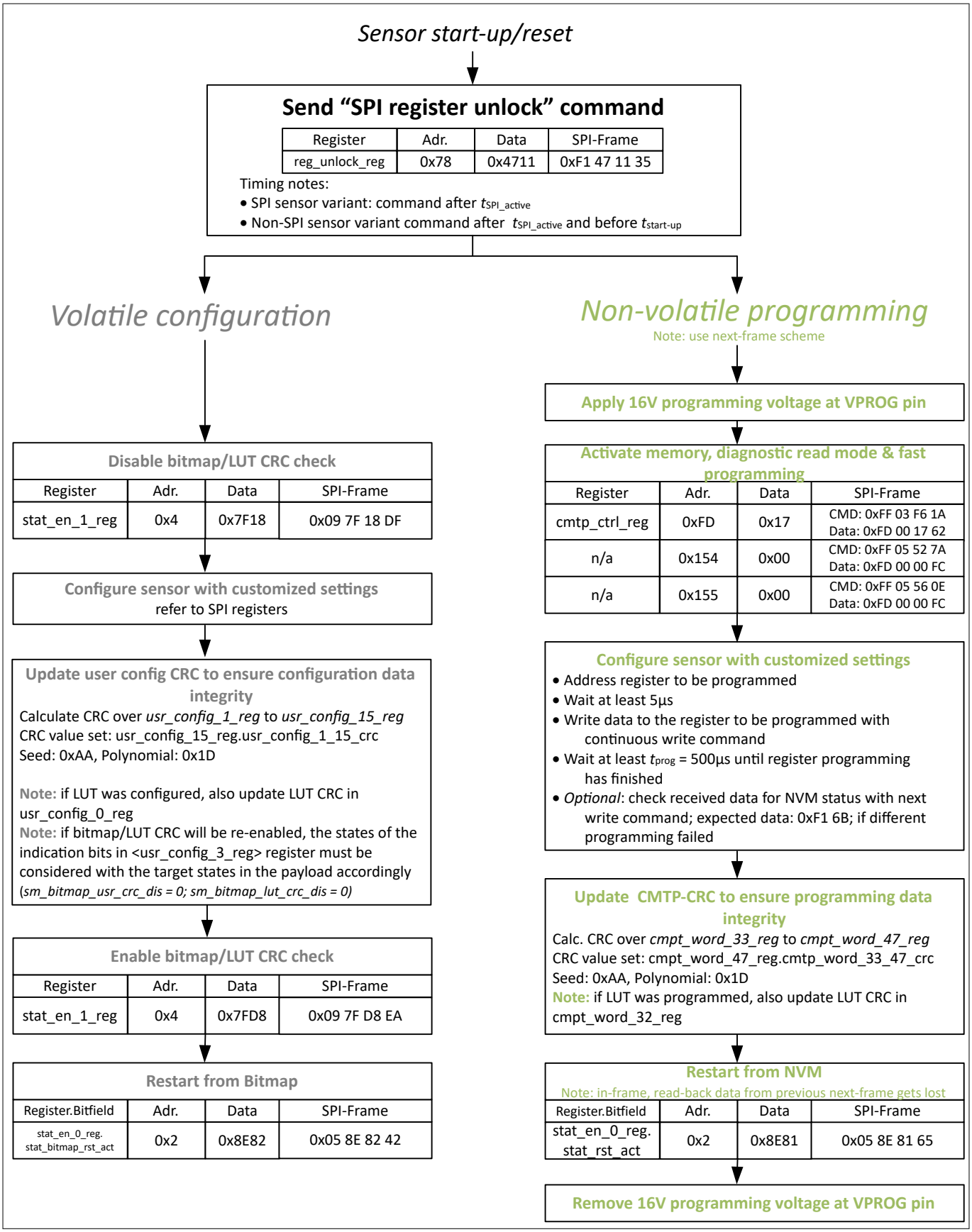


Figure 22 Sensor volatile and non-volatile configuration options

7 Programming and configuration

A non-volatile programming of the sensor requires an additional programming voltage to be applied at the pin VPROG. The programming voltage is specified in the datasheet and must be only applied during non-volatile programming. In the final application or if no non-volatile programming is required, the VPROG pin can be left open or connected to GND. Typically the programming voltage can be re-used from the battery connection of the sensor module or is directly supplied externally via a power supply or a programmer module. The non-volatile configuration set for the sensor can be programmed when flashing the microcontroller. If the sensor is connected with an SPI interface, the application microcontroller can be used for programming the non-volatile configuration set (see Figure 23). If the sensor is not connected via SPI to the application microcontroller, the external program flasher must be applied to the SPI lines of the sensor, e.g. using pogo-pins as indicated in Figure 24.

***Note**: Application diagrams for operational use in application are available in the device datasheet.*

***Note**: The sensor must always be supplied via VDD whenever voltages on the interface pins are applied to avoid reverse sneak paths through the device.*

***Note**: Infineon evaluation kit includes functionality for non-volatile programming. Infineon does not warrant its suitability for use as a production programmer.*

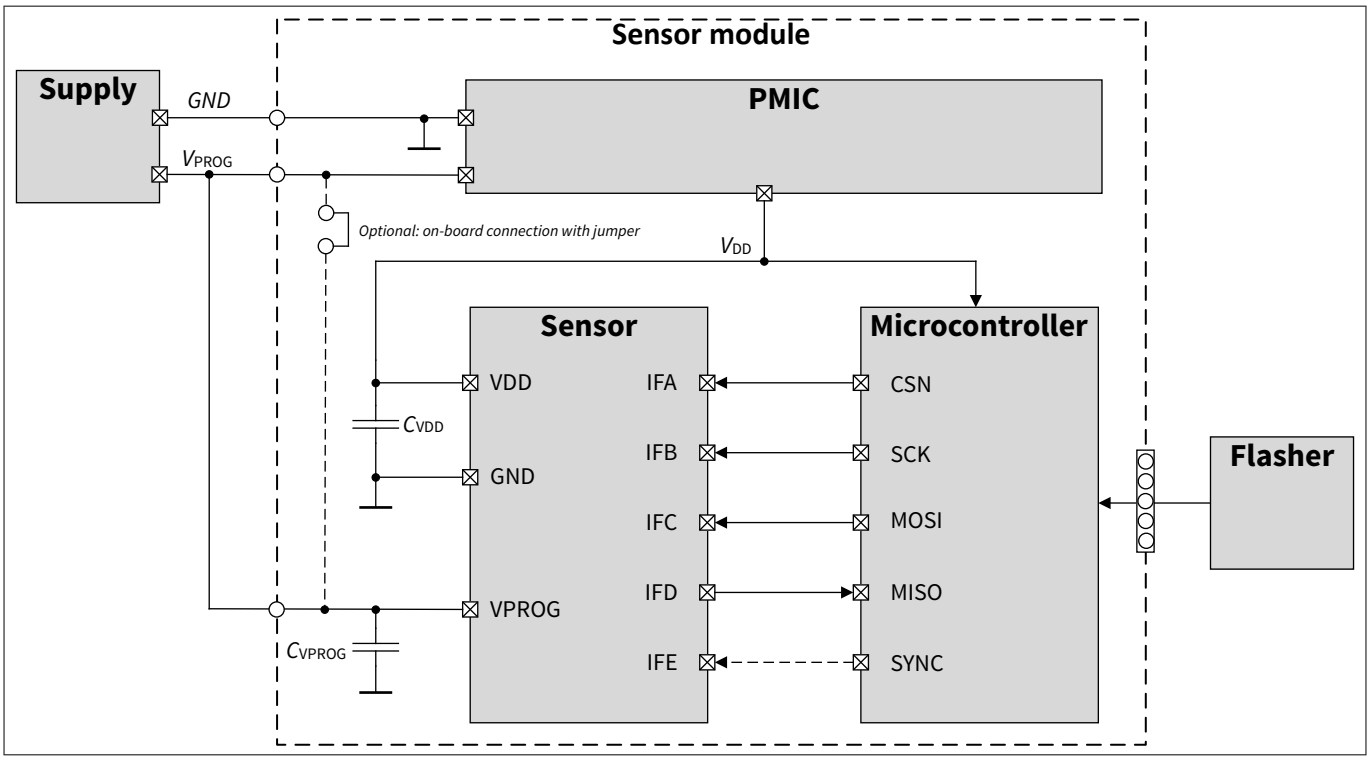


Figure 23 Application diagram for non-volatile programming using the application microcontroller

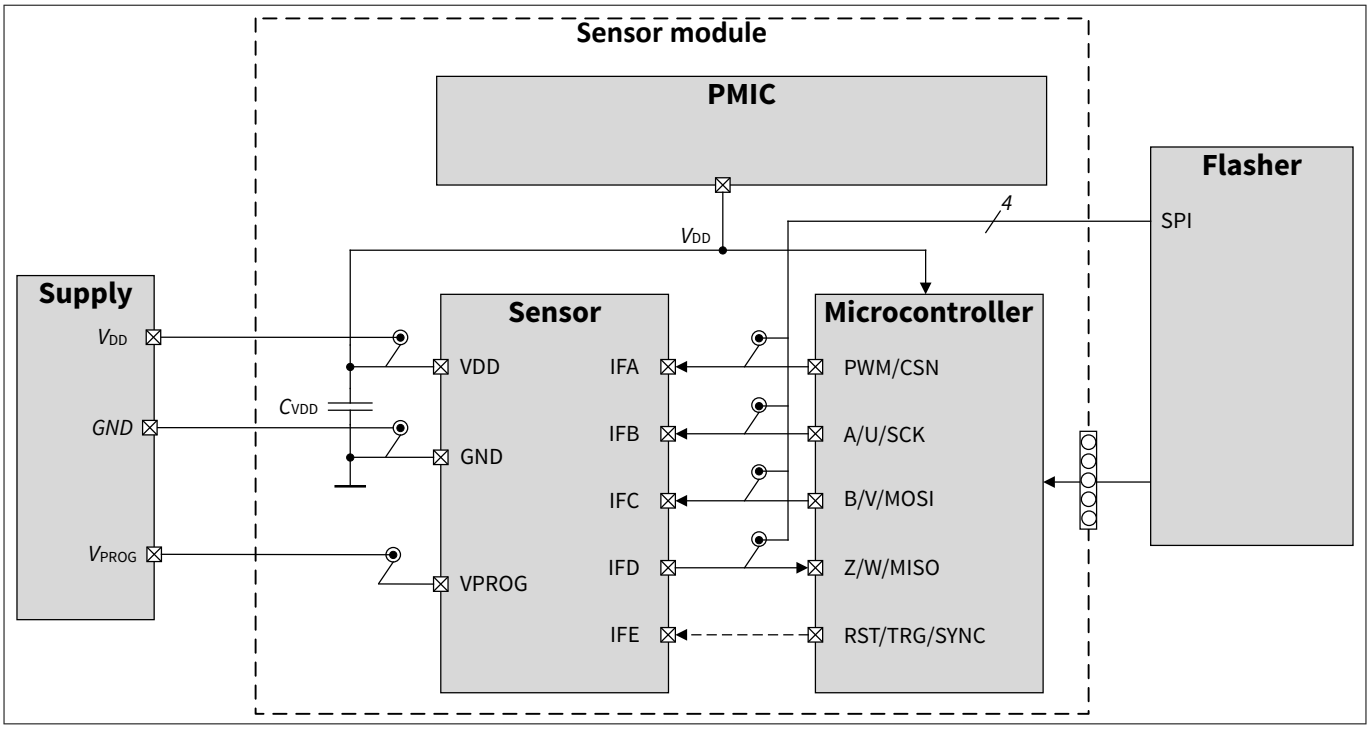


Figure 24 **Application diagram for non-volatile programming using an external programmer and direct access to sensor, e.g. via pogo pins. Care must be taken that pin characteristic of microcontroller and application supply are not violated.**

7.1 CRC

CRC protection ensures reliable data and configuration integrity. The sensor integrates three CRC mechanisms to protect communication, configuration, and correction data.

- SPI-CRC: Ensures end-to-end protection of SPI communication data.
- Config-CRC: Verifies correct sensor configuration in both volatile and non-volatile memory.
- LUT-CRC: Validates lookup table (LUT) correction values to avoid faulty angle measurement

For more information on SPI-CRC please refer to Chapter 7.1.1 or to the sensor datasheet. Each CRC can be disabled to simplify sensor handling. The following table provides an overview of the protected registers and details for setting the correct CRC.

Table 9 **Overview of CRC types and properties**

Type	**SPI CRC**	**Config-CRC**		**LUT-CRC**	
Domain	SPI	volatile config	non-volatile memory	volatile config	non-volatile memory
Payload	data and address of SPI frames, see datasheet	<usr_config_1_reg> -<usr_config_15_reg>	<cmpt_word_33_reg> - <cmpt_word_47_reg>	<cmtp_mpc_lut_0_reg> - <cmtp_mpc_lut_31_reg> **Note**: append 0x00 byte at the end	
Seed	0xFF	0xAA	0xAA	0xAA	0xAA
Polynomial	0x1D	0x1D	0x1D	0x1D	0x1D
CRC register	CRC bitfield in SPI frame	<usr_config_15_reg>	<cmpt_word_47_reg>	<usr_config_0_reg>	<cmtp_word_32_reg>

(table continues...)

Table 9 (continued) Overview of CRC types and properties

Type	SPI CRC	Config-CRC		LUT-CRC	
Disable bitfield	"spi_mosi_crc_dis" "cmtp_spi_mosi_crc_dis"	"sm_bitmap_usr_crc_dis"	n/a	"sm_bitmap_lut_crc_dis"	n/a

7.1.1 SPI CRC

The sensor performs a cyclic redundancy check (CRC) for each received SPI frame. A transmitted SPI frame is secured with a CRC to enable data transfer protection.

The 8-bit CRC value is calculated according to *CRC-8 SAE-J1850* (polynomial: 0x1D, seed value: 0xFF, XOR value: 0xFF).

If the microcontroller detects a CRC error on the MISO line, the data should be discarded and the register access should be repeated.

Each MOSI frame contains an 8 bit CRC field at the end of the frame calculated with 24 MOSI frame payload bits:

- payload-byte 0: MOSI bits [31:24]
- payload-byte 1: MOSI bits [23:16]
- payload-byte 2: MOSI bits [15:8]

Note: The calculation starts with payload-byte 0

Each MISO frame contains an 8 bit CRC field at the end of the frame calculated with 8 MOSI payload bits and 24 MISO frame payload bits:

- payload-byte 0: dummy 0_B [MSB] + MOSI bits [31:25]
- payload-byte 1: MISO bits [31:24]
- payload-byte 2: MISO bits [23:16]
- payload-byte 3: MISO bits [15:8]

Note: The calculation starts with payload-byte 0

If the MOSI frame CRC check is disabled, the data in the MOSI-CRC-bitfield is ignored.

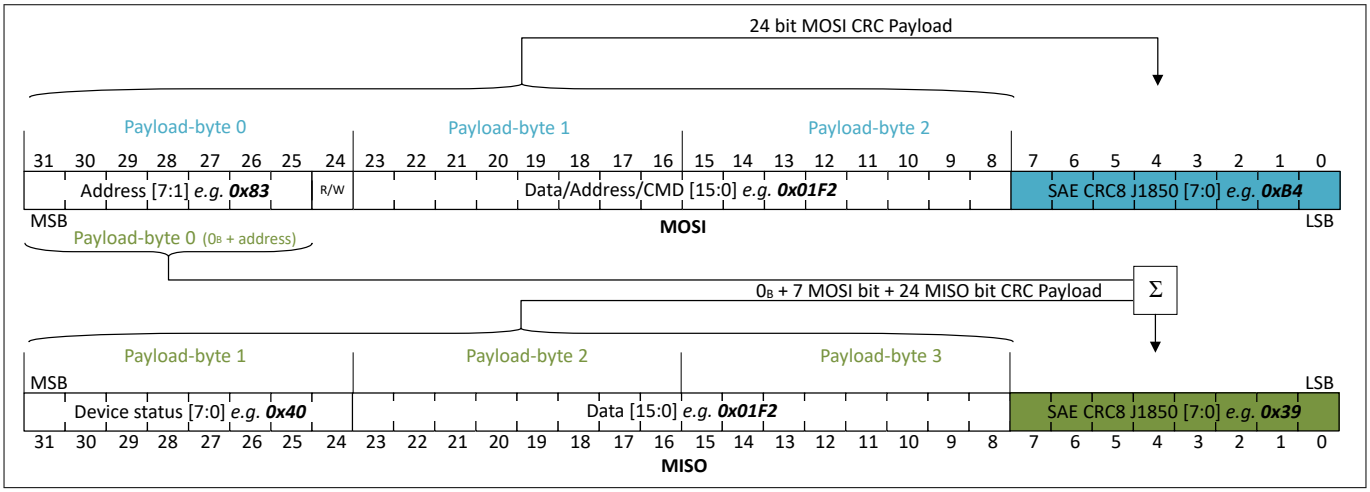


Figure 25 Illustration of SPI CRC payload for MOSI and MISO frame with pseudo data as example (valid for all frame types)

Following Python code snippet calculates the CRC by providing the payload data as input. Please note that Infineon microcontrollers usually also support hardware based CRC engines.

```
function calculate_crc8_sae_j1850(data_bytes):
    polynomial = 0x1D
    crc = 0xFF

    for byte in data_bytes:
        crc ^= byte
        for i in range(8):  // Process each bit
            if crc & 0x80: // If the leftmost (most significant) bit is set
                crc = (crc << 1) ^ polynomial
            else:
                crc = crc << 1
            crc &= 0xFF // Ensure CRC remains an 8-bit value

    crc ^= 0xFF
    return crc
```

7.1.2 LUT CRC

The LUT is protected by CRC. When LUT values change, update the CRC fields in <usr_config_0_reg>. This ensures the sensor acknowledges the programmed LUT values and maintains data integrity. The Python example below demonstrates how to calculate the CRC for the first 16 LUT bytes. Write the resulting value to the bitfield "lut_word_0_15_crc" in register <usr_config_0_reg>. For a linear LUT characteristic - the default setting - the computed CRC is 0xC7.

***Note**: A 0x00 byte must be appended at the end of the payload for a correct LUT CRC calculation, refer to example code.*

7.1.2.1 LUT CRC example code

```
#Python Example Code - LUT CRC calculation

def crc8(data):
    polynomial = 0x1D
    crc = 0xAA

    for byte in data:
        crc ^= byte
        for _ in range(8):
            if crc & 0x80:
                crc = (crc << 1) ^ polynomial
            else:
                crc <<= 1
            crc &= 0xFF  # Ensure CRC remains 8-bit
    crc ^= 0xFF
    return crc

# Payload example for a linear transfer characteristic (default LUT values)
payload = [0x00, 0x00, 0x10, 0x00, 0x20, 0x00, 0x30, 0x00, 0x40, 0x00, 0x50, 0x00, 0x60, 0x00,
0x70, 0x00, 0x80, 0x00, 0x90, 0x00, 0xA0, 0x00, 0xB0, 0x00, 0xC0, 0x00, 0xD0, 0x00, 0xE0, 0x00,
0xF0, 0x00, 0x00]

# Calculate CRC8
crc8_result = crc8(payload)
print(f"LUT-CRC: {crc8_result:#04x}")
```

7.1.2.2 LUT CRC example code - fast

```
//C Example Code - LUT CRC fast code

uint8_t crcTable[256];

uint8_t CalcCRC(uint8_t * buf, uint8_t len, uint8_t seed)
{
  const uint8_t * ptr = buf;
  uint8_t _crc = seed; // seed
  while(len--) _crc = crcTable[_crc ^ *ptr++];
  return ~_crc;
}

void CRCInit(void)
{
  uint8_t _crc;
  for (int i = 0; i < 0x100; i++)
  {
    _crc = i;
    for (uint8_t bit = 0; bit < 8; bit++) _crc = (_crc & 0x80) ? ((_crc << 1) ^ 0x1D) : (_crc
<< 1); // 0x1D polynomial
    crcTable[i] = _crc;
  }
}
```

7.1.3 Config CRC

The sensor’s configuration settings are protected by a Config-CRC to ensure correct programming and to prevent unintended angle processing during operation. When calculating the Config-CRC, it is important to consider the correct state of the <usr_config_3_reg> register, which defines which sensor errors are reported or suppressed. Error reporting is controlled through the status-enable registers stat_en_x_reg (x = 0, 1, 2, 3). Therefore, the Config-CRC must always be disabled via the "stat_bitmap_usr_crc_en" bit in <stat_en_1_reg> before modifying the error-indication settings. After completing the configuration of the desired error reporting, the bitmap CRC check must be re-enabled. When re-activating the bitmap CRC, its reporting state must be correctly reflected in the CRC payload ("sm_bitmap_usr_crc_dis" = 0_B). Please also refer to configuration

procedure outlined in Figure 22. Following code snippet is an extract of Infineon's Arduino example code library for calculating and updating the Config CRC.

```
// USER CONFIGURATION AREA
#define USR_CONFIG_START_ADDRESS  63
#define USR_CONFIG_STOP_ADDRESS   77
#define CONFIG_SEED     0xAA // CRC seed used for configurations/bitmap

uint8_t crcTable[256];
uint8_t CalcCRC(uint8_t * buf, uint8_t len, uint8_t seed)
{
  const uint8_t * ptr = buf;
  uint8_t _crc = seed; // seed
  while(len--) _crc = crcTable[_crc ^ *ptr++];
  return ~_crc;
}
void CRCInit(void)
{
  uint8_t _crc;
  for (int i = 0; i < 0x100; i++)
  {
    _crc = i;
    for (uint8_t bit = 0; bit < 8; bit++) _crc = (_crc & 0x80) ? ((_crc << 1) ^ 0x1D) : (_crc
<< 1); // 0x1D polynomial
    crcTable[i] = _crc;
  }
}

uint32_t CalcUserConfigCRC(uint8_t startAddr, uint8_t stopAddr, bool clearStatus)
{
  if (stopAddr < startAddr)
  {
    return 0; // Empty range
  }

  uint8_t count = (uint8_t)(stopAddr - startAddr + 1);

  uint8_t stream[2 * count - 1]; // Last register contains also the CRC of the block => use
only MSB from register
  uint8_t idx = 0;

  SpiCommandNextFrame(startAddr, AC_READ_INC); // Send next frame read command with auto
increment to the first register address, do not care about response

  for (uint8_t i = 0; i < count; i++)
  {
    uint32_t resp = SpiReadNextFrame(clearStatus); // Read registers with auto increment

    uint8_t msb = (uint8_t)((resp >> 16) & 0xFF); // Take data MSB
    uint8_t lsb = (uint8_t)((resp >>  8) & 0xFF); // Take data LSB

    if (i == (uint8_t)(count - 1))
```

```
      {
        stream[idx++] = msb; // Last word: only MSB
      }
      else
      {
        // All other words: MSB then LSB
        stream[idx++] = msb;
        stream[idx++] = lsb;
      }
    }

     return CalcCRC(stream, idx, CONFIG_SEED);
  }

  void WriteUserConfigCRC()
  {
    uint8_t crc = CalcUserConfigCRC(USR_CONFIG_START_ADDRESS, USR_CONFIG_STOP_ADDRESS, true);
       // Calculate CRC
    uint16_t crcLine = (uint16_t)((SpiReadRegNextFrame(USR_CONFIG_STOP_ADDRESS, true) >> 8) &
  0xFFFF);  // Read CRC line
    crcLine = (uint16_t)((crcLine & 0xFF00) | crc);
       // Replace CRC and keep useful data
    SpiWriteRegNextFrame(USR_CONFIG_STOP_ADDRESS, crcLine);
       // Write new CRC

  }
```

8 Device registers

Figure 26 illustrates the measurement values that can be read back via SPI, based on the register naming. Not defined bitfields represented with a "-" description shall always be written with a 0_B. Bit types denoted with "r" define readable bits, whereas "rw" defines a readable and writeable bit. The reset values of the register are provided in the SPI and CMTP register overview.

***Note**: It is recommended to check the sensor diagnosis status via the SPI device status in every MISO frame. For diagnosis read after start-up, please refer to Reset.*

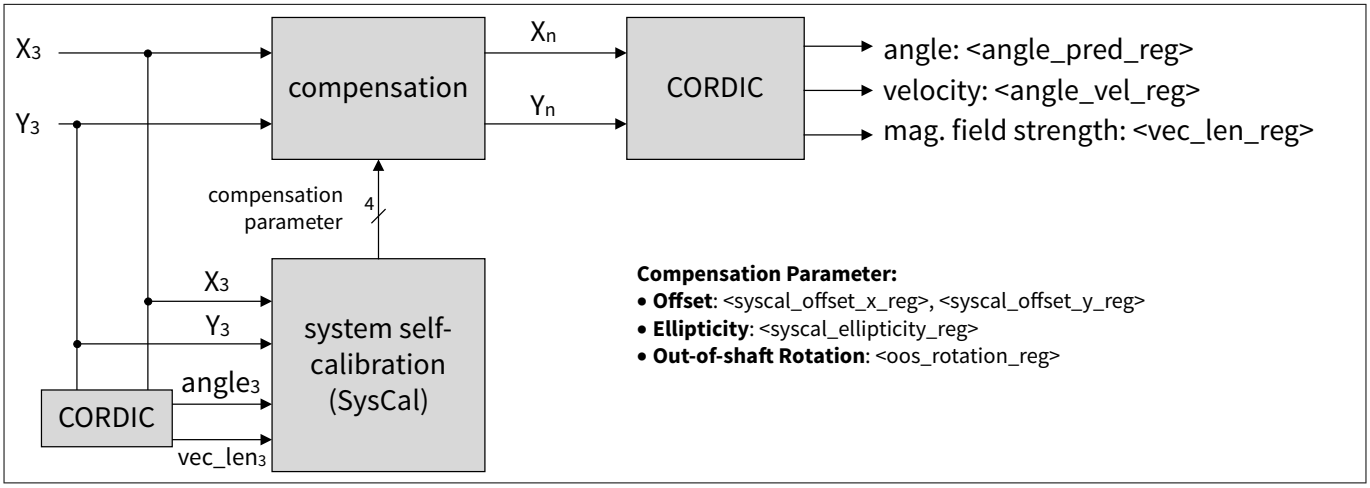


Figure 26 Register denotation illustrated in simplified signal flow chart

Table 10 Register to signal assignment

Signal name	Register name	Description
X3	<x3_reg>	cosine component of magnetic field measured by Hall element X. For direction orientation in package, refer to Datasheet.
Y3	<y3_reg>	sine component of magnetic field measured by Hall element Y. For direction orientation in package, refer to Datasheet.
angle3	<angle3_reg>	calculated angle out of raw magnetic field values
vec_len3	<vec_len3_reg>	magnetic field strength out of raw magnetic field values
Xn	<xn_reg>	system self-calibration compensated magnetic field cosine component
Yn	<yn_reg>	system self-calibration compensated magnetic field sine component
$angle_{raw}$	<angle_raw_reg>	angle value calculated out of system self-calibration compensated field components
vec_len	<vec_len_reg>	magnetic field strength
$angle_{pred}$	<angle_pred_reg>	static and dynamic compensated and linearized angle measurement value
angle_vel	<angle_vel_reg>	compensated and linearized velocity measurement value

Table 11 In-Frame and Next-frame Registers

SPI Frame	Address range [DEC]	Register types
In-frame capable	0-121	SPI registers
Next-frame	all registers	all registers
Next-frame only	253-303	NVM registers

8.1 Register transfer formulas

unsigned representation (0° – 360°):

$$\alpha_{\text{unsigned}} = \frac{360°}{2^{16}} \cdot \text{angle_pred_spi}$$

signed representation (±180°):

if angle_pred_spi $< 2^{15}$:

$$\alpha_{\text{signed}} = \frac{360°}{2^{16}} \cdot \text{angle_pred_spi} \tag{3}$$

else:

$$\alpha_{\text{signed}} = \frac{360°}{2^{16}} \cdot (\text{angle_pred_spi} - 2^{16})$$

unsigned representation (0rpm -90000rpm):

$$\omega_{rpm,\,unsigned} = \left(\frac{60 \cdot 40 \cdot 10^6}{2^{28}} \cdot \text{angle_vel_spi}\right)\text{rpm}$$

signed representation (±45000rpm):

if angle_vel_spi $< 2^{15}$:

$$\omega_{rpm,\,signed} = \left(\frac{60 \cdot 40 \cdot 10^6}{2^{28}} \cdot \text{angle_vel_spi}\right)\text{rpm} \tag{4}$$

else:

$$\omega_{rpm,\,signed} = \left(\frac{60 \cdot 40 \cdot 10^6}{2^{28}} \cdot (\text{angle_vel_spi} - 2^{16})\right)\text{rpm}$$

uncompensate field strength (actual measurement):
$B_{meas} = \text{vec_len3_spi} \cdot 6.4 \quad \mu\text{T}$
$B_{x3} = \text{x3_spi} \cdot 6.4 \quad \mu\text{T}$
$B_{y3} = \text{y3_spi} \cdot 6.4 \quad \mu\text{T}$

compensated field strength:
$B_{comp} = \text{vec_len_spi} \cdot 6.4 \quad \mu\text{T}$
$B_{xn} = \text{xn_spi} \cdot 6.4 \quad \mu\text{T}$
$B_{yn} = \text{yn_spi} \cdot 6.4 \quad \mu\text{T}$

(5)

$$T_J = \left(\frac{\text{t_lin}}{153.52} + 25\right)°\text{C} \tag{6}$$

Table 12 Symbol denotation

Symbol	Description
α_{unsigned}	unsigned angle after static and dyn. compensation and linearization in °
α_{signed}	signed angle after static and dyn. compensation and linearization in °
$\omega_{\text{rpm,unsigned}}$	unsigned rotational velocity in rpm
$\omega_{\text{rpm,signed}}$	signed rotational velocity in rpm
B_{meas}	un-compensated magnetic field vector in µT
B_{x3}	X-component of un-compensated magnetic field vector in µT
B_{y3}	Y-component of un-compensated magnetic field vector in µT
B_{comp}	SysCal compensated magnetic field vector in µT
B_{xn}	SysCal compensated X-component of magnetic field vector in µT
$B\text{yn}$	SysCal compensated Y-component of magnetic field vector in µT
TJ	junction temperature of the sensor in °C

8.2 SPI register

8.2.1 Register overview

Address	Register Name	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	stat_device_reg	stat_spi_next_fra	stat_spi_next_fra	stat_spi_in_frame_	stat_spi_in_frame_	stat_spi_next_fra	stat_spi_mosi_crc	stat_spi_frame	stat_spi_polarity			stat_dev_diag	stat_dev_mag	stat_dev_sup	stat_dev_rst	stat_dev_ot	stat_dev_cal
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	stat_0_reg	stat_test_spi	stat_adc_toggle_s			stat_cordic_overfl	stat_cmtp_rd_spi	stat_cmtp_ecc_sp	stat_dsp_check_sp	stat_dsp_sched_sp	stat_rom_run_spi	stat_rom_su_spi	stat_om_spi		stat_sm_rst_spi	stat_bitmap_rst_sp	stat_rst_spi
	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
2	stat_en_0_reg	stat_test_en				stat_cordic_overfl	stat_cmtp_rd_en	stat_cmtp_ecc_en		stat_dsp_sched_en						stat_bitmap_rst_ac	stat_rst_act
	36480	1	0	0	0	1	1	1	0	1	0	0	0	0	0	0	0
3	stat_1_reg	stat_dyn_comp_li	stat_diff_stress_rn	stat_shear_stress_	stat_t_lin_max_spi	stat_t_lin_min_spi	stat_y_adc_rng_spi	stat_x_adc_rng_spi	stat_bitmap_ifx_cr	stat_bitmap_usr_cr	stat_bitmap_lut_cr	stat_cmtp_ifx_crc	stat_cmtp_usr_crc	stat_cmtp_lut_crc	stat_cmtp_ifx_vir	stat_cmtp_usr_vir	stat_cmtp_lut_vir
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	stat_en_1_reg		stat_diff_stress_rn	stat_shear_stress_	stat_t_lin_max_en	stat_t_lin_min_en	stat_y_adc_rng_en	stat_x_adc_rng_en	stat_bitmap_ifx_cr	stat_bitmap_usr_cr	stat_bitmap_lut_cr		stat_cmtp_usr_crc	stat_cmtp_lut_crc			
	32728	0	1	1	1	1	1	1	1	1	1	0	1	1	0	0	0
5	stat_2_reg	stat_interpolate_l	stat_vec_len_max	stat_vec_len_min_	stat_vhall_bias_rn	stat_vddp_ov_spi	stat_vddp_uv_spi	stat_vddd_ov_spi	stat_vddd_uv_spi	stat_vdda_ov_spi	stat_vdda_uv_spi	stat_vdd2v5d_ov_	stat_vdd2v5d_uv_	stat_vdd2v5a_ov_	stat_vdd2v5a_uv_	stat_tv_0_rng_spi	stat_tv_fs_rng_spi
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
6	stat_en_2_reg	stat_interpolate_l	stat_vec_len_max	stat_vec_len_min_	stat_vhall_bias_rn	stat_vddp_ov_en	stat_vddp_uv_en	stat_vddd_ov_en	stat_vddd_uv_en	stat_vdda_ov_en	stat_vdda_uv_en	stat_vdd2v5d_ov_	stat_vdd2v5d_uv_	stat_vdd2v5a_ov_	stat_vdd2v5a_uv_	stat_tv_0_rng_en	stat_tv_fs_rng_en
	65535	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
10	angle_sync_reg	angle_sync														stat_sync_as	stat_read_as
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
11	angle_vel_sync_reg	angle_vel_sync														stat_sync_avs	stat_read_avs
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
12	angle_pred_reg	angle_pred_spi															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
13	angle_set_reg	angle_set_spi															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
14	angle_vel_reg	angle_vel_spi															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
15	angle_reg	angle_spi															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
16	angle_raw_reg	angle_raw_spi															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
17	vec_len_reg	vec_len_spi															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	xn_reg	xn_spi															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
19	yn_reg	yn_spi															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
20	angle3_reg	angle3_spi															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
21	vec_len3_reg	vec_len3_spi															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
22	x3_reg	x3_spi															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
23	y3_reg	y3_spi															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
37	t_lin_reg	t_lin															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
43	syscal_angle_base_reg	syscal_angle_base															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
44	syscal_offset_x_reg	syscal_offset_x															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
45	syscal_offset_y_reg	syscal_offset_y															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Figure 27 SPI Registermap - part 1

Address	Register Name	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
46	syscal_ellipticity_reg	syscal_ellipticity_scale		syscal_ellipticity													
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
47	syscal_oos_rotation_reg	syscal_oos_rotation															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
56	mpc_lut_idx_reg												mpc_lut_idx				
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
58	mpc_lut_data_0_reg	mpc_lut_data_0															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
59	mpc_lut_data_1_reg	mpc_lut_data_1															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
60	mpc_lut_data_2_reg	mpc_lut_data_2															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
61	mpc_lut_data_3_reg	mpc_lut_data_3															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
62	usr_config_0_reg	lut_word_16_31_crc								lut_word_0_15_crc							
	6425	0	0	0	1	1	0	0	1	0	0	0	1	1	0	0	1
63	usr_config_1_reg	adc_max_rng_en	abz_init_pos_mod	abz_mode	spi_miso_crc_inv_	spi_mosi_crc_dis	sync_edge		sync_mode		pwm_start_edge	pwm_freq			if_mode		
	24	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0
64	usr_config_2_reg	cfg_unused_64		abz_z_mode		abz_uvw_pulses_pp											
	4095	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1
65	usr_config_3_reg	cfg_unused_65		sm_fw_overflow_	sm_adc_overflow	sm_vddp_ov_dis	sm_vddp_uv_dis	sm_vddx_ouv_dis	sm_vdd2v5x_ouv_	sm_tv_rng_dis	sm_diff_stress_rn	sm_shear_stress_r	sm_t_lin_max_dis	sm_t_lin_min_dis	sm_bitmap_usr_cr	sm_bitmap_lut_cr	sm_cmtp_lut_usr_
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
66	usr_config_4_reg	cfg_unused_66											ife_spike_filter_e	ifd_spike_filter_e	ifc_spike_filter_e	ifb_spike_filter_e	ifa_spike_filter_e
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
67	usr_config_5_reg	x3_base															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
68	usr_config_6_reg	y3_base															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
69	usr_config_7_reg	angle_base															angle_dir
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
70	usr_config_8_reg	offset_app_x															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
71	usr_config_9_reg	offset_app_y															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
72	usr_config_10_reg	syscal_angle_base	syscal_mode			lut_mode		exp_tan					exp_rad				
	8192	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
73	usr_config_11_reg	dyn_comp_k2								dyn_comp_k1							
	13062	0	0	1	1	0	0	1	1	0	0	0	0	0	1	1	0
74	usr_config_12_reg	ellipticity								dyn_comp_time							
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
75	usr_config_13_reg	oos_rotation										dyn_comp_mode	dyn_comp_init_samples			dyn_comp_init_k1	dyn_comp_init_co
	20	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0
76	usr_config_14_reg	sm_vec_len_lim_min							sm_vec_len_lim_	dyn_comp_lim			syscal_t_lin_lim		dyn_comp_en	dyn_comp_hyst	
	3302	0	0	0	0	1	1	0	0	1	1	1	0	0	1	1	0
77	usr_config_15_reg	sm_vec_len_lim_max								usr_config_1_15_crc							
	45181	1	0	1	1	0	0	0	0	0	1	1	1	1	1	0	1
104	sm_clk_cnt_reg	sm_clk_cnt															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
120	reg_unlock_reg	reg_unlock															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
121	hw_fw_id_reg									hw_id				fw_id			
	35	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	1

Figure 28 SPI Registermap - part 2

8.2.2 stat_device_reg

Register	stat_device_reg							
Address	0x0							
Bit	15	14	13	12	11	10	9	8
Field	stat_spi_next_frame_wr	stat_spi_next_frame_rd	stat_spi_in_frame_wr	stat_spi_in_frame_rd	stat_spi_next_frame_cmd	stat_spi_mosi_crc	stat_spi_frame	stat_spi_polarity
Bit	7	6	5	4	3	2	1	0
Field	-		stat_dev_diag	stat_dev_mag	stat_dev_sup	stat_dev_rst	stat_dev_ot	stat_dev_cal

Field	Type	Description
stat_dev_cal	r	System self-calibration status indication for SPI device status: 0 = the system self-calibration is disabled or inactive (in initialization phase or failed), 1 = the system self-calibration is enabled and active (succeeded and the determined calibration parameters are used for angle compensation).
stat_dev_ot	r	Overtemperature warning for SPI device status (mirror of stat_t_lin_max status flag): 0 = sensor temperature below overtemperature limit, 1 = sensor operates in overtemperature range. Status bit can only be reset via "Clear request device status" bits in SPI frame.
stat_dev_rst	r	Reset indication for SPI device status (OR combination of all status flags referring to device status "RST" in their description): 0 = no sensor reset occurred, 1 = a sensor reset occurred. Note: It is recommended that you reset this status flag after every intentional reset to enable the detection of unintentional resets. Please check stat_rst_spi and stat_sm_rst_spit status flags first in case of unintended reset to determine its root cause. Status bit can only be reset via "Clear request device status" bits in SPI frame.
stat_dev_sup	r	Extended supply range warning for SPI device status (OR combination of all status flags referring to device status "SUP" in their description): 0 = sensor supply is within functional range, 1 = sensor supply operates in extended supply voltage range. Status bit can only be reset via "Clear request device status" bits in SPI frame.
stat_dev_mag	r	Magnetic field too high or too low warning for SPI device status (OR combination of all status flags referring to device status "MAG" in their description): 0 = magnetic field is within the minimum and maximum magnetic thresholds, 1 = magnetic field exceeded the minimum or maximum magnetic threshold. Status bit can only be reset via "Clear request device status" bits in SPI frame.

Field	Type	Description
stat_dev_diag	r	Sensor diagnosis information for SPI device status (OR combination of all status flags referring to device status "DIAG" in their description): 0 = no fault diagnosed, 1 = fault occurred. Note: The other status registers can be read for more detailed diagnosis information. Status bit is preserved over soft reset from NVM and can only be reset via "Clear request device status" bits in SPI frame.
stat_spi_polarity	r	The SCK signal was high during rising or falling edges of CSN. Status bit can only be reset via "Clear request device status" bit for OFC in SPI frame.
stat_spi_frame	r	The number of received SCK clock pulses was not equal to 32 while CSN was low. Status bit can only be reset via "Clear request device status" bit for OFC in SPI frame.
stat_spi_mosi_crc	r	Incorrect CRC detection in received MOSI frame if MOSI-CRC check is enabled. Status bit can only be reset via "Clear request device status" bit for OFC in SPI frame.
stat_spi_next_frame_cmd	r	The microcontroller sent a next-frame read/write data frame without a previous next-frame command frame. Status bit can only be reset via "Clear request device status" bit for OFC in SPI frame.
stat_spi_in_frame_rd	r	Register read timeout during in-frame read, e.g. the microcontroller accessed a next-frame register with an in-frame request (read access time-out). Status bit can only be reset via "Clear request device status" bit for OFC in SPI frame.
stat_spi_in_frame_wr	r	Register write timeout after in-frame write: the transfer delay time was too short to process the data internally. Status bit can only be reset via "Clear request device status" bit for OFC in SPI frame.
stat_spi_next_frame_rd	r	Register read timeout at the end of a next-frame access: the transfer delay time was too short to process the data internally. Status bit can only be reset via "Clear request device status" bit for OFC in SPI frame.
stat_spi_next_frame_wr	r	Register write timeout at the end of a next-frame access: the transfer delay time was too short to process the data internally. Status bit can only be reset via "Clear request device status" bit for OFC in SPI frame.

8.2.3 stat_0_reg

Register	stat_0_reg							
Address	0x1							
Bit	15	14	13	12	11	10	9	8
Field	stat_test_spi	stat_adc_toggle_spi	-		stat_cordic_overflow_spi	stat_cmtp_rd_spi	stat_cmtp_ecc_spi	stat_dsp_check_spi
Bit	7	6	5	4	3	2	1	0
Field	stat_dsp_sched_spi	stat_rom_run_spi	stat_rom_su_spi	stat_om_spi		stat_sm_rst_spi	stat_bitmap_rst_spi	stat_rst_spi

Field	Type	Description
stat_rst_spi	rw	Power-on hardware reset resp. restart from NVM indication for SPI interface. Device status: RST Indication: error status is preserved until explicit write by 0 or reset by "Clear request device status" bit for RST in SPI frame. Reset value is only valid for true power-on hardware reset!
stat_bitmap_rst_spi	rw	Soft reset indication for SPI interface due to restart from bitmap configuration registers. Device status: RST Indication: error status is preserved until explicit write by 0 or reset by "Clear request device status" bit for RST in SPI frame.
stat_sm_rst_spi	rw	Soft reset indication for SPI interface due to safety mechanism detecting a fault that leads to a soft reset. Device status: RST Indication: error status is preserved until explicit write by 0 or reset by "Clear request device status" bit for RST in SPI frame. Reset value is only valid for true power-on hardware reset!
stat_om_spi	r	Operating mode status: 0 = startup mode, but init in progress, 1 = startup mode and init done, 2 and 3 = running mode.
stat_rom_su_spi	r	DSP ROM BIST startup error. Device status: DIAG Indication: error status is permanently set till reset.

Field	Type	Description
stat_rom_run_spi	rw	DSP ROM EDC and address parity runtime error. Device status: DIAG Indication: error status is either: 1. permanently set (when error occurs during startup) or 2. preserved over soft reset from NVM and afterwards until explicit write by 0 or reset by "Clear request device status" bit for DIAG in SPI frame (when error occurs during running mode).
stat_dsp_sched_spi	rw	DSP firmware scheduling error. Device status: DIAG Indication: error status is either: 1. permanently set (when error occurs during startup) or 2. preserved over soft reset from NVM and afterwards until explicit write by 0 or reset by "Clear request device status" bit for DIAG in SPI frame (when error occurs during running mode).
stat_dsp_check_spi	rw	DSP check error. Device status: DIAG Indication: error status is either: 1. permanently set (when error occurs during startup) or 2. preserved over soft reset from NVM and afterwards until explicit write by 0 or reset by "Clear request device status" bit for DIAG in SPI frame (when error occurs during running mode).
stat_cmtp_ecc_spi	rw	CMTP ECC error. Device status: DIAG Indication: error status is preserved until explicit write by 0 or reset by "Clear request device status" bit for DIAG in SPI frame.
stat_cmtp_rd_spi	rw	CMTP readout error. Device status: DIAG Indication: error status is preserved until explicit write by 0 or reset by "Clear request device status" bit for DIAG in SPI frame.
stat_cordic_overflow_spi	rw	Hardware CORDIC calculation overflow error. Device status: DIAG Indication: error status is preserved until explicit write by 0 or reset by "Clear request device status" bit for DIAG in SPI frame.
stat_adc_toggle_spi	rw	ADC comparator toggle error during sigma-delta and tracking phases. Indication: error status is preserved until explicit write by 0.
stat_test_spi	rw	Testmode activated that may have an effect on the provided angle value. Device status: DIAG Indication: error status is preserved until explicit write by 0 or reset by "Clear request device status" bit for DIAG in SPI frame.

8.2.4 stat_en_0_reg

Register	stat_en_0_reg							
Address	0x2							
Bit	15	14	13	12	11	10	9	8
Field	stat_test_en	-			stat_cordic_overflow_en	stat_cmtp_rd_en	stat_cmtp_ecc_en	-
Bit	7	6	5	4	3	2	1	0
Field	stat_dsp_sched_en	-					stat_bitmap_rst_act	stat_rst_act

Field	Type	Description
stat_rst_act	r	Restart from NVM: 0 = no action, 1 = performs reset of complete digital core. Condition: needs user register unlock to be set.
stat_bitmap_rst_act	r	Restart from bitmap configuration registers: 0 = no action, 1 = performs reset of digital core except configuration registers. Condition: needs user register unlock to be set.
stat_dsp_sched_en	rw	Status flag enable for DSP firmware scheduling error: 0 = error indication suppressed, 1 = error indicated (default). Condition: needs user register unlock to be set.
stat_cmtp_ecc_en	rw	Status flag enable for CMTP ECC error: 0 = error indication suppressed, 1 = error indicated (default). Condition: needs user register unlock to be set.
stat_cmtp_rd_en	rw	Status flag enable for CMTP read error: 0 = error indication suppressed, 1 = error indicated (default). Condition: needs user register unlock to be set.
stat_cordic_overflow_en	rw	Status flag enable for CORDIC overflow error: 0 = error indication suppressed, 1 = error indicated (default). Dependency: sm_fw_overflow_dis Condition: needs user register unlock to be set.

Field	Type	Description
stat_test_en	rw	Status flag enable for active testmode indication. 0 = error indication suppressed, 1 = error indicated (default). Condition: needs IFX register unlock to be set.

8.2.5 stat_1_reg

Register	stat_1_reg							
Address	0x3							
Bit	15	14	13	12	11	10	9	8
Field	stat_dyn_comp_lim_spi	stat_diff_stress_rng_spi	stat_shear_stress_rng_spi	stat_t_lin_max_spi	stat_t_lin_min_spi	stat_y_adc_rng_spi	stat_x_adc_rng_spi	stat_bitmap_ifx_crc_spi
Bit	7	6	5	4	3	2	1	0
Field	stat_bitmap_usr_crc_spi	stat_bitmap_lut_crc_spi	stat_cmtp_ifx_crc_spi	stat_cmtp_usr_crc_spi	stat_cmtp_lut_crc_spi	stat_cmtp_ifx_virgin_spi	stat_cmtp_usr_virgin_spi	stat_cmtp_lut_virgin_spi

Field	Type	Description
stat_cmtp_lut_virgin_spi	r	MPC LUT area of CMTP is fully virgin. Device status: no Indication: error status is permanently set till reset.
stat_cmtp_usr_virgin_spi	r	User config area of CMTP is fully virgin. Device status: no Indication: error status is permanently set till reset.
stat_cmtp_ifx_virgin_spi	r	IFX config area of CMTP is at least partially virgin. Device status: DIAG, but masked in case of restart from bitmap configuration registers. Indication: error status is permanently set till reset.
stat_cmtp_lut_crc_spi	r	CRC error in MPC LUT area of CMTP. Device status: DIAG, but masked if corresponding virgin flag is set and masked in case of restart from bitmap configuration registers. Indication: error status is permanently set till reset.
stat_cmtp_usr_crc_spi	r	CRC error in user config area of CMTP. Device status: DIAG, but masked if corresponding virgin flag is set and masked in case of restart from bitmap configuration registers. Indication: error status is permanently set till reset.
stat_cmtp_ifx_crc_spi	r	CRC error in IFX config area of CMTP. Device status: DIAG, but masked in case of restart from bitmap configuration registers. Indication: error status is permanently set till reset.
stat_bitmap_lut_crc_spi	rw	CRC error in MPC LUT area of bitmap. Device status: DIAG Indication: error status is either: 1. permanently set (when error occurs during startup) or 2. preserved over soft reset from NVM and afterwards until explicit write by 0 or reset by "Clear request device status" bit for DIAG in SPI frame (when error occurs during running mode).

Field	Type	Description
stat_bitmap_usr_crc_spi	rw	CRC error in user config area of bitmap. Device status: DIAG Indication: error status is either: 1. permanently set (when error occurs during startup) or 2. preserved over soft reset from NVM and afterwards until explicit write by 0 or reset by "Clear request device status" bit for DIAG in SPI frame (when error occurs during running mode).
stat_bitmap_ifx_crc_spi	rw	CRC error in IFX config area of bitmap. Device status: DIAG Indication: error status is either: 1. permanently set (when error occurs during startup) or 2. preserved over soft reset from NVM and afterwards until explicit write by 0 or reset by "Clear request device status" bit for DIAG in SPI frame (when error occurs during running mode).
stat_x_adc_rng_spi	rw	X ADC value out of range error. Device status: DIAG Indication: error status is preserved until explicit write by 0 or reset by "Clear request device status" bit for DIAG in SPI frame.
stat_y_adc_rng_spi	rw	Y ADC value out of range error. Device status: DIAG Indication: error status is preserved until explicit write by 0 or reset by "Clear request device status" bit for DIAG in SPI frame.
stat_t_lin_min_spi	rw	Compensated junction temperature below minimum operating temperature. Device status: DIAG Indication: error status is preserved until explicit write by 0 or reset by "Clear request device status" bit for DIAG in SPI frame.
stat_t_lin_max_spi	rw	Compensated junction temperature above maximum operating temperature. Device status: OT Indication: error status is preserved until explicit write by 0 or reset by "Clear request device status" bit for OT in SPI frame.
stat_shear_stress_rng_spi	rw	Shear stress value outside of the expected +/-100MPa. Device status: DIAG Indication: error status is preserved until explicit write by 0 or reset by "Clear request device status" bit for DIAG in SPI frame.
stat_diff_stress_rng_spi	rw	Differential stress value outside of the expected +/-400MPa. Device status: DIAG Indication: error status is preserved until explicit write by 0 or reset by "Clear request device status" bit for DIAG in SPI frame.

Field	Type	Description
stat_dyn_comp_lim_spi	rw	Kalman predict difference exceeds defined limit. Device status: DIAG Indication: error status is preserved until explicit write by 0 or reset by "Clear request device status" bit for DIAG in SPI frame. Indication can be disabled by setting the "dyn_comp_lim" bitfield to disabled.

8.2.6 stat_en_1_reg

Register	stat_en_1_reg							
Address	0x4							
Bit	15	14	13	12	11	10	9	8
Field	-	stat_diff_stress_rng_en	stat_shear_stress_rng_en	stat_t_lin_max_en	stat_t_lin_min_en	stat_y_adc_rng_en	stat_x_adc_rng_en	stat_bitmap_ifx_crc_en
Bit	7	6	5	4	3	2	1	0
Field	stat_bitmap_usr_crc_en	stat_bitmap_lut_crc_en	-	stat_cmtp_usr_crc_en	stat_cmtp_lut_crc_en	-		

Field	Type	Description
stat_cmtp_lut_crc_en	rw	Status flag enable for CRC error in MPC LUT area of CMTP: 0 = error indication suppressed, 1 = error indicated (default). Condition: needs user register unlock to be set. Note that this bit is protected by the bitmap user CRC, as writing this bit also changes "sm_cmtp_lut_usr_crc_dis"! To prevent a soft reset "stat_bitmap_usr_crc_en" should be cleared before writing this bit. This is done to prevent false-negative error indications due to a flipped status enable bit. Also note that this bit has no influence on the download of the CMTP MPC LUT area on a CRC error. It only suppresses error indication via the corresponding SPI status flag.
stat_cmtp_usr_crc_en	rw	Status flag enable for CRC error in user config area of CMTP: 0 = error indication suppressed, 1 = error indicated (default). Condition: needs user register unlock to be set. Note that this bit is protected by the bitmap user CRC, as writing this bit also changes "sm_cmtp_lut_usr_crc_dis"! To prevent a soft reset "stat_bitmap_usr_crc_en" should be cleared before writing this bit. This is done to prevent false-negative error indications due to a flipped status enable bit. Also note that this bit has no influence on the download of the CMTP user area or the startup behavior of the sensor (e.g. transition from OM_startup to OM_running) on a CRC error. It only suppresses error indication via the corresponding SPI status flag.
stat_bitmap_lut_crc_en	rw	Status flag enable for CRC error in MPC LUT area of bitmap: 0 = error indication suppressed, 1 = error indicated (default). Condition: needs user register unlock to be set. Note that this bit is protected by the bitmap user CRC, as writing this bit also changes "sm_bitmap_lut_crc_dis"! To prevent a soft reset "stat_bitmap_usr_crc_en" should be cleared before writing this bit. This is done to prevent false-negative error indications due to a flipped status enable bit.

Field	Type	Description
stat_bitmap_usr_crc_en	rw	Status flag enable for CRC error in user config area of bitmap: 0 = error indication suppressed, 1 = error indicated (default). Condition: needs user register unlock to be set. Note that this bit is partially protected by the bitmap user CRC, as writing this bit also changes "sm_bitmap_usr_crc_dis"! To prevent a soft reset a correct user CRC checksum shall be written before setting this bit. Clearing this bit does not trigger a soft reset.
stat_bitmap_ifx_crc_en	rw	Status flag enable for CRC error in IFX config area of bitmap: 0 = error indication suppressed, 1 = error indicated (default). Condition: needs IFX register unlock to be set.
stat_x_adc_rng_en	rw	Status flag enable for X ADC value out of range error: 0 = error indication suppressed, 1 = error indicated (default). Condition: needs user register unlock to be set.
stat_y_adc_rng_en	rw	Status flag enable for Y ADC value out of range error: 0 = error indication suppressed, 1 = error indicated (default). Condition: needs user register unlock to be set.
stat_t_lin_min_en	rw	Status flag enable for compensated temperature below minimum operating temperature: 0 = error indication suppressed, 1 = error indicated (default). Condition: needs user register unlock to be set. Note that this bit is protected by the bitmap user CRC, as writing this bit also changes "sm_t_lin_min_dis"! To prevent a soft reset "stat_bitmap_usr_crc_en" should be cleared before writing this bit. This is done to prevent false-negative error indications due to a flipped status enable bit.
stat_t_lin_max_en	rw	Status flag enable for compensated temperature above maximum operating temperature: 0 = error indication suppressed, 1 = error indicated (default). Condition: needs user register unlock to be set. Note that this bit is protected by the bitmap user CRC, as writing this bit also changes "sm_t_lin_max_dis"! To prevent a soft reset "stat_bitmap_usr_crc_en" should be cleared before writing this bit. This is done to prevent false-negative error indications due to a flipped status enable bit.

Field	Type	Description
stat_shear_stress_rng_en	rw	Status flag enable for shear stress value outside of the expected +/-100MPa: 0 = error indication suppressed, 1 = error indicated (default). Condition: needs user register unlock to be set. Note that this bit is protected by the bitmap user CRC, as writing this bit also changes "sm_shear_stress_rng_dis"! To prevent a soft reset "stat_bitmap_usr_crc_en" should be cleared before writing this bit. This is done to prevent false-negative error indications due to a flipped status enable bit.
stat_diff_stress_rng_en	rw	Status flag enable for differential stress value outside of the expected +/-400MPa: 0 = error indication suppressed, 1 = error indicated (default). Condition: needs user register unlock to be set. Note that this bit is protected by the bitmap user CRC, as writing this bit also changes "sm_diff_stress_rng_dis"! To prevent a soft reset "stat_bitmap_usr_crc_en" should be cleared before writing this bit. This is done to prevent false-negative error indications due to a flipped status enable bit.

8.2.7 stat_2_reg

Register	stat_2_reg							
Address	0x5							
Bit	15	14	13	12	11	10	9	8
Field	stat_interpolate_lim_spi	stat_vec_len_max_spi	stat_vec_len_min_spi	stat_vhall_bias_rng_spi	stat_vddp_ov_spi	stat_vddp_uv_spi	stat_vddd_ov_spi	stat_vddd_uv_spi
Bit	7	6	5	4	3	2	1	0
Field	stat_vdda_ov_spi	stat_vdda_uv_spi	stat_vdd2v5d_ov_spi	stat_vdd2v5d_uv_spi	stat_vdd2v5a_ov_spi	stat_vdd2v5a_uv_spi	stat_tv_0_rng_spi	stat_tv_fs_rng_spi

Field	Type	Description
stat_tv_fs_rng_spi	rw	Full-scale test vector out of range error. Device status: DIAG Indication: error status is preserved until explicit write by 0 or reset by "Clear request device status" bit for DIAG in SPI frame.
stat_tv_0_rng_spi	rw	Differential 0V test vector out of range error. Device status: DIAG Indication: error status is preserved until explicit write by 0 or reset by "Clear request device status" bit for DIAG in SPI frame.
stat_vdd2v5a_uv_spi	rw	VDD2V5A undervoltage error. Device status: DIAG Indication: error status is preserved until explicit write by 0 or reset by "Clear request device status" bit for DIAG in SPI frame.
stat_vdd2v5a_ov_spi	rw	VDD2V5A overvoltage error. Device status: DIAG Indication: error status is preserved until explicit write by 0 or reset by "Clear request device status" bit for DIAG in SPI frame.
stat_vdd2v5d_uv_spi	rw	VDD2V5D undervoltage error. Device status: DIAG Indication: error status is preserved until explicit write by 0 or reset by "Clear request device status" bit for DIAG in SPI frame.
stat_vdd2v5d_ov_spi	rw	VDD2V5D overvoltage error. Device status: DIAG Indication: error status is preserved until explicit write by 0 or reset by "Clear request device status" bit for DIAG in SPI frame.
stat_vdda_uv_spi	rw	VDDA undervoltage error. Device status: DIAG Indication: error status is preserved until explicit write by 0 or reset by "Clear request device status" bit for DIAG in SPI frame.

Field	Type	Description
stat_vdda_ov_spi	rw	VDDA overvoltage error. Device status: DIAG Indication: error status is preserved until explicit write by 0 or reset by "Clear request device status" bit for DIAG in SPI frame.
stat_vddd_uv_spi	rw	VDDD undervoltage error. Device status: DIAG Indication: error status is preserved until explicit write by 0 or reset by "Clear request device status" bit for DIAG in SPI frame. Note that this status flag is not related to the smi_vddd_uv_check safety mechanism, which directly triggers a chip reset and thus is indicated via stat_rst_spi.
stat_vddd_ov_spi	rw	VDDD overvoltage error. Device status: DIAG Indication: error status is preserved until explicit write by 0 or reset by "Clear request device status" bit for DIAG in SPI frame. Note that this status flag is not related to the smi_vddd_ov_check safety mechanism, which directly triggers a chip reset and thus is indicated via stat_rst_spi.
stat_vddp_uv_spi	rw	VDDP undervoltage error. Device status: SUP Indication: error status is preserved until explicit write by 0 or reset by "Clear request device status" bit for SUP in SPI frame.
stat_vddp_ov_spi	rw	VDDP overvoltage error. Device status: SUP Indication: error status is preserved until explicit write by 0 or reset by "Clear request device status" bit for SUP in SPI frame.
stat_vhall_bias_rng_spi	rw	Vertical Hall bias voltage out of range error. Note that this range check is disabled in firmware! Device status: DIAG Indication: error status is preserved until explicit write by 0 or reset by "Clear request device status" bit for DIAG in SPI frame.
stat_vec_len_min_spi	rw	Vector length below minimum configured limit. Device status: MAG Indication: error status is preserved until explicit write by 0 or reset by "Clear request device status" bit for MAG in SPI frame.
stat_vec_len_max_spi	rw	Vector length above maximum configured limit. Device status: MAG Indication: error status is preserved until explicit write by 0 or reset by "Clear request device status" bit for MAG in SPI frame.

Field	Type	Description
stat_interpolate_lim_spi	rw	Difference between angle_pred and angle_exp exceeds defined limit. Device status: DIAG Indication: error status is preserved until explicit write by 0 or reset by "Clear request device status" bit for DIAG in SPI frame.

8.2.8 stat_en_2_reg

Register	stat_en_2_reg							
Address	0x6							
Bit	15	14	13	12	11	10	9	8
Field	stat_interpolate_lim_en	stat_vec_len_max_en	stat_vec_len_min_en	stat_vhall_bias_rng_en	stat_vddp_ov_en	stat_vddp_uv_en	stat_vddd_ov_en	stat_vddd_uv_en
Bit	7	6	5	4	3	2	1	0
Field	stat_vdda_ov_en	stat_vdda_uv_en	stat_vdd2v5d_ov_en	stat_vdd2v5d_uv_en	stat_vdd2v5a_ov_en	stat_vdd2v5a_uv_en	stat_tv_0_rng_en	stat_tv_fs_rng_en

Field	Type	Description
stat_tv_fs_rng_en	rw	Status enable flag for full-scale test vector out of range error: 0 = error indication suppressed, 1 = error indicated (default). Condition: needs user register unlock to be set. Note that this bit is protected by the bitmap user CRC, as writing this bit also changes "sm_tv_rng_dis"! To prevent a soft reset "stat_bitmap_usr_crc_en" should be cleared before writing this bit. This is done to prevent false-negative error indications due to a flipped status enable bit.
stat_tv_0_rng_en	rw	Status enable flag for differential 0V test vector out of range error: 0 = error indication suppressed, 1 = error indicated (default). Condition: needs user register unlock to be set. Note that this bit is protected by the bitmap user CRC, as writing this bit also changes "sm_tv_rng_dis"! To prevent a soft reset "stat_bitmap_usr_crc_en" should be cleared before writing this bit. This is done to prevent false-negative error indications due to a flipped status enable bit.
stat_vdd2v5a_uv_en	rw	Status enable flag for VDD2V5A undervoltage error: 0 = error indication suppressed, 1 = error indicated (default). Condition: needs user register unlock to be set. Note that this bit is protected by the bitmap user CRC, as writing this bit also changes "sm_vdd2v5x_ouv_dis"! To prevent a soft reset "stat_bitmap_usr_crc_en" should be cleared before writing this bit. This is done to prevent false-negative error indications due to a flipped status enable bit.

Field	Type	Description
stat_vdd2v5a_ov_en	rw	Status enable flag for VDD2V5A overvoltage error: 0 = error indication suppressed, 1 = error indicated (default). Condition: needs user register unlock to be set. Note that this bit is protected by the bitmap user CRC, as writing this bit also changes "sm_vdd2v5x_ouv_dis"! To prevent a soft reset "stat_bitmap_usr_crc_en" should be cleared before writing this bit. This is done to prevent false-negative error indications due to a flipped status enable bit.
stat_vdd2v5d_uv_en	rw	Status enable flag for VDD2V5D undervoltage error: 0 = error indication suppressed, 1 = error indicated (default). Condition: needs user register unlock to be set. Note that this bit is protected by the bitmap user CRC, as writing this bit also changes "sm_vdd2v5x_ouv_dis"! To prevent a soft reset "stat_bitmap_usr_crc_en" should be cleared before writing this bit. This is done to prevent false-negative error indications due to a flipped status enable bit.
stat_vdd2v5d_ov_en	rw	Status enable flag for VDD2V5D overvoltage error: 0 = error indication suppressed, 1 = error indicated (default). Condition: needs user register unlock to be set. Note that this bit is protected by the bitmap user CRC, as writing this bit also changes "sm_vdd2v5x_ouv_dis"! To prevent a soft reset "stat_bitmap_usr_crc_en" should be cleared before writing this bit. This is done to prevent false-negative error indications due to a flipped status enable bit.
stat_vdda_uv_en	rw	Status enable flag for VDDA undervoltage error: 0 = error indication suppressed, 1 = error indicated (default). Condition: needs user register unlock to be set. Note that this bit is protected by the bitmap user CRC, as writing this bit also changes "sm_vddx_ouv_dis"! To prevent a soft reset "stat_bitmap_usr_crc_en" should be cleared before writing this bit. This is done to prevent false-negative error indications due to a flipped status enable bit.
stat_vdda_ov_en	rw	Status enable flag for VDDA overvoltage error: 0 = error indication suppressed, 1 = error indicated (default). Condition: needs user register unlock to be set. Note that this bit is protected by the bitmap user CRC, as writing this bit also changes "sm_vddx_ouv_dis"! To prevent a soft reset "stat_bitmap_usr_crc_en" should be cleared before writing this bit. This is done to prevent false-negative error indications due to a flipped status enable bit.

Field	Type	Description
stat_vddd_uv_en	rw	Status enable flag for VDDD undervoltage error: 0 = error indication suppressed, 1 = error indicated (default). Condition: needs user register unlock to be set. Note that this bit is protected by the bitmap user CRC, as writing this bit also changes "sm_vddx_ouv_dis"! To prevent a soft reset "stat_bitmap_usr_crc_en" should be cleared before writing this bit. This is done to prevent false-negative error indications due to a flipped status enable bit.
stat_vddd_ov_en	rw	Status enable flag for VDDD overvoltage error: 0 = error indication suppressed, 1 = error indicated (default). Condition: needs user register unlock to be set. Note that this bit is protected by the bitmap user CRC, as writing this bit also changes "sm_vddx_ouv_dis"! To prevent a soft reset "stat_bitmap_usr_crc_en" should be cleared before writing this bit. This is done to prevent false-negative error indications due to a flipped status enable bit.
stat_vddp_uv_en	rw	Status enable flag for VDDP undervoltage error: 0 = error indication suppressed, 1 = error indicated (default). Condition: needs user register unlock to be set. Note that this bit is protected by the bitmap user CRC, as writing this bit also changes "sm_vddp_uv_dis"! To prevent a soft reset "stat_bitmap_usr_crc_en" should be cleared before writing this bit. This is done to prevent false-negative error indications due to a flipped status enable bit.
stat_vddp_ov_en	rw	Status enable flag for VDDP overvoltage error: 0 = error indication suppressed, 1 = error indicated (default). Condition: needs user register unlock to be set. Note that this bit is protected by the bitmap user CRC, as writing this bit also changes "sm_vddp_ov_dis"! To prevent a soft reset "stat_bitmap_usr_crc_en" should be cleared before writing this bit. This is done to prevent false-negative error indications due to a flipped status enable bit.
stat_vhall_bias_rng_en	rw	Status enable flag for vertical Hall bias voltage out of range error: 0 = error indication suppressed, 1 = error indicated (default). Condition: needs user register unlock to be set.
stat_vec_len_min_en	rw	Status enable flag for vector length below minimum configured limit: 0 = error indication suppressed, 1 = error indicated (default). Condition: needs user register unlock to be set.

Field	Type	Description
stat_vec_len_max_en	rw	Status enable flag for vector length above maximum configured limit: 0 = error indication suppressed, 1 = error indicated (default). Condition: needs user register unlock to be set.
stat_interpolate_lim_en	rw	Status enable flag for difference between angle_pred and angle_exp exceeding defined limit: 0 = error indication suppressed, 1 = error indicated (default). Condition: needs user register unlock to be set.

8.2.9 angle_sync_reg

Register	angle_sync_reg							
Address	0xA							
Bit	15	14	13	12	11	10	9	8
Field	angle_sync...							
Bit	7	6	5	4	3	2	1	0
Field	...angle_sync						stat_sync_as	stat_read_as

Field	Type	Description
stat_read_as	r	Read status of predicted (interpolated) angle value: 0 = fresh value, 1 = value already read without new sync trigger. Allows in-frame readout.
stat_sync_as	r	Sync status of predicted (interpolated) angle value: 0 = single or no sync trigger since last readout. 1 = more than one sync trigger since last readout. Allows in-frame readout.
angle_sync	r	Synchronized predicted (interpolated) angle value (synchronized to IFA or IFE according to "sync_mode" and "sync_edge" settings). Allows in-frame readout.

8.2.10 angle_vel_sync_reg

Register	angle_vel_sync_reg							
Address	0xB							
Bit	15	14	13	12	11	10	9	8
Field	angle_vel_sync...							
Bit	7	6	5	4	3	2	1	0
Field	...angle_vel_sync						stat_sync_avs	stat_read_avs

Field	Type	Description
stat_read_avs	r	Read status of synchronized estimated angle velocity value: 0 = fresh value, 1 = value already read without new sync trigger. Allows in-frame readout.
stat_sync_avs	r	Sync status of synchronized estimated angle velocity value: 0 = single or no sync trigger since last readout. 1 = more than one sync trigger since last readout. Allows in-frame readout.
angle_vel_sync	r	Synchronized estimated angle velocity value in units of 0.0280 LSB14/RPM (synchronized to IFA or IFE according to "sync_mode" and "sync_edge" settings). Allows in-frame readout.

8.2.11 angle_pred_reg

Register	angle_pred_reg							
Address	0xC							
Bit	15	14	13	12	11	10	9	8
Field	angle_pred_spi...							
Bit	7	6	5	4	3	2	1	0
Field	...angle_pred_spi							

Field	Type	Description
angle_pred_spi	r	Predicted (interpolated) angle value. Allows in-frame readout.

8.2.12 angle_set_reg

Register	angle_set_reg							
Address	0xD							
Bit	15	14	13	12	11	10	9	8
Field	angle_set_spi...							
Bit	7	6	5	4	3	2	1	0
Field	...angle_set_spi							

Field	Type	Description
angle_set_spi	r	Estimated (latency predicted) angle value as input for interpolation. Allows in-frame readout.

8.2.13 angle_vel_reg

Register	angle_vel_reg							
Address	0xE							
Bit	15	14	13	12	11	10	9	8
Field	angle_vel_spi...							
Bit	7	6	5	4	3	2	1	0
Field	...angle_vel_spi							

Field	Type	Description
angle_vel_spi	r	Estimated angle velocity value register in units of 0.112 LSB16/RPM. Allows in-frame readout.

8.2.14 angle_reg

Register	angle_reg							
Address	0xF							
Bit	15	14	13	12	11	10	9	8
Field	angle_spi...							
Bit	7	6	5	4	3	2	1	0
Field	...angle_spi							

Field	Type	Description
angle_spi	r	Angle value compensated by multi-point calibration LUT. Allows in-frame readout.

8.2.15 angle_raw_reg

Register	angle_raw_reg							
Address	0x10							
Bit	15	14	13	12	11	10	9	8
Field	angle_raw_spi...							
Bit	7	6	5	4	3	2	1	0
Field	...angle_raw_spi							

Field	Type	Description
angle_raw_spi	r	Raw angle value based on Xn/Yn (compensated for application imperfections). Allows in-frame readout.

8.2.16 vec_len_reg

Register	vec_len_reg							
Address	0x11							
Bit	15	14	13	12	11	10	9	8
Field	vec_len_spi...							
Bit	7	6	5	4	3	2	1	0
Field	...vec_len_spi							

Field	Type	Description
vec_len_spi	r	Vector length value based on Xn/Yn (compensated for application imperfections). Allows in-frame readout.

8.2.17 xn_reg

Register	xn_reg							
Address	0x12							
Bit	15	14	13	12	11	10	9	8
Field	xn_spi...							
Bit	7	6	5	4	3	2	1	0
Field	...xn_spi							

Field	Type	Description
xn_spi	r	Compensated Xn value (compensated for application imperfections). Allows in-frame readout. Needed for system calibration!

8.2.18 yn_reg

Register	yn_reg							
Address	0x13							
Bit	15	14	13	12	11	10	9	8
Field	yn_spi...							
Bit	7	6	5	4	3	2	1	0
Field	...yn_spi							

Field	Type	Description
yn_spi	r	Compensated Yn value (compensated for application imperfections). Allows in-frame readout. Needed for system calibration!

8.2.19 angle3_reg

Register	angle3_reg							
Address	0x14							
Bit	15	14	13	12	11	10	9	8
Field	angle3_spi...							
Bit	7	6	5	4	3	2	1	0
Field	...angle3_spi							

Field	Type	Description
angle3_spi	r	Angle value 3 based on X3/Y3 (compensated for signal path inaccuracies only). Allows in-frame readout. Needed for system calibration!

8.2.20 vec_len3_reg

Register	vec_len3_reg							
Address	0x15							
Bit	15	14	13	12	11	10	9	8
Field	vec_len3_spi...							
Bit	7	6	5	4	3	2	1	0
Field	...vec_len3_spi							

Field	Type	Description
vec_len3_spi	r	Vector length 3 value based on X3/Y3 (compensated for signal path inaccuracies only). Allows in-frame readout. Needed for system calibration!

8.2.21 x3_reg

Register	x3_reg							
Address	0x16							
Bit	15	14	13	12	11	10	9	8
Field	x3_spi...							
Bit	7	6	5	4	3	2	1	0
Field	...x3_spi							

Field	Type	Description
x3_spi	r	Linearly compensated X3 value (compensating signal path inaccuracies only). Allows in-frame readout. Needed for system calibration!

8.2.22 y3_reg

Register	y3_reg							
Address	0x17							
Bit	15	14	13	12	11	10	9	8
Field	y3_spi...							
Bit	7	6	5	4	3	2	1	0
Field	...y3_spi							

Field	Type	Description
y3_spi	r	Linearly compensated Y3 value (compensating signal path inaccuracies only). Allows in-frame readout. Needed for system calibration!

8.2.23 t_lin_reg

Register	t_lin_reg							
Address	0x25							
Bit	15	14	13	12	11	10	9	8
Field	t_lin...							
Bit	7	6	5	4	3	2	1	0
Field	...t_lin							

Field	Type	Description
t_lin	r	Calibrated linearized temperature.

8.2.24 syscal_angle_base_reg

Register	syscal_angle_base_reg							
Address	0x2B							
Bit	15	14	13	12	11	10	9	8
Field	syscal_angle_base...							
Bit	7	6	5	4	3	2	1	0
Field	...syscal_angle_base							-

Field	Type	Description
syscal_angle_base	r	Angle base between 0 to 360 degree updated by system self-calibration with a resolution of (360 degree / 32768 LSB) = 0.011 degree/LSB.

8.2.25 syscal_offset_x_reg

Register	syscal_offset_x_reg							
Address	0x2C							
Bit	15	14	13	12	11	10	9	8
Field	syscal_offset_x...							
Bit	7	6	5	4	3	2	1	0
Field	...syscal_offset_x							

Field	Type	Description
syscal_offset_x	r	X offset compensation value updated by system self-calibration.

8.2.26 syscal_offset_y_reg

Register	syscal_offset_y_reg							
Address	0x2D							
Bit	15	14	13	12	11	10	9	8
Field	syscal_offset_y...							
Bit	7	6	5	4	3	2	1	0
Field	...syscal_offset_y							

Field	Type	Description
syscal_offset_y	r	Y offset compensation value updated by system self-calibration.

8.2.27 syscal_ellipticity_reg

Register	syscal_ellipticity_reg							
Address	0x2E							
Bit	15	14	13	12	11	10	9	8
Field	syscal_ellipticity_scale		syscal_ellipticity...					
Bit	7	6	5	4	3	2	1	0
Field	...syscal_ellipticity							

Field	Type	Description
syscal_ellipticity	r	Ellipticity compensation value for system self-calibration. Ellipticity value calculated as: ellipticity = syscal_ellipticity >> (13 - syscal_ellipticity_scale)
syscal_ellipticity_scale	r	Scaling factor for "syscal_ellipticity".

8.2.28 syscal_oos_rotation_reg

Register	syscal_oos_rotation_reg							
Address	0x2F							
Bit	15	14	13	12	11	10	9	8
Field	syscal_oos_rotation...							
Bit	7	6	5	4	3	2	1	0
Field	...syscal_oos_rotation							-

Field	Type	Description
syscal_oos_rotation	r	OoS rotation from system self-calibration for linear compensation of application imperfections in the range of −90 degree ≤ oos_rotation < +90 degree: −16384 = -90 degree, 0 = 0 degree, 1 = 180 degree/2^15 = ca. 0.0055 degree, 16383 = +90 degree - 180 degree/2^15 = ca. 89.995 degree.

8.2.29 mpc_lut_idx_reg

Register	mpc_lut_idx_reg							
Address	0x38							
Bit	15	14	13	12	11	10	9	8
Field	-							
Bit	7	6	5	4	3	2	1	0
Field	-			mpc_lut_idx				

Field	Type	Description
mpc_lut_idx	rw	MPC LUT index register for equidistant mode. Also allows to read current index in non-equidistant mode.

8.2.30 mpc_lut_data_0_reg

Register	mpc_lut_data_0_reg							
Address	0x3A							
Bit	15	14	13	12	11	10	9	8
Field	mpc_lut_data_0...							
Bit	7	6	5	4	3	2	1	0
Field	...mpc_lut_data_0							

Field	Type	Description
mpc_lut_data_0	rw	Provides lower calibration value in equidistant mode at the given index over the whole LUT or the lower supporting point in non-equidistant mode for the given angle in the lower half of the LUT.

8.2.31 mpc_lut_data_1_reg

Register	mpc_lut_data_1_reg							
Address	0x3B							
Bit	15	14	13	12	11	10	9	8
Field	mpc_lut_data_1...							
Bit	7	6	5	4	3	2	1	0
Field	...mpc_lut_data_1							

Field	Type	Description
mpc_lut_data_1	rw	Provides upper calibration value in equidistant mode at the given index+1 over the whole LUT or the upper supporting point in non-equidistant mode for the given angle in the lower half of the LUT.

8.2.32 mpc_lut_data_2_reg

Register	mpc_lut_data_2_reg							
Address	0x3C							
Bit	15	14	13	12	11	10	9	8
Field	mpc_lut_data_2...							
Bit	7	6	5	4	3	2	1	0
Field	...mpc_lut_data_2							

Field	Type	Description
mpc_lut_data_2	rw	Provides calibration value at index+2 in equidistant mode or lower calibration value for the given angle in non-equidistant mode.

8.2.33 mpc_lut_data_3_reg

Register	mpc_lut_data_3_reg							
Address	0x3D							
Bit	15	14	13	12	11	10	9	8
Field	mpc_lut_data_3...							
Bit	7	6	5	4	3	2	1	0
Field	...mpc_lut_data_3							

Field	Type	Description
mpc_lut_data_3	rw	Provides calibration value at index+3 in equidistant mode or upper calibration value for the given angle in non-equidistant mode.

8.2.34 usr_config_0_reg

Register	usr_config_0_reg							
Address	0x3E							
Bit	15	14	13	12	11	10	9	8
Field	lut_word_16_31_crc							
Bit	7	6	5	4	3	2	1	0
Field	lut_word_0_15_crc							

Field	Type	Description
lut_word_0_15_crc	rw	CRC checksum over lower 16 LUT entries.
lut_word_16_31_crc	rw	CRC checksum over upper 16 LUT entries.

8.2.35 usr_config_1_reg

Register	usr_config_1_reg							
Address	0x3F							
Bit	15	14	13	12	11	10	9	8
Field	adc_max_rng_en	abz_init_pos_mode	abz_mode	spi_miso_crc_inv_dis	spi_mosi_crc_dis	sync_edge		sync_mode...
Bit	7	6	5	4	3	2	1	0
Field	...sync_mode	pwm_start_edge	pwm_freq			if_mode		

Field	Type	Description
if_mode	rw	Interface mode depending on chip variant: - TLE49012-S0001: 0 = SPI, 1 = ABZ, 2 = UVW, 3 = UVW, 4 = PWM only, 5 = ABZ + PWM, 6 = UVW + PWM, 7 = UVW + PWM. - chip variant = TLE49012-S0002: 0 = ABZ + PWM, 1 = ABZ, 2 = ABZ, 3 = ABZ, 4 = PWM only, 5 = ABZ + PWM, 6 = ABZ + PWM, 7 = ABZ + PWM. - chip variant = TLE49012-E0001: 0 = UVW + PWM, 1 = UVW, 2 = UVW, 3 = UVW, 4 = PWM only, 5 = UVW + PWM, 6 = UVW + PWM, 7 = UVW + PWM. - chip variant = TLI49012-E0001: 0 = SPI, 1 = ABZ, 2 = UVW, 3 = UVW, 4 = PWM only, 5 = ABZ + PWM, 6 = UVW + PWM, 7 = UVW + PWM. - chip variant = TLI49012-E0002: 0 = ABZ + PWM, 1 = ABZ, 2 = ABZ,

Field	Type	Description
		3 = ABZ, 4 = PWM only, 5 = ABZ + PWM, 6 = ABZ + PWM, 7 = ABZ + PWM.
pwm_freq	rw	PWM frequency: 0 = 250Hz, 1 = 500Hz, 2 = 1000Hz, 3 = 2000Hz (default), 4 = 2500Hz, 5 = 3000Hz, 6 = 3500Hz, 7 = 4000Hz.
pwm_start_edge	rw	PWM starting edge: 0 = falling edge, 1 = rising edge.
sync_mode	rw	Synchronization pin mode: 0 = measurement synchronization on configured edge of IFE for SPI or PWM (if active), 1 = trigger initial pulses for ABZ (if active) on configured edge of IFE or measurement synchronization on falling edge of IFA (CSN) only for SPI, 2...3 = trigger sensor reset on configured edge of IFE for ABZ, UVW or PWM.
sync_edge	rw	Synchronization pin trigger edge: 0 = disable, 1 = rising edge, 2 = falling edge, 3 = any edge.
spi_mosi_crc_dis	rw	SPI MOSI CRC check disable: 0 = MOSI CRC check enabled, 1 = MOSI CRC check disabled.
spi_miso_crc_inv_dis	rw	SPI MISO CRC inversion disable on read timeout: 0 = MISO CRC inverted on read timeout, 1 = MISO CRC not inverted on read timeout.
abz_mode	rw	ABZ interface mode. 0 = A/B mode, 1 = step/direction mode.
abz_init_pos_mode	rw	ABZ initial position mode: 0 = initialization pulses after startup resp. absolute position mode, 1 = no initialization pulses resp. relative position mode.

Field	Type	Description
adc_max_rng_en	rw	Enable large ADC range: 0 = small range (default), 1 = large range.

8.2.36 usr_config_2_reg

Register	usr_config_2_reg							
Address	0x40							
Bit	15	14	13	12	11	10	9	8
Field	cfg_unused_64		abz_z_mode		abz_uvw_pulses_pp...			
Bit	7	6	5	4	3	2	1	0
Field	...abz_uvw_pulses_pp							

Field	Type	Description
abz_uvw_pulses_pp	rw	ABZ/UVW interface pulses resp. pole-pairs. For UVW interface, the least-significant 4 bits define the number of pole-pairs.
abz_z_mode	rw	ABZ index/Z pulse mode: 0 = ungated for tz (typ. 5 us), 1 = A/B gated (only for A/B mode, fallback to ungated for step/dir mode), 2 = A gated, 3 = B gated (only for A/B mode, fallback to ungated for step/dir mode).
cfg_unused_64	rw	

8.2.37 usr_config_3_reg

Register	usr_config_3_reg							
Address	0x41							
Bit	15	14	13	12	11	10	9	8
Field	cfg_unused_65		sm_fw_overflow_dis	sm_adc_overflow_ob_dis	sm_vddp_ov_dis	sm_vddp_uv_dis	sm_vddx_ouv_dis	sm_vdd2v5x_ouv_dis
Bit	7	6	5	4	3	2	1	0
Field	sm_tv_rng_dis	sm_diff_stress_rng_dis	sm_shear_stress_rng_dis	sm_t_lin_max_dis	sm_t_lin_min_dis	sm_bitmap_usr_crc_dis	sm_bitmap_lut_crc_dis	sm_cmtp_lut_usr_crc_dis

Field	Type	Description
sm_cmtp_lut_usr_crc_dis	r	Combined status flag disable for CRC errors in MPC LUT and user config area of CMTP: 0 = error indicated (default), 1 = error indication suppressed. Note that this bit has no influence on the download of the CMTP user area, as the corresponding CRC checksum has to be checked before this bit is downloaded. Hence CRC errors in the CMTP user area cannot be masked with this bit to change the startup behavior of the sensor (e.g. transition from OM_startup to OM_running). The same basically applies to CRC errors in the CMTP MPC LUT area. As a result, the bit only suppresses error indications via the corresponding SPI status flags.
sm_bitmap_lut_crc_dis	r	Status flag disable for CRC error in MPC LUT area of bitmap: 0 = error indicated (default), 1 = error indication suppressed.
sm_bitmap_usr_crc_dis	r	Status flag disable for CRC error in user config area of bitmap: 0 = error indicated (default), 1 = error indication suppressed.
sm_t_lin_min_dis	r	Status flag disable for compensated temperature below minimum operating temperature: 0 = error indicated (default), 1 = error indication suppressed.
sm_t_lin_max_dis	r	Status flag disable for compensated temperature above maximum operating temperature: 0 = error indicated (default), 1 = error indication suppressed.
sm_shear_stress_rng_dis	r	Status flag disable for shear stress value outside of the expected +/-100MPa: 0 = error indicated (default), 1 = error indication suppressed.

Field	Type	Description
sm_diff_stress_rng_dis	r	Status flag disable for differential stress value outside of the expected +/-400MPa: 0 = error indicated (default), 1 = error indication suppressed.
sm_tv_rng_dis	r	Combined status disable flag for test vectors out of range errors: 0 = error indicated (default), 1 = error indication suppressed.
sm_vdd2v5x_ouv_dis	r	Combined status disable flag for VDD2V5x over- and undervoltage errors: 0 = error indicated (default), 1 = error indication suppressed.
sm_vddx_ouv_dis	r	Combined status disable flag for VDDA/VDDD over- and undervoltage errors: 0 = error indicated (default), 1 = error indication suppressed.
sm_vddp_uv_dis	r	Status disable flag for VDDP undervoltage error: 0 = error indicated (default), 1 = error indication suppressed.
sm_vddp_ov_dis	r	Status disable flag for VDDP overvoltage error: 0 = error indicated (default), 1 = error indication suppressed.
sm_adc_overflow_oob_dis	r	Combined status flag disable for overflow and out of bound errors X, Y and GP ADC channels: 0 = error indicated (default), 1 = error indication suppressed.
sm_fw_overflow_dis	r	Combined status flag disable for firmware and CORDIC overflow errors: 0 = error indicated (default), 1 = error indication suppressed.
cfg_unused_65	r	

8.2.38 usr_config_4_reg

Register	usr_config_4_reg							
Address	0x42							
Bit	15	14	13	12	11	10	9	8
Field	cfg_unused_66...							
Bit	7	6	5	4	3	2	1	0
Field	...cfg_unused_66			ife_spike_filter_en	ifd_spike_filter_en	ifc_spike_filter_en	ifb_spike_filter_en	ifa_spike_filter_en

Field	Type	Description
ifa_spike_filter_en	rw	IFA input spike filter enable: 0 = disabled, 1 = enabled.
ifb_spike_filter_en	rw	IFB input spike filter enable: 0 = disabled, 1 = enabled.
ifc_spike_filter_en	rw	IFC input spike filter enable: 0 = disabled, 1 = enabled.
ifd_spike_filter_en	rw	IFD input spike filter enable: 0 = disabled, 1 = enabled.
ife_spike_filter_en	rw	IFE input spike filter enable: 0 = disabled, 1 = enabled.
cfg_unused_66	rw	

8.2.39 usr_config_5_reg

Register	usr_config_5_reg							
Address	0x43							
Bit	15	14	13	12	11	10	9	8
Field	x3_base...							
Bit	7	6	5	4	3	2	1	0
Field	...x3_base							

Field	Type	Description
x3_base	rw	X3 value corresponding to programmed angle base (required for updating angle base during system self-calibration).

8.2.40 usr_config_6_reg

Register	usr_config_6_reg							
Address	0x44							
Bit	15	14	13	12	11	10	9	8
Field	y3_base...							
Bit	7	6	5	4	3	2	1	0
Field	...y3_base							

Field	Type	Description
y3_base	rw	Y3 value corresponding to programmed angle base (required for updating angle base during system self-calibration).

8.2.41 usr_config_7_reg

Register	usr_config_7_reg							
Address	0x45							
Bit	15	14	13	12	11	10	9	8
Field	angle_base...							
Bit	7	6	5	4	3	2	1	0
Field	...angle_base							angle_dir

Field	Type	Description
angle_dir	rw	Angle direction setting: 0 = CCW: angle and angle velocity values not inverted, 1 = CW: two's complement of angle and angle velocity values.
angle_base	rw	Angle base resp. zero angle calibration setting in the range of 0 to 360 degree with a resolution of (360 degree / 32768 LSB) = 0.011 degree/LSB.

8.2.42 usr_config_8_reg

Register	usr_config_8_reg							
Address	0x46							
Bit	15	14	13	12	11	10	9	8
Field	offset_app_x...							
Bit	7	6	5	4	3	2	1	0
Field	...offset_app_x							

Field	Type	Description
offset_app_x	rw	Application X channel offset value.

8.2.43 usr_config_9_reg

Register	usr_config_9_reg							
Address	0x47							
Bit	15	14	13	12	11	10	9	8
Field	offset_app_y...							
Bit	7	6	5	4	3	2	1	0
Field	...offset_app_y							

Field	Type	Description
offset_app_y	rw	Application Y channel offset value.

8.2.44 usr_config_10_reg

Register	usr_config_10_reg							
Address	0x48							
Bit	15	14	13	12	11	10	9	8
Field	syscal_angle_base_en	syscal_mode			lut_mode		-	
Bit	7	6	5	4	3	2	1	0
Field	-							

Field	Type	Description
lut_mode	rw	Multipoint calibration mode: 0 and 1 = no multipoint calibration, 2 = equidistant mode with 32 calibration points, 3 = non-equidistant mode with 16 calibration points.
syscal_mode	rw	System self-calibration setting: 0 = syscal disabled, 1 = vector length calibration, filter disabled, 2 = quadrature calibration, offset disabled, 3 = defaults to 0, 4 = quadrature calibration, offset for strayfield, 5 = vector length calibration, filter enabled, 6...7 = defaults to 0.
syscal_angle_base_en	rw	Angle base update by system self-calibration: 0 = disabled, 1 = enabled.

8.2.45 usr_config_11_reg

Register	usr_config_11_reg							
Address	0x49							
Bit	15	14	13	12	11	10	9	8
Field	dyn_comp_k2							
Bit	7	6	5	4	3	2	1	0
Field	dyn_comp_k1							

Field	Type	Description
dyn_comp_k1	rw	Dynamic compensation factor K1, influencing effective bandwidth of predicted angle values. See user manual for further explanations. Range from 0.02 to 1.00 in steps of 0.02: 0 = 0.02 = relaxed dynamic compensation, reduced angle noise, increased dynamic errors due to acceleration, 6 = 0.14 = compromise for fast and agile applications (default), 11 = 0.24 = very responsive dynamic compensation, quick response to acceleration, increased angle noise, 255 = 1.00 = max effective bandwidth, no averaging, quickest response to acceleration, excessive angle noise.
dyn_comp_k2	rw	Dynamic compensation factor K2, influencing effective bandwidth of angle velocity values. See user manual for further explanations. Range from 4.88e-4 to 0.125 in steps of 4.88e-4: 0 = 4.88e-4 = relaxed dynamic compensation, reduced angle noise, increased dynamic errors due to acceleration, 51 = 0.025 = compromise for fast and agile applications (default), 108 = 0.053 = very responsive dynamic compensation, quick response to acceleration, increased angle noise, 255 = 0.125 = max effective bandwidth, quickest response to acceleration, excessive angle (velocity) noise. Note that a value of 1.00 would correspond to no averaging at all for calculating the angle velocity. Such large values are only used upon initialization of the dynamic compensation, c.f. register fields starting with "dyn_comp_init_".

8.2.46 usr_config_12_reg

Register	usr_config_12_reg							
Address	0x4A							
Bit	15	14	13	12	11	10	9	8
Field	ellipticity							
Bit	7	6	5	4	3	2	1	0
Field	dyn_comp_time							

Field	Type	Description
dyn_comp_time	rw	Time adjustment for the dynamic compensation in units of 0.55 us/LSB. Positive values result in over-compensation of signal path latency, i.e. predicted angle values lead the magnetic angle. Negative values result in under-compensation of signal path latency, i.e. predicted angle values lag behind the magnetic angle.
ellipticity	rw	Ellipticity setting calculated as (1.00 + 0.01 * N). Calibrated by customer. 0 = 1.00, 1 = 1.01, ... 255 = 3.55.

8.2.47 usr_config_13_reg

Register	usr_config_13_reg							
Address	0x4B							
Bit	15	14	13	12	11	10	9	8
Field	oos_rotation...							
Bit	7	6	5	4	3	2	1	0
Field	...oos_rotation		dyn_comp_mode	dyn_comp_init_samples			dyn_comp_init_k1	dyn_comp_init_cond

Field	Type	Description
dyn_comp_init_cond	rw	Conditions upon which to initialize the dynamic compensation: 0 = only on startup, 1 = on startup and whenever prediction plausibility fails (according to the value of "dyn_comp_lim").
dyn_comp_init_k1	rw	Gradual adjustment of K1 during initialization: 0 = K1 remains fixed at 1.0 during initialization, 1 = gradually adjust K1 during initialization. Note that K2 is always gradually adjusted during initialization. Gradual adjustment means K value is reduced by 1/2 for each sample (e.g. start with 1.0, then 0.5, 0.25, 0.125, etc.).
dyn_comp_init_samples	rw	(N-1) number of samples for initialization of dynamic compensation with (K1, K2) in brackets: 0 = initialization without speed: (1,0), 1 = initialization with speed: (1,0), (1,1), 2 = start like in mode 1, then 1 sample gradual adjustment: 2^-1, 3 = start like in mode 1, then 2 samples gradual adjustment: 2^-1, 2^-2, N = start like in mode 1, then (N-1) samples gradual adjustment: 2^-1, 2^-2, ... 2^-(N-1), 7 = start like in mode 1, then 6 samples gradual adjustment: 2^-1, 2^-2, ... 2^-6. After initialization, K1 and K2 are given by "cmtp_dyn_comp_k1" and "cmtp_dyn_comp_k2" settings.
dyn_comp_mode	rw	Dynamic compensation mode: 0 = continuous (default for all interfaces), 1 = discontinuous (recommended for SPI for optimal velocity noise).
oos_rotation	rw	OoS rotation for linear compensation of application imperfections in the range of −90 degree ≤ oos_rotation < +90 degree: −512 = -90 degree, 0 = 0 degree, 1 = 180 degree/2^10 = ca. 0.176 degree, 511 = +90 degree - 180 degree/2^10 = ca. 89.824 degree.

8.2.48 usr_config_14_reg

Register	usr_config_14_reg							
Address	0x4C							
Bit	15	14	13	12	11	10	9	8
Field	sm_vec_len_lim_min							sm_vec_len_lim_mode
Bit	7	6	5	4	3	2	1	0
Field	dyn_comp_lim			syscal_t_lin_lim		dyn_comp_en	dyn_comp_hyst	

Field	Type	Description
dyn_comp_hyst	rw	Dynamic compensation hysteresis: 0 = no hysteresis emulation (recommended for SPI to be able to resolve already the tiniest movements), 1 = +/-0.2 degree, 2 = +/-0.4 degree, 3 = +/-0.8 degree.
dyn_comp_en	rw	Dynamic compensation enable: 0 = disabled, 1 = enabled (default).
syscal_t_lin_lim	rw	Restart any ongoing data sampling of system self-calibration if "t_lin" changes by more than the configured limit: 0 = 1536LSB16 (ca. 10degC), 1 = 768LSB16 (ca. 5degC), 2 = disabled, 3 = 3072LSB16 (ca. 20degC).
dyn_comp_lim	rw	Limit for Kalman prediction plausibility, |(angle_spi - angle_expected)|. 0 = 0.5 degree, 1 = 1.0 degree, 2 = 2.0 degree, 3 = 4.0 degree, 4 = 8.0 degree, 5 = 16.0 degree, 6 = 32.0 degree and 7 = disabled (default).
sm_vec_len_lim_mode	rw	Vector length value used for safety mechanism smi_vhall_vector_length_check: 0 = fully compensated vector length (VECLEN), 1 = amplitude of raw magnetic field (VECLEN3).
sm_vec_len_lim_min	rw	Vector length minimum limit for smi_vhall_vector_length_check. Value is multiplied with 256 to match scaling of vector length value.

8.2.49 usr_config_15_reg

Register	usr_config_15_reg							
Address	0x4D							
Bit	15	14	13	12	11	10	9	8
Field	sm_vec_len_lim_max							
Bit	7	6	5	4	3	2	1	0
Field	usr_config_1_15_crc							

Field	Type	Description
usr_config_1_15_crc	rw	CRC checksum over user config registers 1 to 15 starting with register 1. Equal to CRC checksum over CMTP words 33 to 47 starting with word 33.
sm_vec_len_lim_max	rw	Vector length maximum limit for smi_vhall_vector_length_check. Value is multiplied with 128 to match scaling of vector length value.

8.2.50 sm_clk_cnt_reg

Register	sm_clk_cnt_reg							
Address	0x68							
Bit	15	14	13	12	11	10	9	8
Field	sm_clk_cnt...							
Bit	7	6	5	4	3	2	1	0
Field	...sm_clk_cnt							

Field	Type	Description
sm_clk_cnt	r	Clock counter running at 20MHz for external safety mechanism to perform clock period measurement at least once within tsafety.

8.2.51 reg_unlock_reg

Register	reg_unlock_reg							
Address	0x78							
Bit	15	14	13	12	11	10	9	8
Field	reg_unlock...							
Bit	7	6	5	4	3	2	1	0
Field	...reg_unlock							

Field	Type	Description
reg_unlock	rw	Unlock password for interface write access to registers and CMTP. Password: 0x4711

8.2.52 hw_fw_id_reg

Register	hw_fw_id_reg							
Address	0x79							
Bit	15	14	13	12	11	10	9	8
Field	-							
Bit	7	6	5	4	3	2	1	0
Field	hw_id				fw_id			

Field	Type	Description
fw_id	r	Firmware ID for readout by interface: 0 = A11, 1 = B11, 2 = B12, 3 = B13.
hw_id	r	Hardware ID for readout by interface: 0 = A11, 1 = B11, 2 = B12 and B13.

8.3 NVM registers

8.3.1 Register overview

Address	Register Name	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
253	cmtp_ctrl_reg	cmtp_test_prog_vr	cmtp_test_vref_to										cmtp_diagnostic_r	cmtp_prefetch	cmtp_selcustom	cmtp_write_en	cmtp_read_en
	12	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0
254	cmtp_stat_reg	cmtp_read_access	cmtp_vdd2v5_2	cmtp_vdd2v5_1	cmtp_program_access					cmtp_access_grant	cmtp_cell_virgin	cmtp_ecc_stat		cmtp_search_valid	cmtp_data_valid	cmtp_ahb_err_spi	cmtp_busy
	4	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
255	cmtp_stat_2_reg						cmtp_locked	cmtp_vddprog_2	cmtp_vddprog_1	cmtp_seq_code					cmtp_lock_fail_n	cmtp_out_of_ate	cmtp_data_check
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
256	cmtp_word_0_reg	cmtp_mpc_lut_0															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
257	cmtp_word_1_reg	cmtp_mpc_lut_1															
	4096	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
258	cmtp_word_2_reg	cmtp_mpc_lut_2															
	8192	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
259	cmtp_word_3_reg	cmtp_mpc_lut_3															
	12288	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0
260	cmtp_word_4_reg	cmtp_mpc_lut_4															
	16384	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
261	cmtp_word_5_reg	cmtp_mpc_lut_5															
	20480	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0
262	cmtp_word_6_reg	cmtp_mpc_lut_6															
	24576	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0
263	cmtp_word_7_reg	cmtp_mpc_lut_7															
	28672	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0
264	cmtp_word_8_reg	cmtp_mpc_lut_8															
	32768	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
265	cmtp_word_9_reg	cmtp_mpc_lut_9															
	36864	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
266	cmtp_word_10_reg	cmtp_mpc_lut_10															
	40960	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
267	cmtp_word_11_reg	cmtp_mpc_lut_11															
	45056	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0
268	cmtp_word_12_reg	cmtp_mpc_lut_12															
	49152	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
269	cmtp_word_13_reg	cmtp_mpc_lut_13															
	53248	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0
270	cmtp_word_14_reg	cmtp_mpc_lut_14															
	57344	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0
271	cmtp_word_15_reg	cmtp_mpc_lut_15															
	61440	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0
272	cmtp_word_16_reg	cmtp_mpc_lut_16															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
273	cmtp_word_17_reg	cmtp_mpc_lut_17															
	4096	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
274	cmtp_word_18_reg	cmtp_mpc_lut_18															
	8192	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
275	cmtp_word_19_reg	cmtp_mpc_lut_19															
	12288	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0
276	cmtp_word_20_reg	cmtp_mpc_lut_20															
	16384	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
277	cmtp_word_21_reg	cmtp_mpc_lut_21															
	20480	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0
278	cmtp_word_22_reg	cmtp_mpc_lut_22															
	24576	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0

Figure 29 NVM Registermap - part 1

Address	Register Name	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
279	cmtp_word_23_reg	cmtp_mpc_lut_23															
	28672	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0
280	cmtp_word_24_reg	cmtp_mpc_lut_24															
	32768	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
281	cmtp_word_25_reg	cmtp_mpc_lut_25															
	36864	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
282	cmtp_word_26_reg	cmtp_mpc_lut_26															
	40960	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
283	cmtp_word_27_reg	cmtp_mpc_lut_27															
	45056	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0
284	cmtp_word_28_reg	cmtp_mpc_lut_28															
	49152	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
285	cmtp_word_29_reg	cmtp_mpc_lut_29															
	53248	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0
286	cmtp_word_30_reg	cmtp_mpc_lut_30															
	57344	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0
287	cmtp_word_31_reg	cmtp_mpc_lut_31															
	61440	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0
288	cmtp_word_32_reg	cmtp_word_16_31_crc								cmtp_word_0_15_crc							
	51143	1	1	0	0	0	1	1	1	1	1	0	0	0	1	1	1
289	cmtp_word_33_reg	cmtp_adc_max_rn	cmtp_abz_init_pos	cmtp_abz_mode		cmtp_spi_miso_cr	cmtp_spi_mosi_cr	cmtp_sync_edge		cmtp_sync_mode		cmtp_pwm_start_	cmtp_pwm_freq			cmtp_if_mode	
	24	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0
290	cmtp_word_34_reg	cmtp_unused_290		cmtp_abz_z_mode		cmtp_abz_uvw_pulses_pp											
	4095	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1
291	cmtp_word_35_reg	cmtp_unused_291		cmtp_sm_fw_over	cmtp_sm_adc_ove	cmtp_sm_vddp_o	cmtp_sm_vddp_u	cmtp_sm_vddx_o	cmtp_sm_vdd2v5x	cmtp_sm_tv_rng_	cmtp_sm_diffstr	cmtp_sm_shear_st	cmtp_sm_t_lin_m	cmtp_sm_t_lin_mi	cmtp_sm_bitmap_	cmtp_sm_bitmap_	cmtp_sm_cmtp_lu
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
292	cmtp_word_36_reg	cmtp_unused_292											cmtp_ife_spike_fi	cmtp_ifd_spike_fi	cmtp_ifc_spike_fi	cmtp_ifb_spike_fi	cmtp_ifa_spike_fi
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
293	cmtp_word_37_reg	cmtp_x3_base															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
294	cmtp_word_38_reg	cmtp_y3_base															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
295	cmtp_word_39_reg	cmtp_angle_base															cmtp_angle_dir
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
296	cmtp_word_40_reg	cmtp_offset_app_x															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
297	cmtp_word_41_reg	cmtp_offset_app_y															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
298	cmtp_word_42_reg	cmtp_syscal_angle	cmtp_syscal_mode			cmtp_lut_mode		cmtp_exp_tan					cmtp_exp_rad				
	8192	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
299	cmtp_word_43_reg	cmtp_dyn_comp_k2								cmtp_dyn_comp_k1							
	13062	0	0	1	1	0	0	1	1	0	0	0	0	0	1	1	0
300	cmtp_word_44_reg	cmtp_ellipticity								cmtp_dyn_comp_tim							
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
301	cmtp_word_45_reg	cmtp_oos_rotation										cmtp_dyn_comp_	cmtp_dyn_comp_init_samples			cmtp_dyn_comp_	cmtp_dyn_comp_
	20	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0
302	cmtp_word_46_reg	cmtp_sm_vec_len_lim_min							cmtp_sm_vec_len	cmtp_dyn_comp_lim			cmtp_syscal_t_lin_lim		cmtp_dyn_comp_	cmtp_dyn_comp_hyst	
	3302	0	0	0	0	1	1	0	0	1	1	1	0	0	1	1	0
303	cmtp_word_47_reg	cmtp_sm_vec_len_lim_max								cmtp_word_33_47_crc							
	45181	1	0	1	1	0	0	0	0	0	1	1	1	1	1	0	1

Figure 30 NVM Registermap - part 2

8.3.2 cmtp_ctrl_reg

Register	cmtp_ctrl_reg							
Address	0xFD							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_test_prog_vreg_mon	cmtp_test_vref_to_ext	-					
Bit	7	6	5	4	3	2	1	0
Field	-			cmtp_diagnostic_read	cmtp_prefetch	cmtp_selcustom	cmtp_write_en	cmtp_read_en

Field	Type	Description
cmtp_read_en	rw	Read enable signal for CMTP: 0 = cmtp_disabled 1 = cmtp_enabled
cmtp_write_en	rw	Write enable signal for CMTP: 0 = cmtp_write_protected 1 = cmtp_write_enabled
cmtp_selcustom	rw	In customer mode just customer cells are programmable: 0 = cmtp_infineon_mode 1 = cmtp_customer_mode
cmtp_prefetch	rw	When CMTP prefetch is enabled the CMTP wrapper performs an prereadout of address + 1 to be faster: 0 = cmtp_disabled 1 = cmtp_enabled
cmtp_diagnostic_read	rw	Read of diagnostic information from AHB bus: 0 = cmtp_normal (16 LSB of 32 bit AHB data bus) 1 = cmtp_diagnostic (16 MSB of 32 bit AHB data bus) Note: It is recommended to set this bit for programming and testing the CMTP via SPI to get the correct status information instead of reading cmtp_stat_reg and cmtp_stat_2_reg registers. Status information is then provided in the MISO data of the subsequent next-frame access.
cmtp_test_vref_to_ext	rw	Analog parameter testing.
cmtp_test_prog_vreg_mon	rw	Analog parameter testing.

8.3.3 cmtp_stat_reg

Register	cmtp_stat_reg							
Address	0xFE							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_read_access	cmtp_vdd2v5_2	cmtp_vdd2v5_1	cmtp_program_access				
Bit	7	6	5	4	3	2	1	0
Field	cmtp_access_granted	cmtp_cell_virgin	cmtp_ecc_stat		cmtp_search_valid	cmtp_data_valid	cmtp_ahb_err_spi	cmtp_busy

Field	Type	Description
cmtp_busy	r	Status of CMTP: 0 = cmtp_ok 1 = cmtp_busy
cmtp_ahb_err_spi	r	AHB bus error indication (sticky bit): 0 = cmtp_ok 1 = cmtp_ahb_err Error indication is preserved till readout even if reads for other CMTP words do not trigger an AHB bus error!
cmtp_data_valid	r	Data valid indication: 0 = cmtp_data_not_valid 1 = cmtp_data_valid
cmtp_search_valid	r	Search valid indication: 0 = cmtp_data_not_valid 1 = cmtp_data_valid
cmtp_ecc_stat	r	0 = no_error 1 = one_error 2 = two_errors 3 = three_errors
cmtp_cell_virgin	r	Indication if the cell got programmed: 0 = cell_programmed 1 = cell_virgin
cmtp_access_granted	r	Indication if access is denied: 0 = access_denied 1 = access_granted
cmtp_program_access	r	Still available further program access.
cmtp_vdd2v5_1	r	Voltage VDD2V5 in ok.

Field	Type	Description
cmtp_vdd2v5_2	r	Voltage VDD2V5 out ok.
cmtp_read_access	r	Read acces of CMTP: 0 = not a read access 1 = read access

8.3.4 cmtp_stat_2_reg

Register	cmtp_stat_2_reg							
Address	0xFF							
Bit	15	14	13	12	11	10	9	8
Field	-					cmtp_locked	cmtp_vddprog_2	cmtp_vddprog_1
Bit	7	6	5	4	3	2	1	0
Field	cmtp_seq_code					cmtp_lock_fail_n	cmtp_out_of_attempt_n	cmtp_data_check

Field	Type	Description
cmtp_data_check	r	Indication that embedded data verification was successful: 0 = fail 1 = pass
cmtp_out_of_attempt_n	r	Programming failed due to already programmed cell: 0 = out of attempts, programming failed 1 = programming was possible
cmtp_lock_fail_n	r	Indication of impossibility to programm due to write protection: 0 = CMTP write protected, programming was not possible 1 = programming was possible
cmtp_seq_code	r	Sequence code: 00010 after successful programming.
cmtp_vddprog_1	r	Status prgramming voltage: 0 = invalid voltage 1 = valid voltage
cmtp_vddprog_2	r	Status prgramming voltage: 0 = invalid voltage 1 = valid voltage
cmtp_locked	r	CMTP lock status: 0 = cmtp_unlocked 1 = cmtp_locked

8.3.5 cmtp_word_0_reg

Register	cmtp_word_0_reg							
Address	0x100							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_mpc_lut_0...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_mpc_lut_0							

Field	Type	Description
cmtp_mpc_lut_0	r	Supporting point value for non-equidistant mode with 16 LUT values or supporting point's calibration value for equidistant mode with 32 LUT values.

8.3.6 cmtp_word_1_reg

Register	cmtp_word_1_reg							
Address	0x101							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_mpc_lut_1...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_mpc_lut_1							

Field	Type	Description
cmtp_mpc_lut_1	r	Supporting point value for non-equidistant mode with 16 LUT values or supporting point's calibration value for equidistant mode with 32 LUT values.

8.3.7 cmtp_word_2_reg

Register	cmtp_word_2_reg							
Address	0x102							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_mpc_lut_2...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_mpc_lut_2							

Field	Type	Description
cmtp_mpc_lut_2	r	Supporting point value for non-equidistant mode with 16 LUT values or supporting point's calibration value for equidistant mode with 32 LUT values.

8.3.8 cmtp_word_3_reg

Register	cmtp_word_3_reg							
Address	0x103							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_mpc_lut_3...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_mpc_lut_3							

Field	Type	Description
cmtp_mpc_lut_3	r	Supporting point value for non-equidistant mode with 16 LUT values or supporting point's calibration value for equidistant mode with 32 LUT values.

8.3.9 cmtp_word_4_reg

Register	cmtp_word_4_reg							
Address	0x104							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_mpc_lut_4...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_mpc_lut_4							

Field	Type	Description
cmtp_mpc_lut_4	r	Supporting point value for non-equidistant mode with 16 LUT values or supporting point's calibration value for equidistant mode with 32 LUT values.

8.3.10 cmtp_word_5_reg

Register	cmtp_word_5_reg							
Address	0x105							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_mpc_lut_5...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_mpc_lut_5							

Field	Type	Description
cmtp_mpc_lut_5	r	Supporting point value for non-equidistant mode with 16 LUT values or supporting point's calibration value for equidistant mode with 32 LUT values.

8.3.11 cmtp_word_6_reg

Register	cmtp_word_6_reg							
Address	0x106							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_mpc_lut_6...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_mpc_lut_6							

Field	Type	Description
cmtp_mpc_lut_6	r	Supporting point value for non-equidistant mode with 16 LUT values or supporting point's calibration value for equidistant mode with 32 LUT values.

8.3.12 cmtp_word_7_reg

Register	cmtp_word_7_reg							
Address	0x107							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_mpc_lut_7...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_mpc_lut_7							

Field	Type	Description
cmtp_mpc_lut_7	r	Supporting point value for non-equidistant mode with 16 LUT values or supporting point's calibration value for equidistant mode with 32 LUT values.

8.3.13 cmtp_word_8_reg

Register	cmtp_word_8_reg							
Address	0x108							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_mpc_lut_8...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_mpc_lut_8							

Field	Type	Description
cmtp_mpc_lut_8	r	Supporting point value for non-equidistant mode with 16 LUT values or supporting point's calibration value for equidistant mode with 32 LUT values.

8.3.14 cmtp_word_9_reg

Register	cmtp_word_9_reg							
Address	0x109							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_mpc_lut_9...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_mpc_lut_9							

Field	Type	Description
cmtp_mpc_lut_9	r	Supporting point value for non-equidistant mode with 16 LUT values or supporting point's calibration value for equidistant mode with 32 LUT values.

8.3.15 cmtp_word_10_reg

Register	cmtp_word_10_reg							
Address	0x10A							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_mpc_lut_10...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_mpc_lut_10							

Field	Type	Description
cmtp_mpc_lut_10	r	Supporting point value for non-equidistant mode with 16 LUT values or supporting point's calibration value for equidistant mode with 32 LUT values.

8.3.16 cmtp_word_11_reg

Register	cmtp_word_11_reg							
Address	0x10B							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_mpc_lut_11...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_mpc_lut_11							

Field	Type	Description
cmtp_mpc_lut_11	r	Supporting point value for non-equidistant mode with 16 LUT values or supporting point's calibration value for equidistant mode with 32 LUT values.

8.3.17 cmtp_word_12_reg

Register	cmtp_word_12_reg							
Address	0x10C							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_mpc_lut_12...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_mpc_lut_12							

Field	Type	Description
cmtp_mpc_lut_12	r	Supporting point value for non-equidistant mode with 16 LUT values or supporting point's calibration value for equidistant mode with 32 LUT values.

8.3.18 cmtp_word_13_reg

Register	cmtp_word_13_reg							
Address	0x10D							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_mpc_lut_13...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_mpc_lut_13							

Field	Type	Description
cmtp_mpc_lut_13	r	Supporting point value for non-equidistant mode with 16 LUT values or supporting point's calibration value for equidistant mode with 32 LUT values.

8.3.19 cmtp_word_14_reg

Register	cmtp_word_14_reg							
Address	0x10E							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_mpc_lut_14...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_mpc_lut_14							

Field	Type	Description
cmtp_mpc_lut_14	r	Supporting point value for non-equidistant mode with 16 LUT values or supporting point's calibration value for equidistant mode with 32 LUT values.

8.3.20 cmtp_word_15_reg

Register	cmtp_word_15_reg							
Address	0x10F							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_mpc_lut_15...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_mpc_lut_15							

Field	Type	Description
cmtp_mpc_lut_15	r	Supporting point value for non-equidistant mode with 16 LUT values or supporting point's calibration value for equidistant mode with 32 LUT values.

8.3.21 cmtp_word_16_reg

Register	cmtp_word_16_reg							
Address	0x110							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_mpc_lut_16...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_mpc_lut_16							

Field	Type	Description
cmtp_mpc_lut_16	r	Supporting point's calibration value for both equidistant and non-equidistant mode with 32 or 16 LUT values.

8.3.22 cmtp_word_17_reg

Register	cmtp_word_17_reg							
Address	0x111							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_mpc_lut_17...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_mpc_lut_17							

Field	Type	Description
cmtp_mpc_lut_17	r	Supporting point's calibration value for both equidistant and non-equidistant mode with 32 or 16 LUT values.

8.3.23 cmtp_word_18_reg

Register	cmtp_word_18_reg							
Address	0x112							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_mpc_lut_18...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_mpc_lut_18							

Field	Type	Description
cmtp_mpc_lut_18	r	Supporting point's calibration value for both equidistant and non-equidistant mode with 32 or 16 LUT values.

8.3.24 cmtp_word_19_reg

Register	cmtp_word_19_reg							
Address	0x113							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_mpc_lut_19...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_mpc_lut_19							

Field	Type	Description
cmtp_mpc_lut_19	r	Supporting point's calibration value for both equidistant and non-equidistant mode with 32 or 16 LUT values.

8.3.25 cmtp_word_20_reg

Register	cmtp_word_20_reg							
Address	0x114							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_mpc_lut_20...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_mpc_lut_20							

Field	Type	Description
cmtp_mpc_lut_20	r	Supporting point's calibration value for both equidistant and non-equidistant mode with 32 or 16 LUT values.

8.3.26 cmtp_word_21_reg

Register	cmtp_word_21_reg							
Address	0x115							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_mpc_lut_21...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_mpc_lut_21							

Field	Type	Description
cmtp_mpc_lut_21	r	Supporting point's calibration value for both equidistant and non-equidistant mode with 32 or 16 LUT values.

8.3.27 cmtp_word_22_reg

Register	cmtp_word_22_reg							
Address	0x116							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_mpc_lut_22...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_mpc_lut_22							

Field	Type	Description
cmtp_mpc_lut_22	r	Supporting point's calibration value for both equidistant and non-equidistant mode with 32 or 16 LUT values.

8.3.28 cmtp_word_23_reg

Register	cmtp_word_23_reg							
Address	0x117							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_mpc_lut_23...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_mpc_lut_23							

Field	Type	Description
cmtp_mpc_lut_23	r	Supporting point's calibration value for both equidistant and non-equidistant mode with 32 or 16 LUT values.

8.3.29 cmtp_word_24_reg

Register	cmtp_word_24_reg							
Address	0x118							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_mpc_lut_24...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_mpc_lut_24							

Field	Type	Description
cmtp_mpc_lut_24	r	Supporting point's calibration value for both equidistant and non-equidistant mode with 32 or 16 LUT values.

8.3.30 cmtp_word_25_reg

Register	cmtp_word_25_reg							
Address	0x119							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_mpc_lut_25...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_mpc_lut_25							

Field	Type	Description
cmtp_mpc_lut_25	r	Supporting point's calibration value for both equidistant and non-equidistant mode with 32 or 16 LUT values.

8.3.31 cmtp_word_26_reg

Register	cmtp_word_26_reg							
Address	0x11A							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_mpc_lut_26...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_mpc_lut_26							

Field	Type	Description
cmtp_mpc_lut_26	r	Supporting point's calibration value for both equidistant and non-equidistant mode with 32 or 16 LUT values.

8.3.32 cmtp_word_27_reg

Register	cmtp_word_27_reg							
Address	0x11B							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_mpc_lut_27...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_mpc_lut_27							

Field	Type	Description
cmtp_mpc_lut_27	r	Supporting point's calibration value for both equidistant and non-equidistant mode with 32 or 16 LUT values.

8.3.33 cmtp_word_28_reg

Register	cmtp_word_28_reg							
Address	0x11C							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_mpc_lut_28...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_mpc_lut_28							

Field	Type	Description
cmtp_mpc_lut_28	r	Supporting point's calibration value for both equidistant and non-equidistant mode with 32 or 16 LUT values.

8.3.34 cmtp_word_29_reg

Register	cmtp_word_29_reg							
Address	0x11D							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_mpc_lut_29...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_mpc_lut_29							

Field	Type	Description
cmtp_mpc_lut_29	r	Supporting point's calibration value for both equidistant and non-equidistant mode with 32 or 16 LUT values.

8.3.35 cmtp_word_30_reg

Register	cmtp_word_30_reg							
Address	0x11E							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_mpc_lut_30...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_mpc_lut_30							

Field	Type	Description
cmtp_mpc_lut_30	r	Supporting point's calibration value for both equidistant and non-equidistant mode with 32 or 16 LUT values.

8.3.36 cmtp_word_31_reg

Register	cmtp_word_31_reg							
Address	0x11F							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_mpc_lut_31...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_mpc_lut_31							

Field	Type	Description
cmtp_mpc_lut_31	r	Supporting point's calibration value for both equidistant and non-equidistant mode with 32 or 16 LUT values.

8.3.37 cmtp_word_32_reg

Register	cmtp_word_32_reg							
Address	0x120							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_word_16_31_crc							
Bit	7	6	5	4	3	2	1	0
Field	cmtp_word_0_15_crc							

Field	Type	Description
cmtp_word_0_15_crc	r	CRC checksum over lower 16 LUT entries.
cmtp_word_16_31_crc	r	CRC checksum over upper 16 LUT entries.

8.3.38 cmtp_word_33_reg

Register	cmtp_word_33_reg							
Address	0x121							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_adc_max_rng_en	cmtp_abz_init_pos_mode	cmtp_abz_mode	cmtp_spi_miso_crc_inv_dis	cmtp_spi_mosi_crc_dis	cmtp_sync_edge		cmtp_sync_mode...
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_sync_mode	cmtp_pwm_start_edge	cmtp_pwm_freq			cmtp_if_mode		

Field	Type	Description
cmtp_if_mode	r	Interface mode depending on chip variant: - TLE49012-S0001: 0 = SPI, 1 = ABZ, 2 = UVW, 3 = UVW, 4 = PWM only, 5 = ABZ + PWM, 6 = UVW + PWM, 7 = UVW + PWM. - chip variant = TLE49012-S0002: 0 = ABZ + PWM, 1 = ABZ, 2 = ABZ, 3 = ABZ, 4 = PWM only, 5 = ABZ + PWM, 6 = ABZ + PWM, 7 = ABZ + PWM. - chip variant = TLE49012-E0001: 0 = UVW + PWM, 1 = UVW, 2 = UVW, 3 = UVW, 4 = PWM only, 5 = UVW + PWM, 6 = UVW + PWM, 7 = UVW + PWM. - chip variant = TLI49012-E0001: 0 = SPI, 1 = ABZ, 2 = UVW, 3 = UVW, 4 = PWM only, 5 = ABZ + PWM, 6 = UVW + PWM, 7 = UVW + PWM. - chip variant = TLI49012-E0002: 0 = ABZ + PWM, 1 = ABZ, 2 = ABZ,

Field	Type	Description
		3 = ABZ, 4 = PWM only, 5 = ABZ + PWM, 6 = ABZ + PWM, 7 = ABZ + PWM.
cmtp_pwm_freq	r	PWM frequency: 0 = 250Hz, 1 = 500Hz, 2 = 1000Hz, 3 = 2000Hz (default), 4 = 2500Hz, 5 = 3000Hz, 6 = 3500Hz, 7 = 4000Hz.
cmtp_pwm_start_edge	r	PWM starting edge: 0 = falling edge, 1 = rising edge.
cmtp_sync_mode	r	Synchronization pin mode: 0 = measurement synchronization on configured edge of IFE for SPI or PWM (if active), 1 = trigger initial pulses for ABZ (if active) on configured edge of IFE or measurement synchronization on falling edge of IFA (CSN) only for SPI, 2...3 = trigger sensor reset on configured edge of IFE for ABZ, UVW or PWM.
cmtp_sync_edge	r	Synchronization pin trigger edge: 0 = disable, 1 = rising edge, 2 = falling edge, 3 = any edge.
cmtp_spi_mosi_crc_dis	r	SPI MOSI CRC check disable: 0 = MOSI CRC check enabled, 1 = MOSI CRC check disabled.
cmtp_spi_miso_crc_inv_dis	r	SPI MISO CRC inversion disable on read timeout: 0 = MISO CRC inverted on read timeout, 1 = MISO CRC not inverted on read timeout.
cmtp_abz_mode	r	ABZ interface mode. 0 = A/B mode, 1 = step/direction mode.
cmtp_abz_init_pos_mode	r	ABZ initial position mode: 0 = initialization pulses after startup resp. absolute position mode, 1 = no initialization pulses resp. relative position mode.

Field	Type	Description
cmtp_adc_max_rng_en	r	Enable large ADC range: 0 = small range (default), 1 = large range.

8.3.39 cmtp_word_34_reg

Register	cmtp_word_34_reg							
Address	0x122							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_unused_290		cmtp_abz_z_mode		cmtp_abz_uvw_pulses_pp...			
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_abz_uvw_pulses_pp							

Field	Type	Description
cmtp_abz_uvw_pulses_pp	r	ABZ/UVW interface pulses resp. pole-pairs. For UVW interface, the least-significant 4 bits define the number of pole-pairs.
cmtp_abz_z_mode	r	ABZ index/Z pulse mode: 0 = ungated for tz (typ. 5 us), 1 = A/B gated (only for A/B mode, fallback to ungated for step/dir mode), 2 = A gated, 3 = B gated (only for A/B mode, fallback to ungated for step/dir mode).
cmtp_unused_290	r	

8.3.40 cmtp_word_35_reg

Register	cmtp_word_35_reg							
Address	0x123							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_unused_291		cmtp_sm_fw_overflow_dis	cmtp_sm_adc_overflow_oob_dis	cmtp_sm_vddp_ov_dis	cmtp_sm_vddp_uv_dis	cmtp_sm_vddx_ouv_dis	cmtp_sm_vdd2v5x_ouv_dis
Bit	7	6	5	4	3	2	1	0
Field	cmtp_sm_tv_rng_dis	cmtp_sm_diff_stress_rng_dis	cmtp_sm_shear_stress_rng_dis	cmtp_sm_t_lin_max_dis	cmtp_sm_t_lin_min_dis	cmtp_sm_bitmap_usr_crc_dis	cmtp_sm_bitmap_lut_crc_dis	cmtp_sm_cmtp_lut_usr_crc_dis

Field	Type	Description
cmtp_sm_cmtp_lut_usr_crc_dis	r	Combined status flag disable for CRC errors in MPC LUT and user config area of CMTP: 0 = error indicated (default), 1 = error indication suppressed. Note that this bit has no influence on the download of the CMTP user area, as the corresponding CRC checksum has to be checked before this bit can be downloaded. Hence CRC errors in the CMTP user area cannot be masked with this bit to change the startup behavior of the sensor (e.g. transition from OM_startup to OM_running). The same basically applies to CRC errors in the CMTP MPC LUT area. As a result, the bit only suppresses error indications via the corresponding SPI status flags.
cmtp_sm_bitmap_lut_crc_dis	r	Status flag disable for CRC error in MPC LUT area of bitmap: 0 = error indicated (default), 1 = error indication suppressed.
cmtp_sm_bitmap_usr_crc_dis	r	Status flag disable for CRC error in user config area of bitmap: 0 = error indicated (default), 1 = error indication suppressed.
cmtp_sm_t_lin_min_dis	r	Status flag disable for compensated temperature below minimum operating temperature: 0 = error indicated (default), 1 = error indication suppressed.
cmtp_sm_t_lin_max_dis	r	Status flag disable for compensated temperature above maximum operating temperature: 0 = error indicated (default), 1 = error indication suppressed.
cmtp_sm_shear_stress_rng_dis	r	Status flag disable for shear stress value outside of the expected +/-100MPa: 0 = error indicated (default), 1 = error indication suppressed.

Field	Type	Description
cmtp_sm_diff_stress_rng_dis	r	Status flag disable for differential stress value outside of the expected +/-400MPa: 0 = error indicated (default), 1 = error indication suppressed.
cmtp_sm_tv_rng_dis	r	Combined status disable flag for test vectors out of range errors: 0 = error indicated (default), 1 = error indication suppressed.
cmtp_sm_vdd2v5x_ouv_dis	r	Combined status disable flag for VDD2V5x over- and undervoltage errors: 0 = error indicated (default), 1 = error indication suppressed.
cmtp_sm_vddx_ouv_dis	r	Combined status disable flag for VDDA/VDDD over- and undervoltage errors: 0 = error indicated (default), 1 = error indication suppressed.
cmtp_sm_vddp_uv_dis	r	Status disable flag for VDDP undervoltage error: 0 = error indicated (default), 1 = error indication suppressed.
cmtp_sm_vddp_ov_dis	r	Status disable flag for VDDP overvoltage error: 0 = error indicated (default), 1 = error indication suppressed.
cmtp_sm_adc_overflow_oob_dis	r	Combined status flag disable for overflow and out of bound errors X, Y and GP ADC channels: 0 = error indicated (default), 1 = error indication suppressed.
cmtp_sm_fw_overflow_dis	r	Combined status flag disable for firmware and CORDIC overflow errors: 0 = error indicated (default), 1 = error indication suppressed.
cmtp_unused_291	r	

8.3.41 cmtp_word_36_reg

Register	cmtp_word_36_reg							
Address	0x124							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_unused_292...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_unused_292			cmtp_ife_spike_filter_en	cmtp_ifd_spike_filter_en	cmtp_ifc_spike_filter_en	cmtp_ifb_spike_filter_en	cmtp_ifa_spike_filter_en

Field	Type	Description
cmtp_ifa_spike_filter_en	r	IFA input spike filter enable: 0 = disabled, 1 = enabled.
cmtp_ifb_spike_filter_en	r	IFB input spike filter enable: 0 = disabled, 1 = enabled.
cmtp_ifc_spike_filter_en	r	IFC input spike filter enable: 0 = disabled, 1 = enabled.
cmtp_ifd_spike_filter_en	r	IFD input spike filter enable: 0 = disabled, 1 = enabled.
cmtp_ife_spike_filter_en	r	IFE input spike filter enable: 0 = disabled, 1 = enabled.
cmtp_unused_292	r	

8.3.42 cmtp_word_37_reg

Register	cmtp_word_37_reg							
Address	0x125							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_x3_base...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_x3_base							

Field	Type	Description
cmtp_x3_base	r	X3 value corresponding to programmed angle base (required for updating angle base during system self-calibration).

8.3.43 cmtp_word_38_reg

Register	cmtp_word_38_reg							
Address	0x126							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_y3_base...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_y3_base							

Field	Type	Description
cmtp_y3_base	r	Y3 value corresponding to programmed angle base (required for updating angle base during system self-calibration).

8.3.44 cmtp_word_39_reg

Register	cmtp_word_39_reg							
Address	0x127							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_angle_base...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_angle_base							cmtp_angle_dir

Field	Type	Description
cmtp_angle_dir	r	Angle direction setting: 0 = CCW: angle and angle velocity values not inverted, 1 = CW: two's complement of angle and angle velocity values.
cmtp_angle_base	r	Angle base resp. zero angle calibration setting in the range of 0 to 360 degree with a resolution of (360 degree / 32768 LSB) = 0.011 degree/LSB.

8.3.45 cmtp_word_40_reg

Register	cmtp_word_40_reg							
Address	0x128							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_offset_app_x...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_offset_app_x							

Field	Type	Description
cmtp_offset_app_x	r	Application X channel offset value.

8.3.46 cmtp_word_41_reg

Register	cmtp_word_41_reg							
Address	0x129							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_offset_app_y...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_offset_app_y							

Field	Type	Description
cmtp_offset_app_y	r	Application Y channel offset value.

8.3.47 cmtp_word_42_reg

Register	cmtp_word_42_reg							
Address	0x12A							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_syscal_angle_base_en	cmtp_syscal_mode			cmtp_lut_mode		-	
Bit	7	6	5	4	3	2	1	0
Field	-							

Field	Type	Description
cmtp_lut_mode	r	Multipoint calibration mode: 0 and 1 = no multipoint calibration, 2 = equidistant mode with 32 calibration points, 3 = non-equidistant mode with 16 calibration points.
cmtp_syscal_mode	r	System self-calibration setting: 0 = syscal disabled, 1 = vector length calibration, filter disabled, 2 = quadrature calibration, offset disabled, 3 = defaults to 0, 4 = quadrature calibration, offset for strayfield, 5 = vector length calibration, filter enabled, 6...7 = defaults to 0.
cmtp_syscal_angle_base_en	r	Angle base update by system self-calibration: 0 = disabled, 1 = enabled.

8.3.48 cmtp_word_43_reg

Register	cmtp_word_43_reg							
Address	0x12B							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_dyn_comp_k2							
Bit	7	6	5	4	3	2	1	0
Field	cmtp_dyn_comp_k1							

Field	Type	Description
cmtp_dyn_comp_k1	r	Dynamic compensation factor K1, influencing effective bandwidth of predicted angle values. See user manual for further explanations. Range from 0.02 to 1.00 in steps of 0.02: 0 = 0.02 = relaxed dynamic compensation, reduced angle noise, increased dynamic errors due to acceleration, 6 = 0.14 = compromise for fast and agile applications (default), 11 = 0.24 = very responsive dynamic compensation, quick response to acceleration, increased angle noise, 255 = 1.00 = max effective bandwidth, no averaging, quickest response to acceleration, excessive angle noise.
cmtp_dyn_comp_k2	r	Dynamic compensation factor K2, influencing effective bandwidth of angle velocity values. See user manual for further explanations. Range from 4.88e-4 to 0.125 in steps of 4.88e-4: 0 = 4.88e-4 = relaxed dynamic compensation, reduced angle noise, increased dynamic errors due to acceleration, 51 = 0.025 = compromise for fast and agile applications (default), 108 = 0.053 = very responsive dynamic compensation, quick response to acceleration, increased angle noise, 255 = 0.125 = max effective bandwidth, quickest response to acceleration, excessive angle (velocity) noise. Note that a value of 1.00 would correspond to no averaging at all for calculating the angle velocity. Such large values are only used upon initialization of the dynamic compensation, c.f. register fields starting with "dyn_comp_init_".

8.3.49 cmtp_word_44_reg

Register	cmtp_word_44_reg							
Address	0x12C							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_ellipticity							
Bit	7	6	5	4	3	2	1	0
Field	cmtp_dyn_comp_time							

Field	Type	Description
cmtp_dyn_comp_time	r	Time adjustment for the dynamic compensation in units of 0.55 us/LSB. Positive values result in over-compensation of signal path latency, i.e. predicted angle values lead the magnetic angle. Negative values result in under-compensation of signal path latency, i.e. predicted angle values lag behind the magnetic angle.
cmtp_ellipticity	r	Ellipticity setting calculated as (1.00 + 0.01 * N). Calibrated by customer. 0 = 1.00, 1 = 1.01, ... 255 = 3.55.

8.3.50 cmtp_word_45_reg

Register	cmtp_word_45_reg							
Address	0x12D							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_oos_rotation...							
Bit	7	6	5	4	3	2	1	0
Field	...cmtp_oos_rotation		cmtp_dyn_comp_mode	cmtp_dyn_comp_init_samples			cmtp_dyn_comp_init_k1	cmtp_dyn_comp_init_cond

Field	Type	Description
cmtp_dyn_comp_init_cond	r	Conditions upon which to initialize the dynamic compensation: 0 = only on startup, 1 = on startup and whenever prediction plausibility fails (according to the value of "dyn_comp_lim").
cmtp_dyn_comp_init_k1	r	Gradual adjustment of K1 during initialization: 0 = K1 remains fixed at 1.0 during initialization, 1 = gradually adjust K1 during initialization. Note that K2 is always gradually adjusted during initialization. Gradual adjustment means K value is reduced by 1/2 for each sample (e.g. start with 1.0, then 0.5, 0.25, 0.125, etc.).
cmtp_dyn_comp_init_samples	r	(N-1) number of samples for initialization of dynamic compensation with (K1, K2) in brackets: 0 = initialization without speed: (1,0), 1 = initialization with speed: (1,0), (1,1), 2 = start like in mode 1, then 1 sample gradual adjustment: 2^-1, 3 = start like in mode 1, then 2 samples gradual adjustment: 2^-1, 2^-2, N = start like in mode 1, then (N-1) samples gradual adjustment: 2^-1, 2^-2, ... 2^-(N-1), 7 = start like in mode 1, then 6 samples gradual adjustment: 2^-1, 2^-2, ... 2^-6. After initialization, K1 and K2 are given by "cmtp_dyn_comp_k1" and "cmtp_dyn_comp_k2" settings.
cmtp_dyn_comp_mode	r	Dynamic compensation mode: 0 = continuous (default for all interfaces), 1 = discontinuous (recommended for SPI for optimal velocity noise).

Field	Type	Description
cmtp_oos_rotation	r	OoS rotation for linear compensation of application imperfections in the range of -90 degree ≤ oos_rotation < +90 degree: −512 = -90 degree, 0 = 0 degree, 1 = 180 degree/2^10 = ca. 0.176 degree, 511 = +90 degree - 180 degree/2^10 = ca. 89.824 degree.

8.3.51 cmtp_word_46_reg

Register	cmtp_word_46_reg							
Address	0x12E							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_sm_vec_len_lim_min							cmtp_sm_vec_len_lim_mode
Bit	7	6	5	4	3	2	1	0
Field	cmtp_dyn_comp_lim			cmtp_syscal_t_lin_lim		cmtp_dyn_comp_en	cmtp_dyn_comp_hyst	

Field	Type	Description
cmtp_dyn_comp_hyst	r	Dynamic compensation hysteresis: 0 = no hysteresis emulation (recommended for SPI to be able to resolve already the tiniest movements), 1 = +/-0.2 degree, 2 = +/-0.4 degree, 3 = +/-0.8 degree.
cmtp_dyn_comp_en	r	Dynamic compensation enable: 0 = disabled, 1 = enabled (default).
cmtp_syscal_t_lin_lim	r	Restart any ongoing data sampling of system self-calibration if "t_lin" changes by more than the configured limit: 0 = 1536LSB16 (ca. 10degC), 1 = 768LSB16 (ca. 5degC), 2 = disabled, 3 = 3072LSB16 (ca. 20degC).
cmtp_dyn_comp_lim	r	Limit for Kalman prediction plausibility, |(angle_spi - angle_expected)|. 0 = 0.5 degree, 1 = 1.0 degree, 2 = 2.0 degree, 3 = 4.0 degree, 4 = 8.0 degree, 5 = 16.0 degree, 6 = 32.0 degree and 7 = disabled (default).
cmtp_sm_vec_len_lim_mode	r	Vector length value used for safety mechanism smi_vhall_vector_length_check: 0 = fully compensated vector length (VECLEN), 1 = amplitude of raw magnetic field (VECLEN3).

Field	Type	Description
cmtp_sm_vec_len_lim_min	r	Vector length minimum limit for smi_vhall_vector_length_check. Value is multiplied with 256 to match scaling of vector length value.

8.3.52 cmtp_word_47_reg

Register	cmtp_word_47_reg							
Address	0x12F							
Bit	15	14	13	12	11	10	9	8
Field	cmtp_sm_vec_len_lim_max							
Bit	7	6	5	4	3	2	1	0
Field	cmtp_word_33_47_crc							

Field	Type	Description
cmtp_word_33_47_crc	r	CRC checksum over CMTP words 33 to 47 starting with word 33.
cmtp_sm_vec_len_lim_max	r	Vector length maximum limit for smi_vhall_vector_length_check. Value is multiplied with 128 to match scaling of vector length value.

9 Glossary

Table 13 Glossary of used abbreviations

Abbreviation	Description
ABZ	incremental interface
ADC	analog to digital converter
AHB bus	Advanced High-performance Bus used to access the CMTP (non volatile memory), refer to cmtp_ctrl_reg and cmtp_stat_reg
angle_expected	expected angle value according to the dynamic filter, refer to usr_config_14_reg
BIST	built-in self-test
CMTP	customer multi-time programmable (nonvolatile memory)
CORDIC	Coordinate Rotation Digital Computer - Algorithm for angle calc.
CRC	cyclic redundancy check
DIAG	diagnosis
DSP	digital signal processing
ECC	error correction code for memory protection
FOC	field orientated control loop
FW	firmware
GP ADC	General purpose ADC - part of internal ADC architecture, measuring temperatures, stresses, and supply-voltages.
HMI	human machine interface
IFX	Infineon
IFx	interface pin (x = A, B, C, D, E)
LUT	look-up table
MPC	multi-point calibration
NVM	non volatile memory
OM_running	Sensor running operation mode, refer to usr_config_3_reg
OM_startup	Sensor start-up operating mode, refer to usr_config_3_reg
OT	over temperature
PWM	pulse width modulated interface
SAR	successive-approximation ADC - part of internal ADC architecture
SPI	serial peripheral interface
SysCal	System self-calibration feature of TLx49012 sensor family
UVW	hall switch/trapezoidal interface
VDD	Voltage level on VDD pin
VDD2V5A	sensor internal voltage domain
VDD2V5D	sensor internal voltage domain

(table continues...)

Table 13 (continued) Glossary of used abbreviations

Abbreviation	Description
VDDA	sensor internal voltage domain
VDDD	sensor internal voltage domain
VDDP	sensor supply voltage domain, equal to VDD
VHall	vertical Hall probes, refer to e.g. stat_3_reg
X3/Y3	represents non-SysCal-compensated magnetic field components
Xn/Yn	represents SysCal-compensated magnetic field components

10 Revision history

Revision number	Date of release	Description of changes
1.00	2026-07-16	Initial release

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